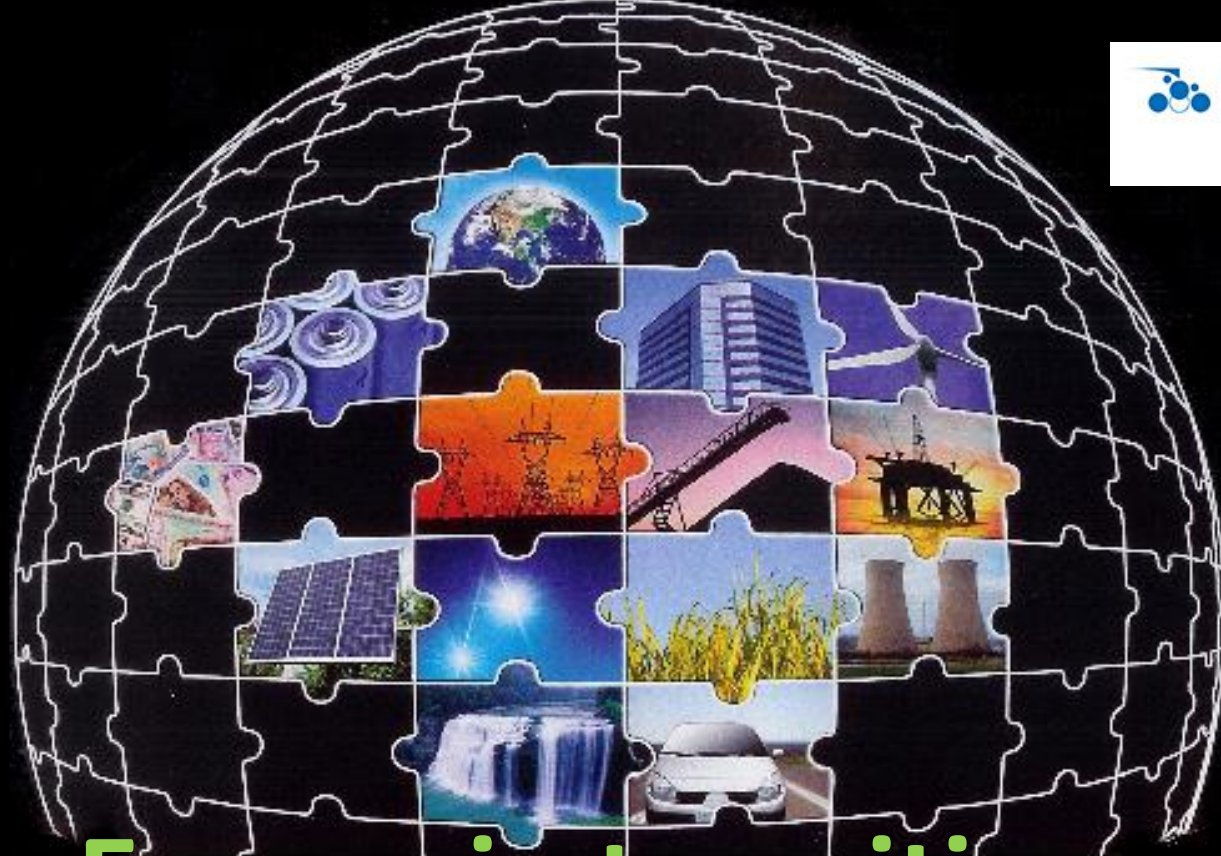




Energy in transition.

A case for Science, Technology and Society

Prof. Pedro Gómez-Romero

Institut Català de Nanociència i Nanotecnologia, ICN2 (CSIC-BIST), Bellaterra (Barcelona)
pedro.gomez@icn2.cat

SEVERO OCHOA SUMMER SCHOOL IN MATERIALS FOR ENERGY
ICMAB, Bellaterra, 17-20 Sep 2018



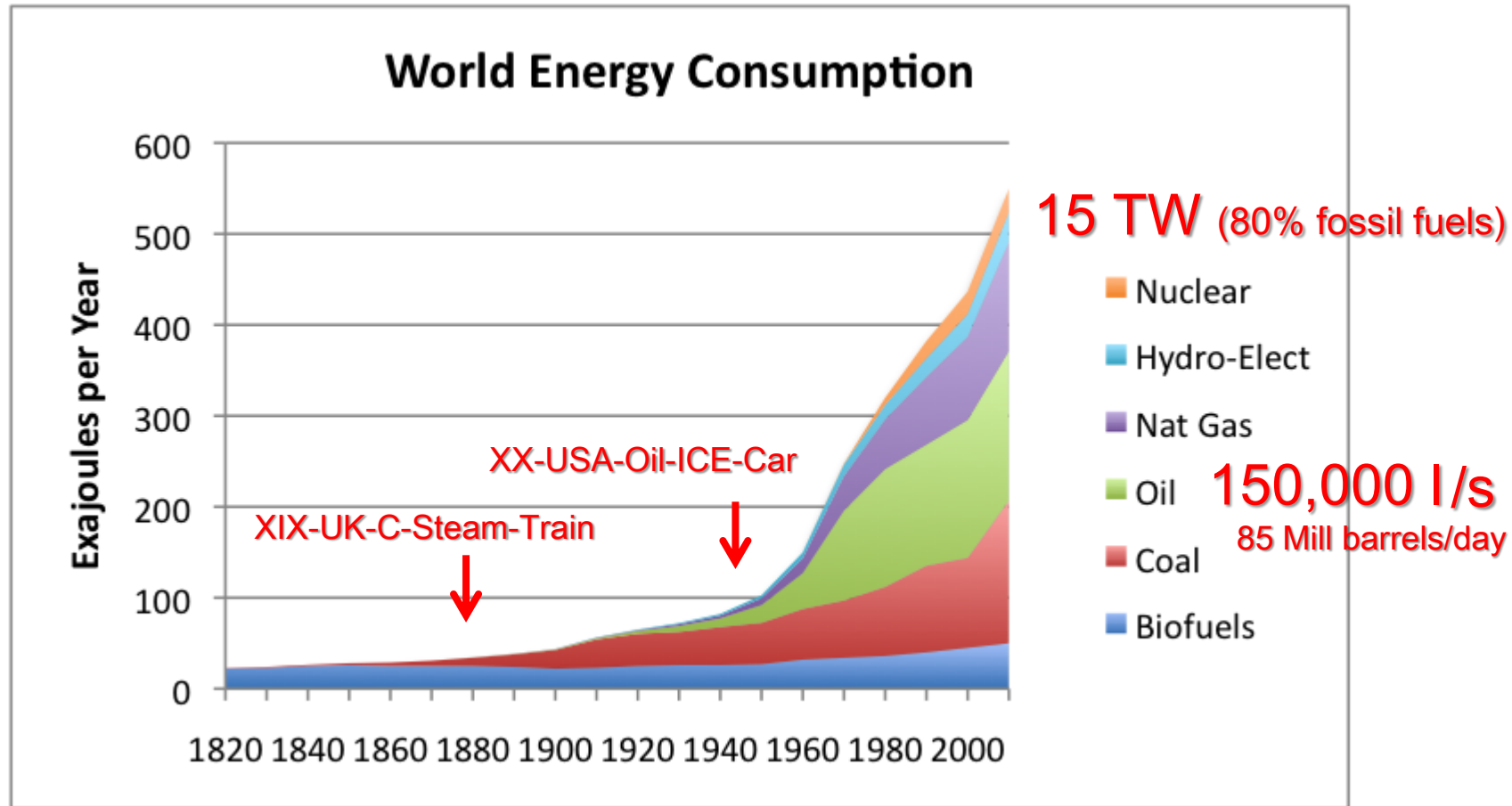
The energy conundrum

20th Century: wasted energy



an alien point of view

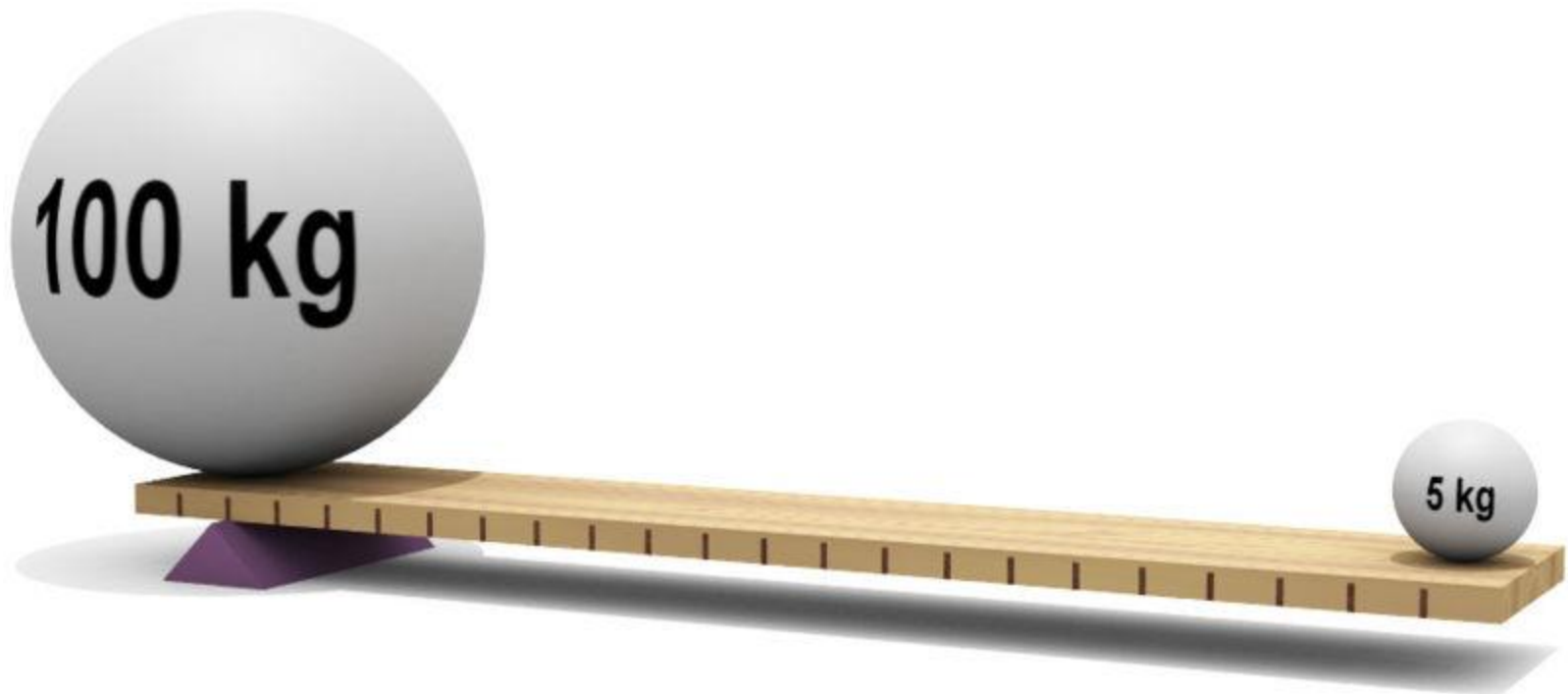
Energy Evolution. The past

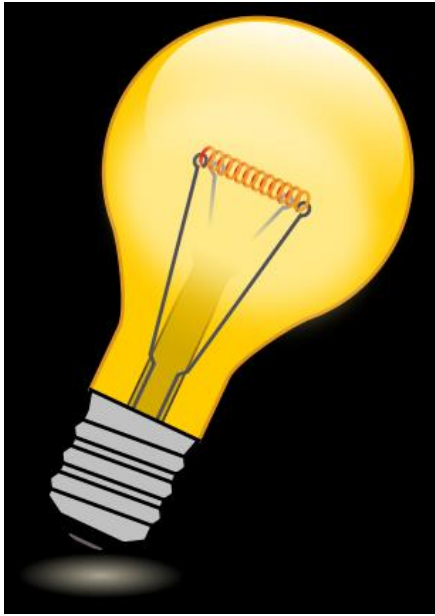


Based on Vaclav Smil estimates from Energy Transitions: History, Requirements and Prospects together with BP Statistical Data for 1965 and subsequent

Human power vs. human needs

A story of self(in)sufficiency





20th Century light bulb

100 W





20th Century CRT TV

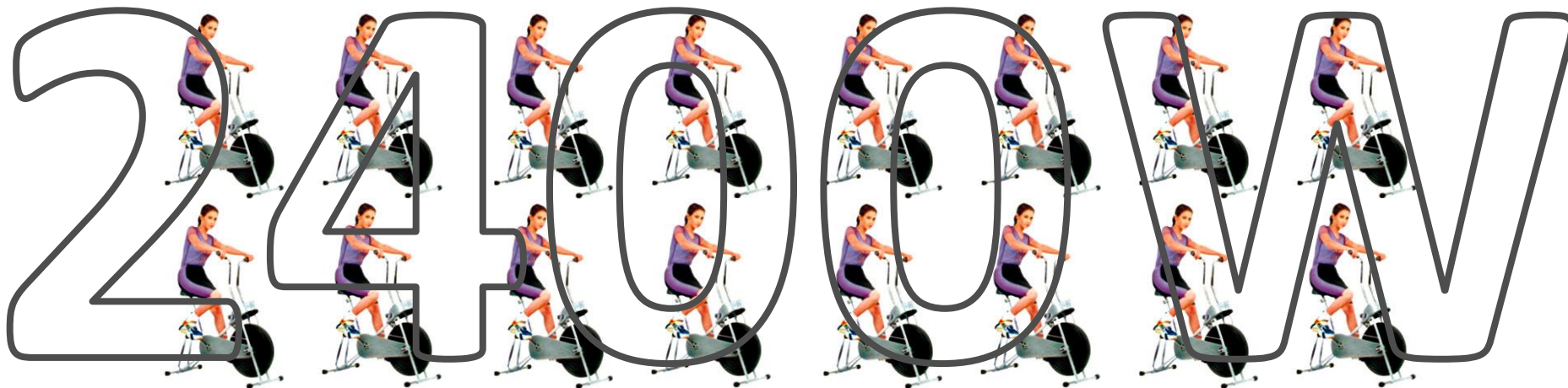
400 W





1000 W





100CV 73600 W



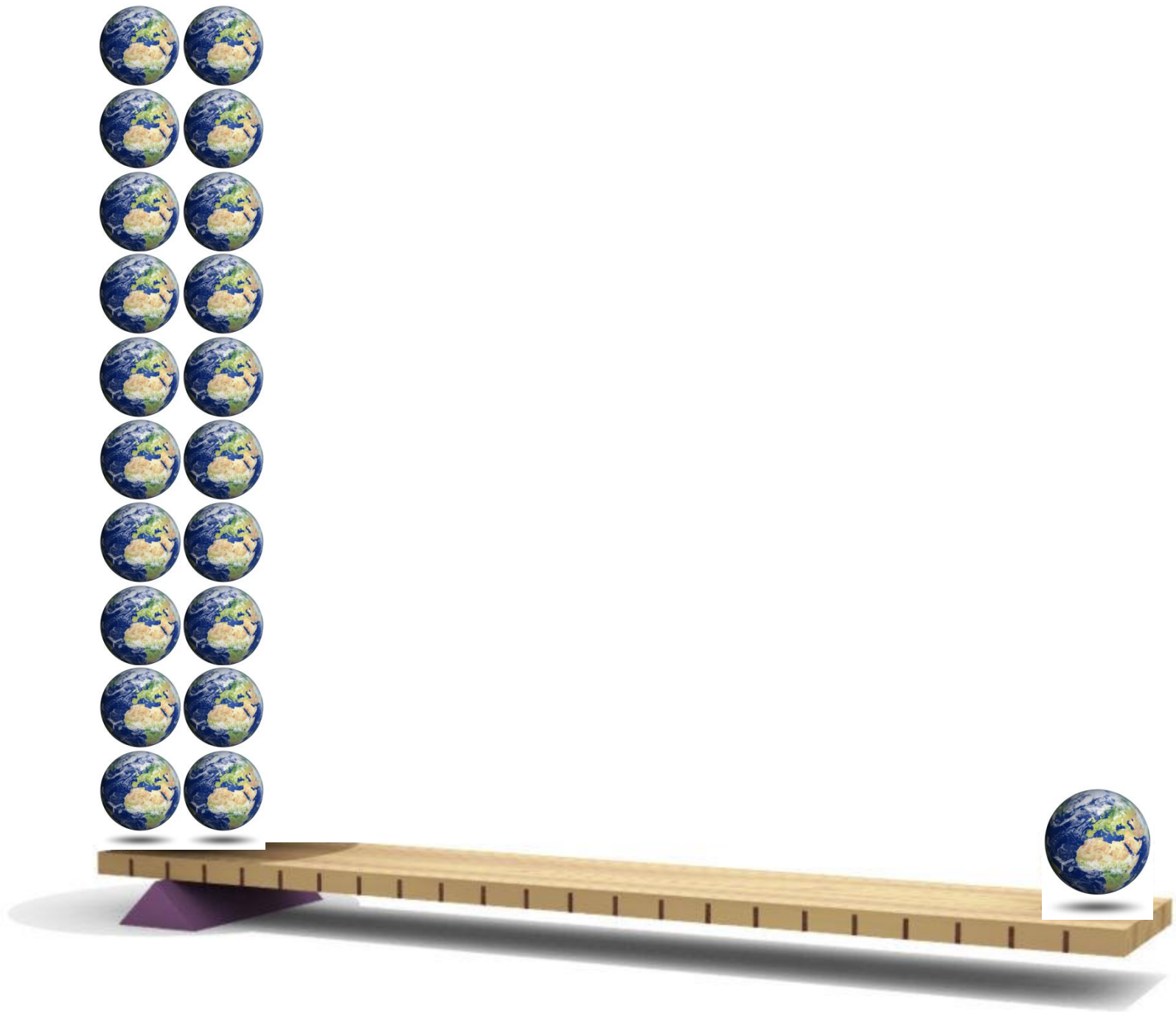
15TW = 15,000,000,000,000 watts

100 watts/human

150,000,000,000 humans

7,300,000,000 humans/Earth

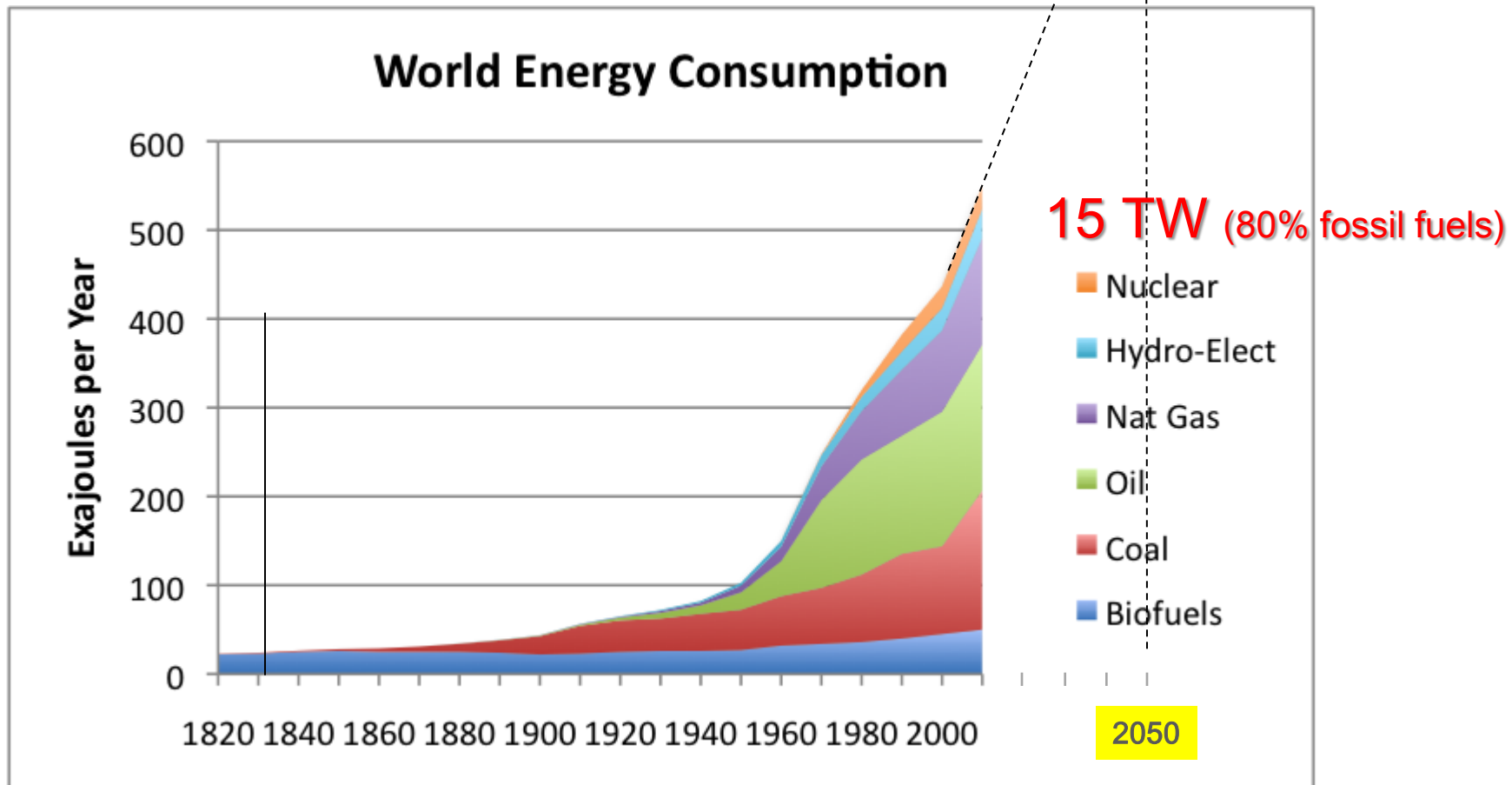
20 planet Earths



Energy Evolution. The future?

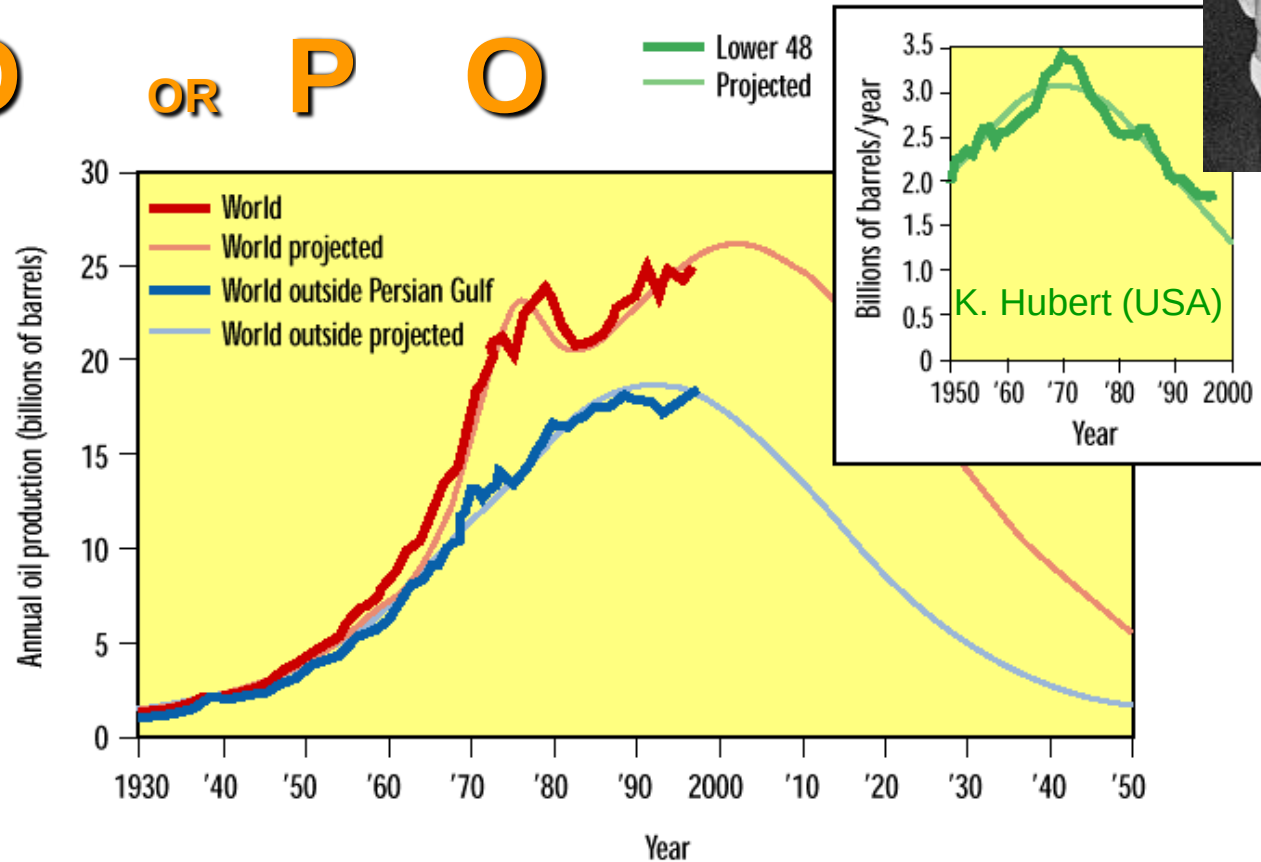
28 TW

% fossil fuels?

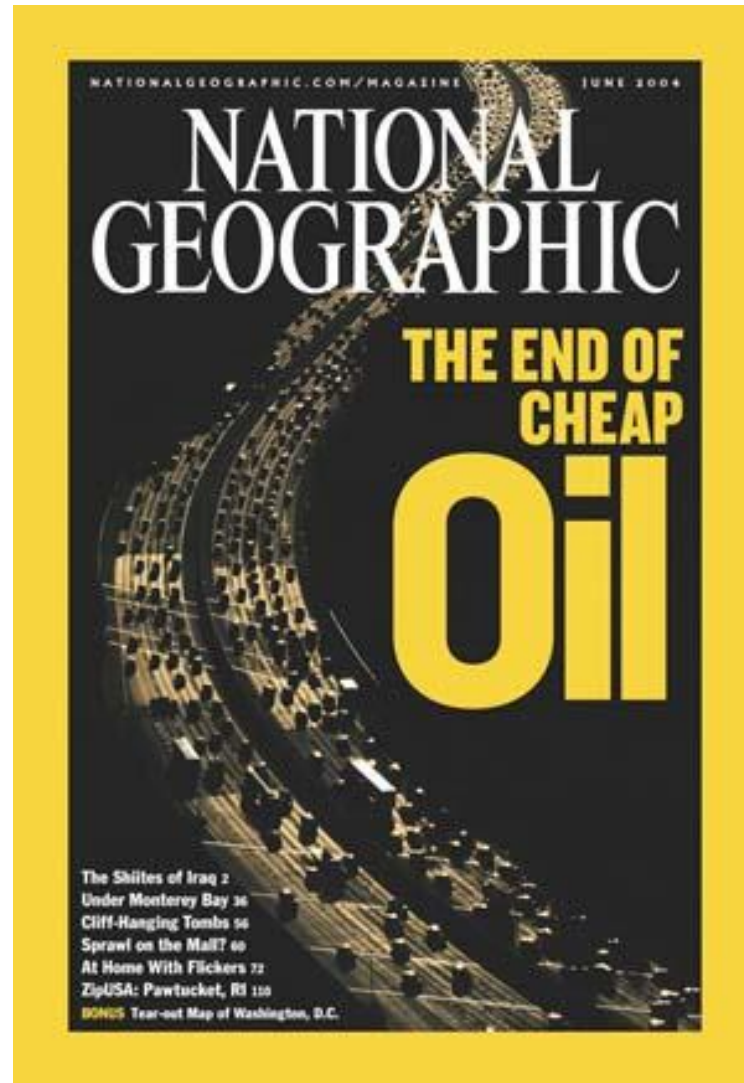


Hubbert's Peak Oil

P O OR P O



C. Campbell & J. Laherrere, Scientific American, March 1998
See also Science, vol. 281, Aug. 21, 1998, p.1128



Junio 2004, Volumen 205, Numero 6

Deepwater Horizon oil spill. 2010

largest marine oil spill in history,



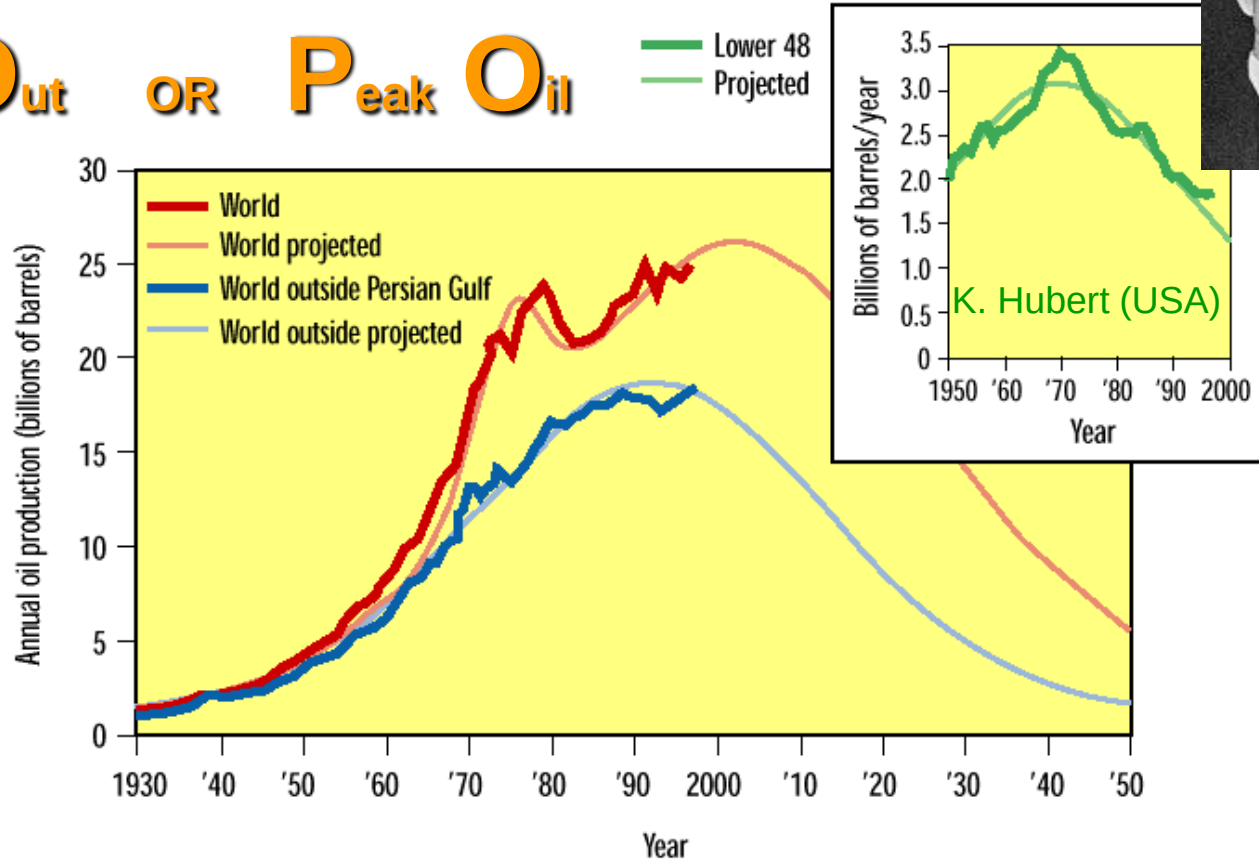
Who is to blame?



© "El Roto" at El País

Hubbert's Peak Oil

Phase Out OR Peak Oil



C. Campbell & J. Laherrere, Scientific American, March 1998
See also Science, vol. 281, Aug. 21, 1998, p.1128

It has happened before

The peak-buffalo story (I mean...history)



From buffalo-bullies ...to cow-boys

From gatherers to farmers A historical perspective

Kiss good bye your “primary” fuel
and
pick your favorite energy vectors



Bio-vectors: Biofuels

Biodiesel (20% bio 80% diesel), (oileaginous)

Bioethanol (Gasoline)(carbohydrates) (Brazil)

Electro-vectors: Batteries. Supercaps / Electric Vehicle

Chem-vectors: H_2 Fuel Cells, CO_2 red. products, solar fuels

How to change? What to change?

RENEWABLE
energy sources

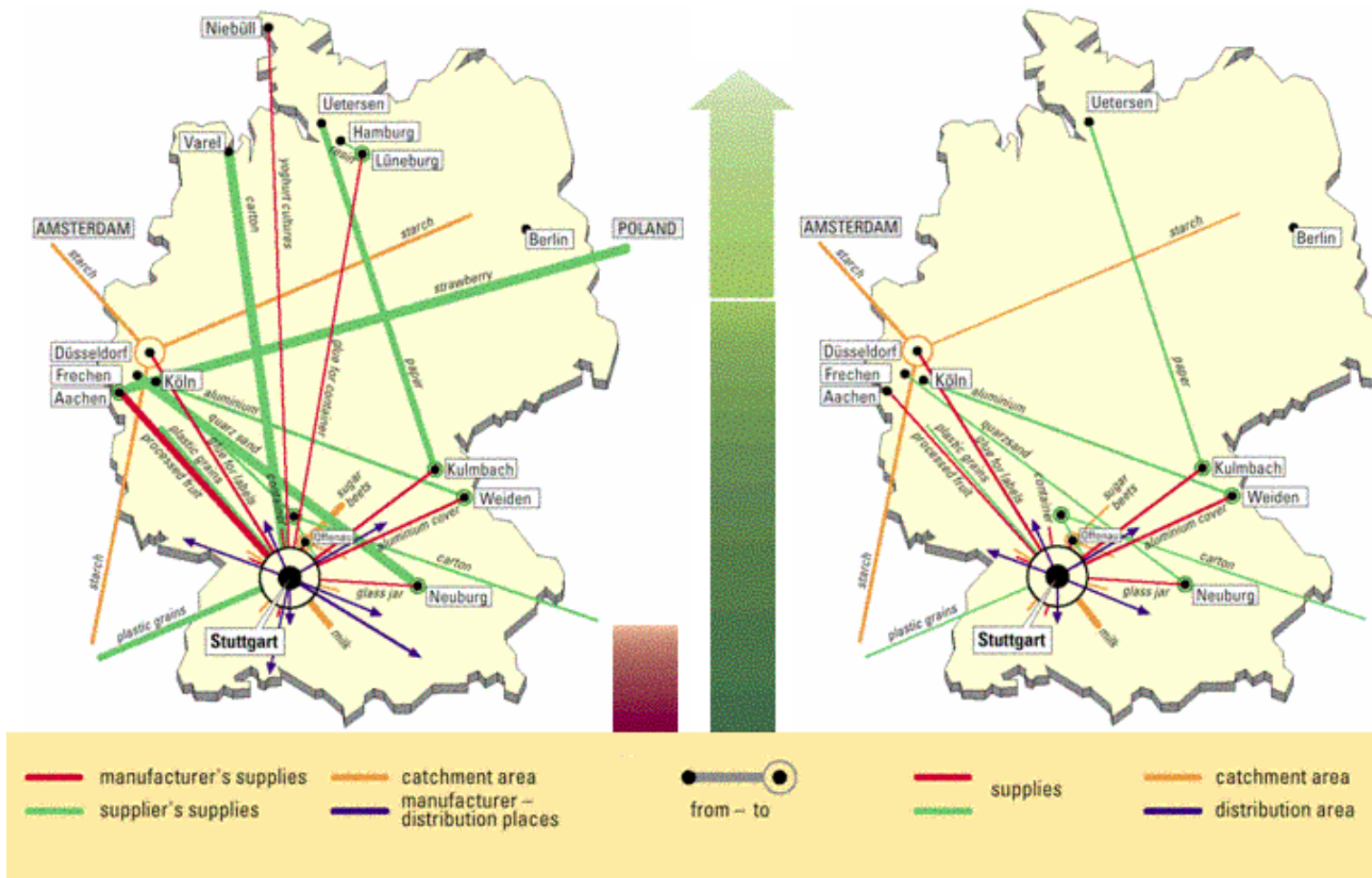


CLEANER
fuels
vectors

ENERGY
EFFICIENCY &
CONSERVATION

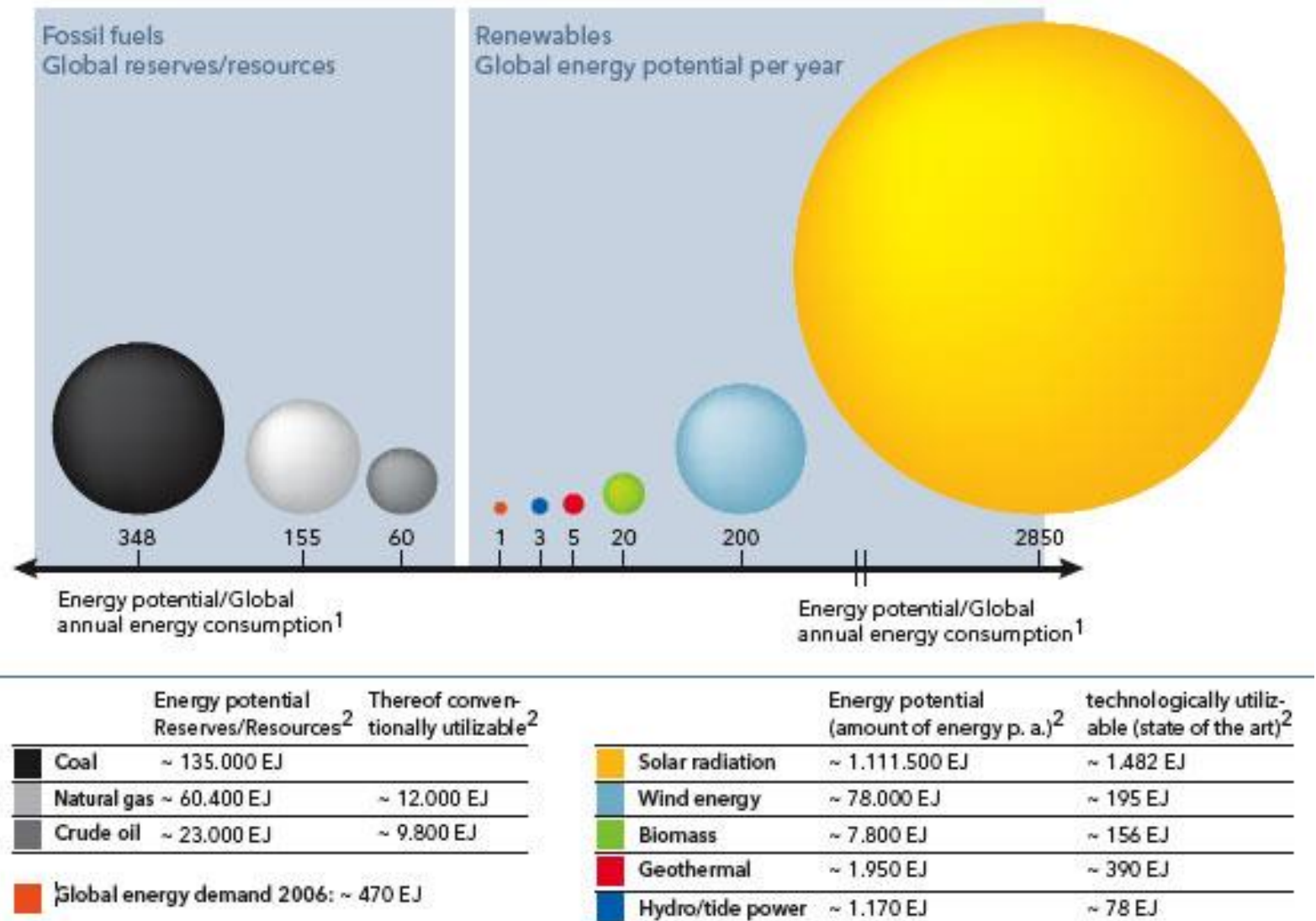
Eficiencia energética / Cadena de suministro

Strawberry yoghurt



Fuente: E.U. von Weizsäcker. "China: Building a Resource-Efficient Society" Beijing, 25 June, 2005

Primary sources: archaic or present-time Sun?

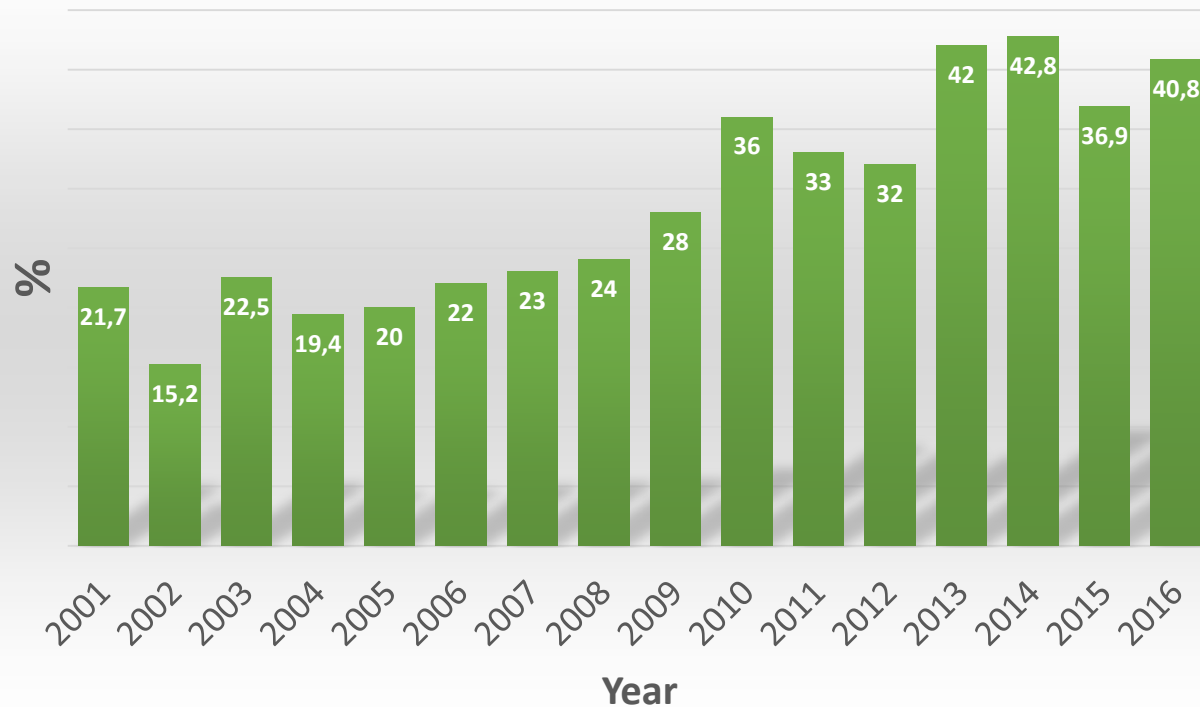


1 EJ = 1 exajoule or 10^{18} joules or ~163 million barrels of oil.

Data source: German Federal Institute for Geosciences and Natural Resources

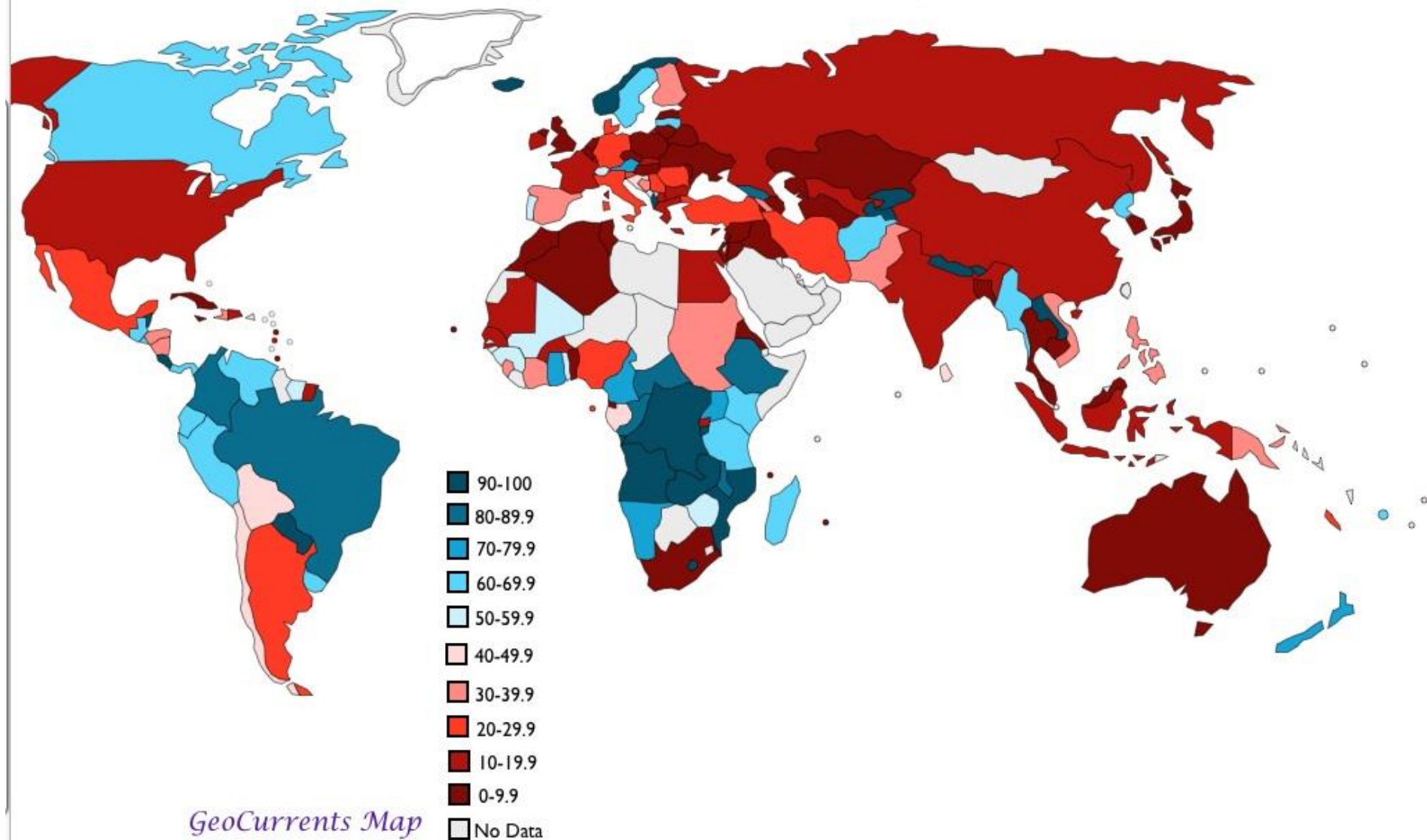
The new Energy Landscape

Percent renewable electricity in Spain



Percentage of Electricity Generation from Renewable Sources

(Hydro, Geothermal, Solar, Biomass, Wind)



Data Source: http://en.wikipedia.org/wiki/List_of_countries_by_electricity_production_from_renewable_sources

pros and cons of renewable energies

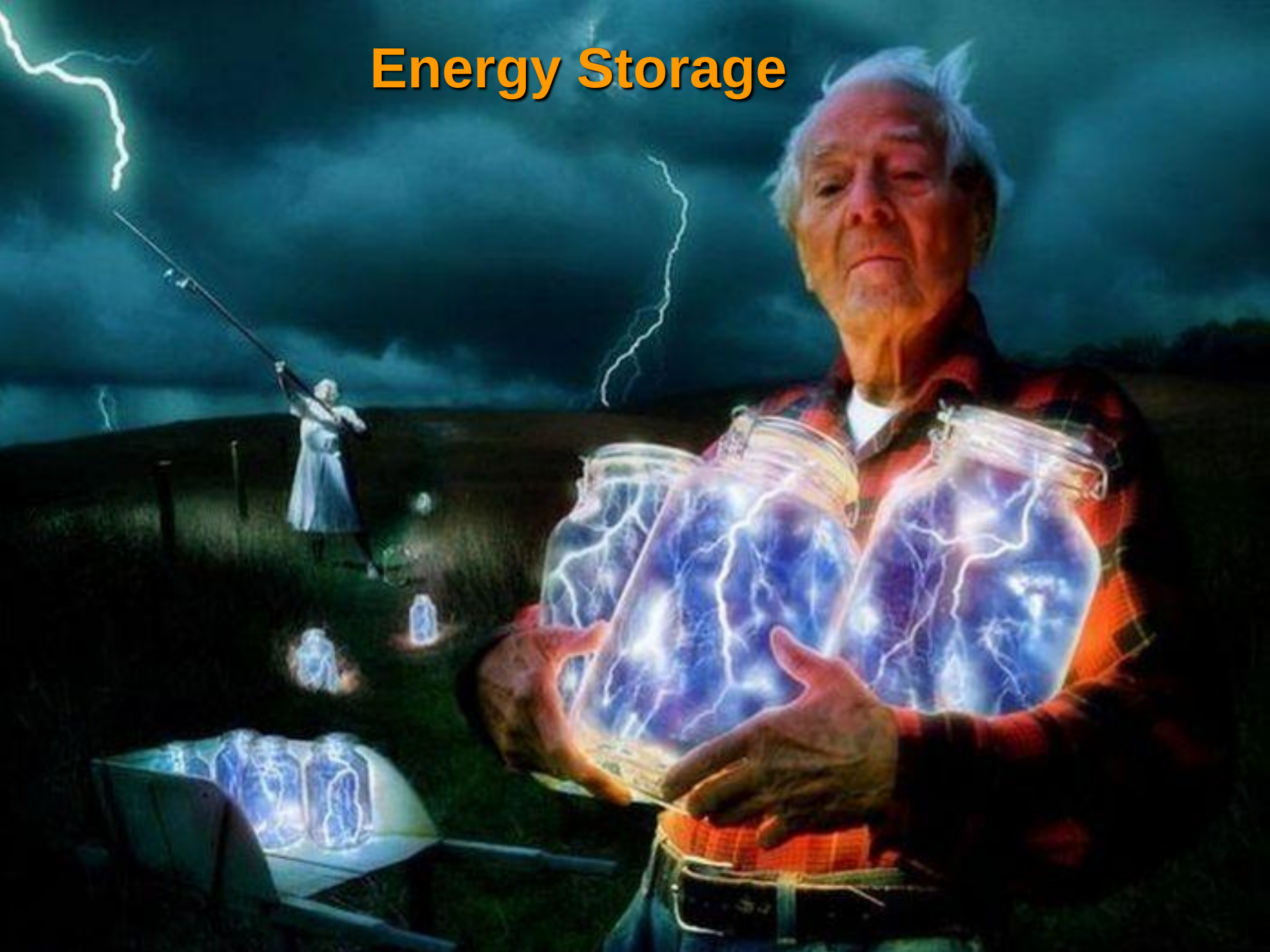
cons

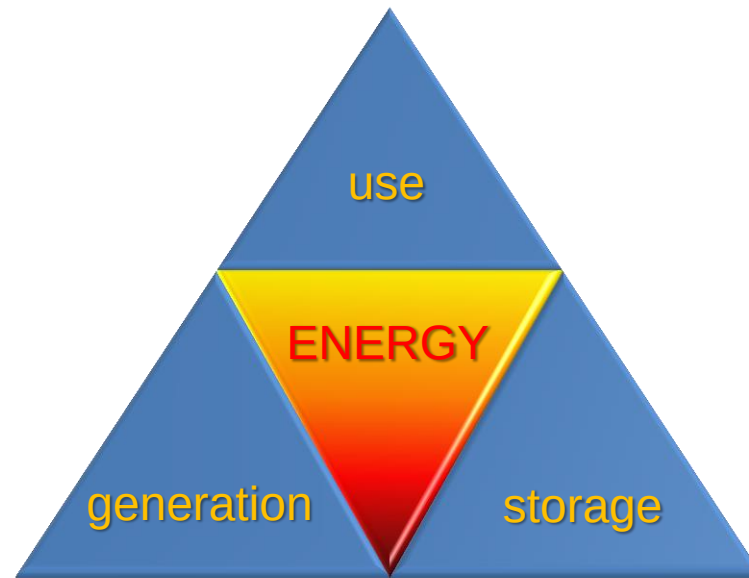
Solar	Does NOT work at night
Wind	Does NOT work in the absence of wind

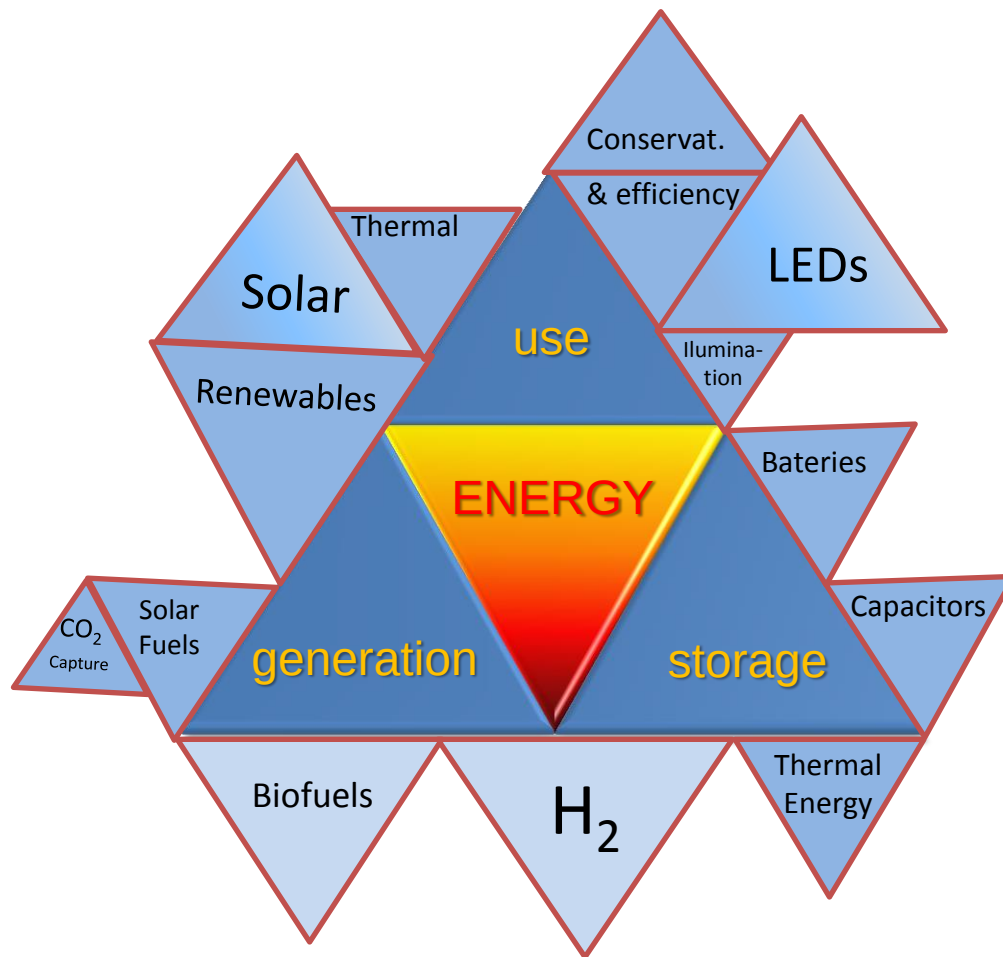
pros

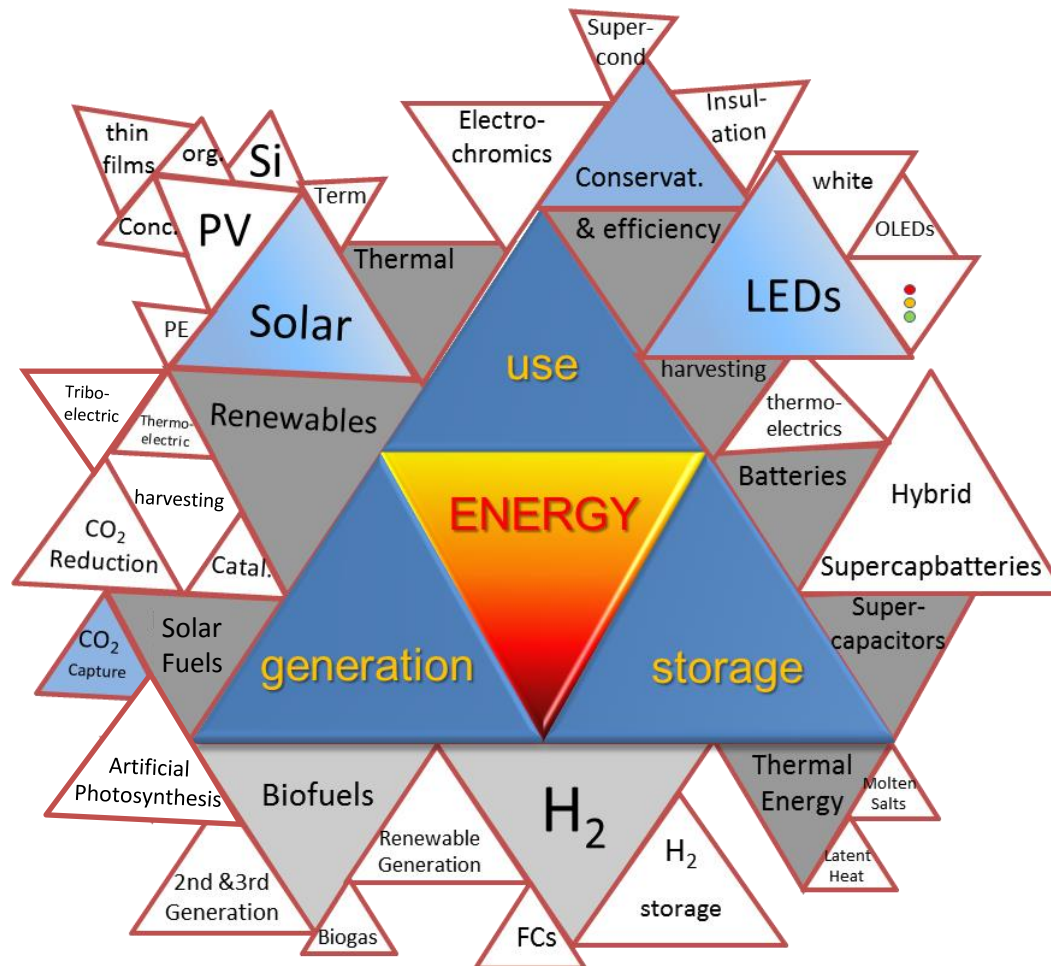
Solar	It DOES work in the absence of wind
Wind	It DOES work at night

Energy Storage



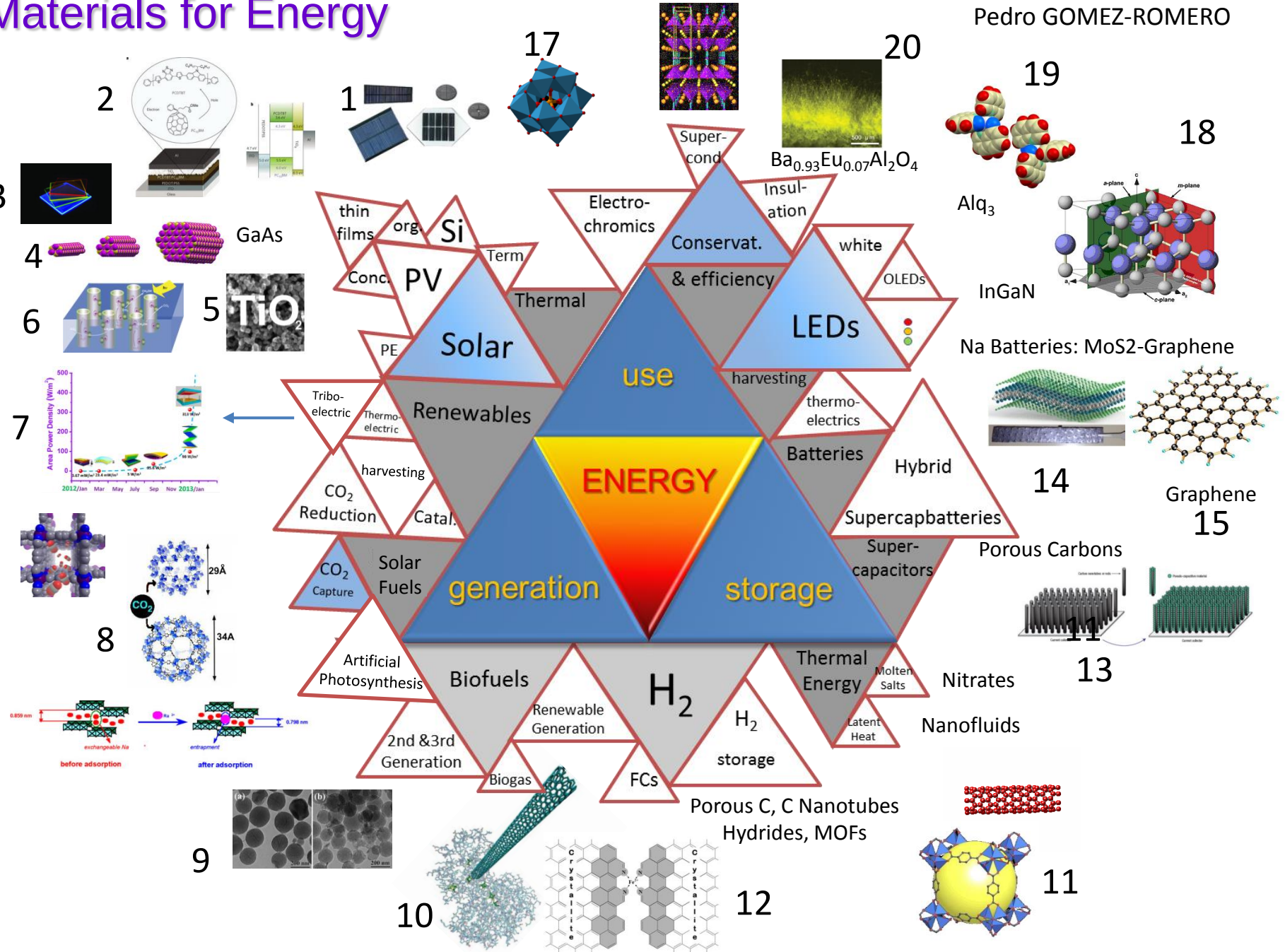




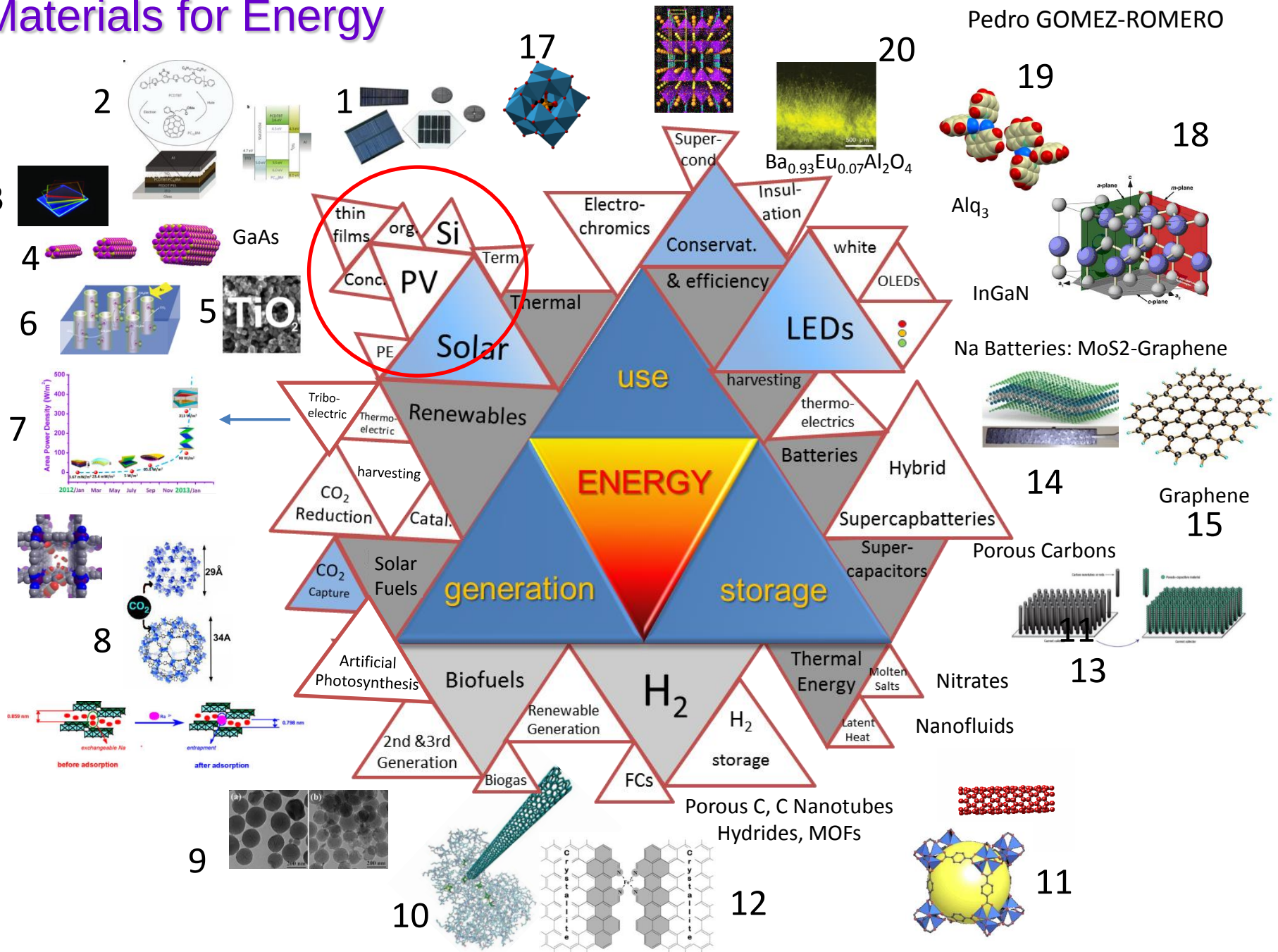


Materials for Energy

Pedro GOMEZ-ROMERO

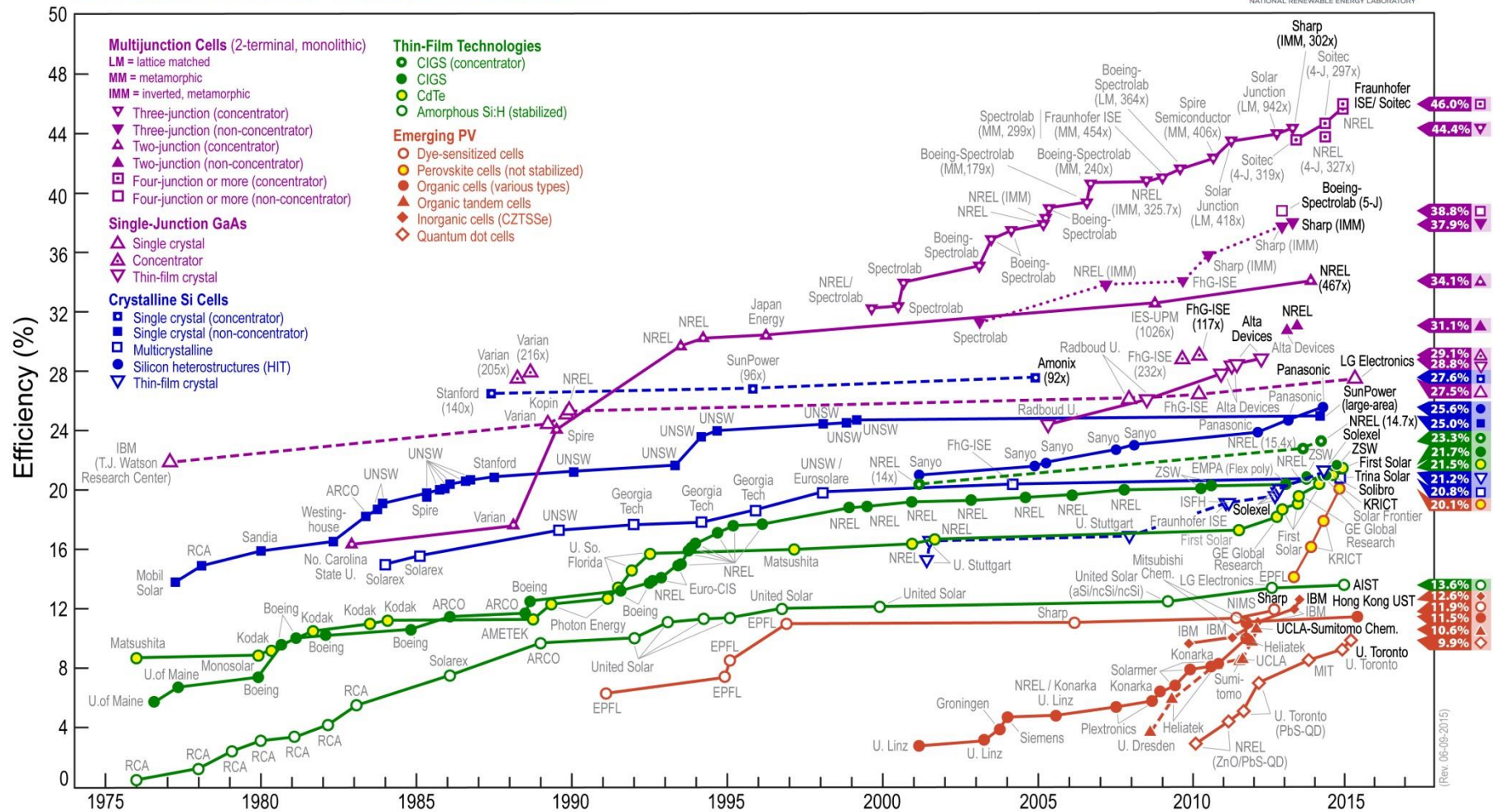


Pedro GOMEZ-ROMERO



Eficiencias record Solar PV

Best Research-Cell Efficiencies



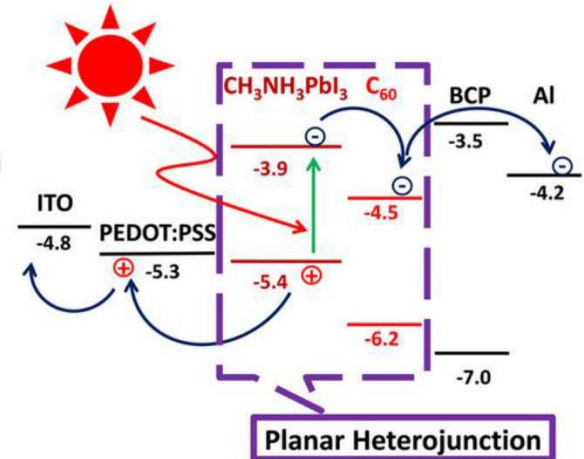
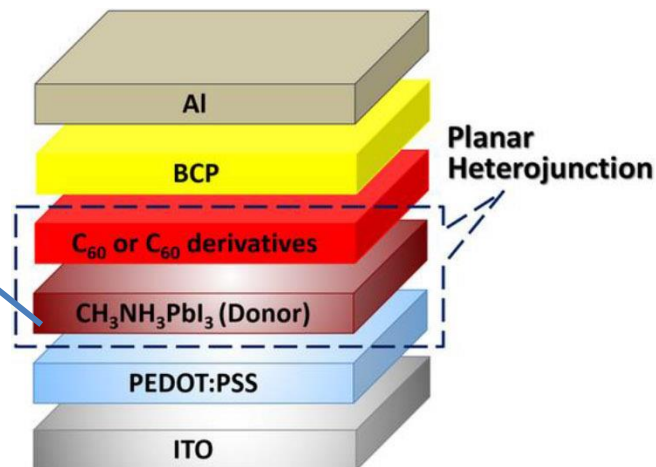
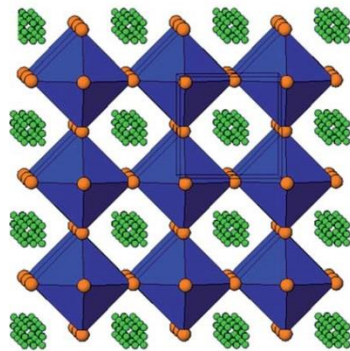
Hybrid Perovskites for Solar PV Cells

Methylammonium Lead Iodide perovskite

2009: 4.9%

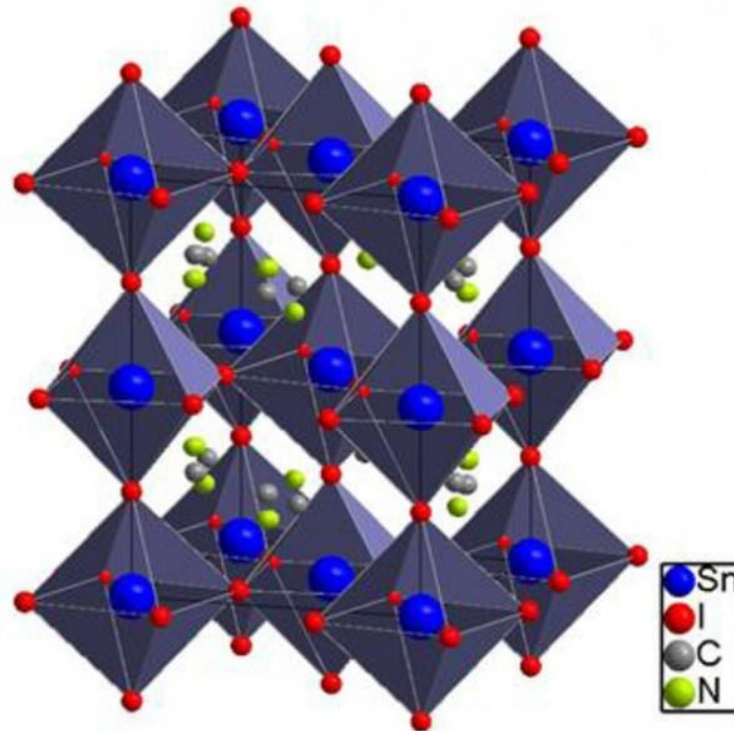


2015: 20%



A. Kojima, K. Teshima, Y. Shirai, T. Miyasaka, J. Am. Chem. Soc., 131 (2009) 6050

Lead-free Solar Cells



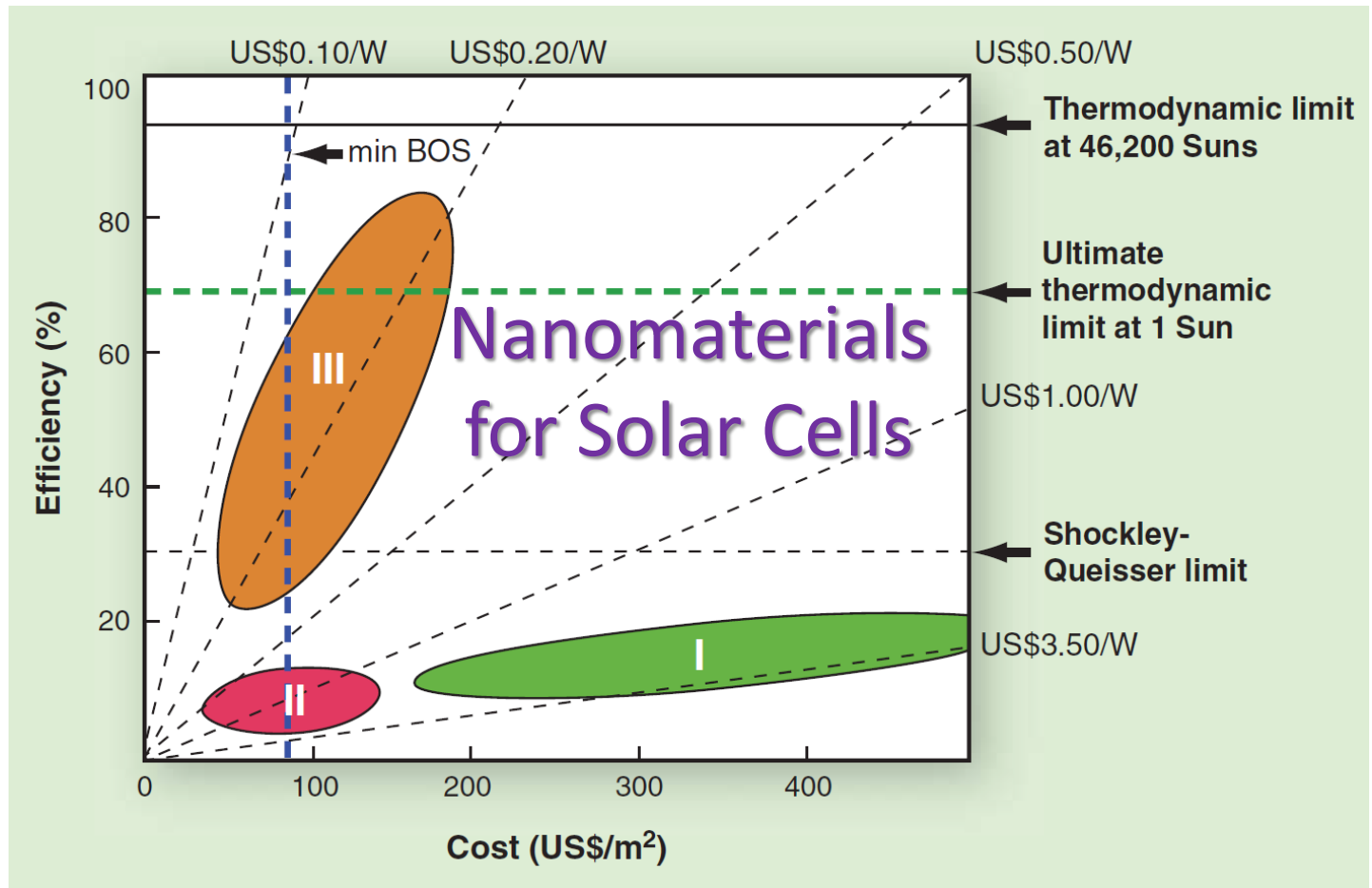
6.4 %

Lead-free organic–inorganic tin halide perovskites for photovoltaic applications
Nakita K. Noel et al. *Energy Environ. Sci.*, **2014**, 7, 3061-3068

Conducting tin halides with a layered organic-based perovskite structure
David Mitzi *et al.*, *Nature* **1994**, 369, 467-469 (1994)

From Materials to Nanomaterials

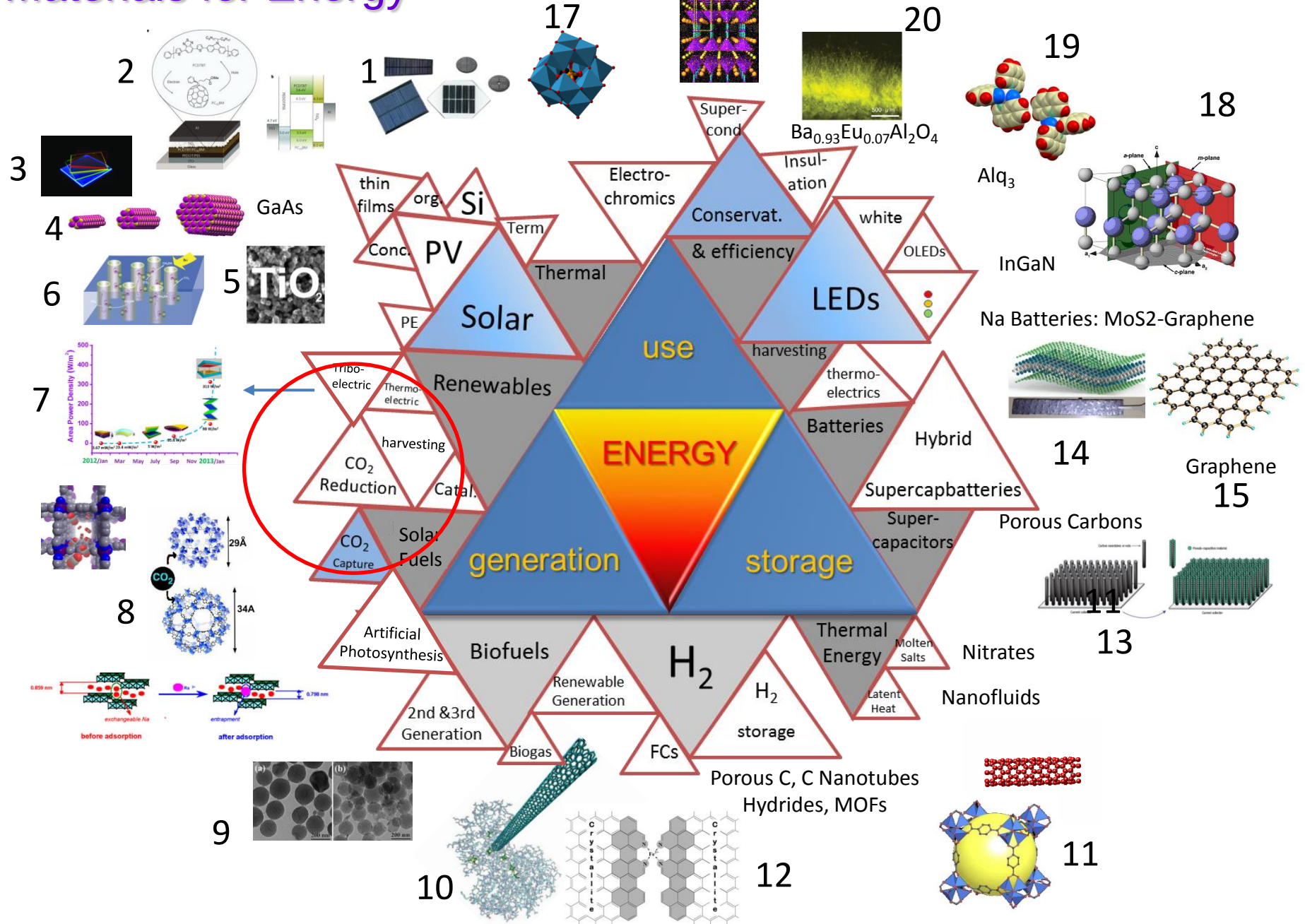
Intrinsic changes in properties



Nathan S. Lewis, et al. Toward Cost-Effective Solar Energy Use. Science 315, 798 (2007)

Materials for Energy

Pedro GOMEZ-ROMERO



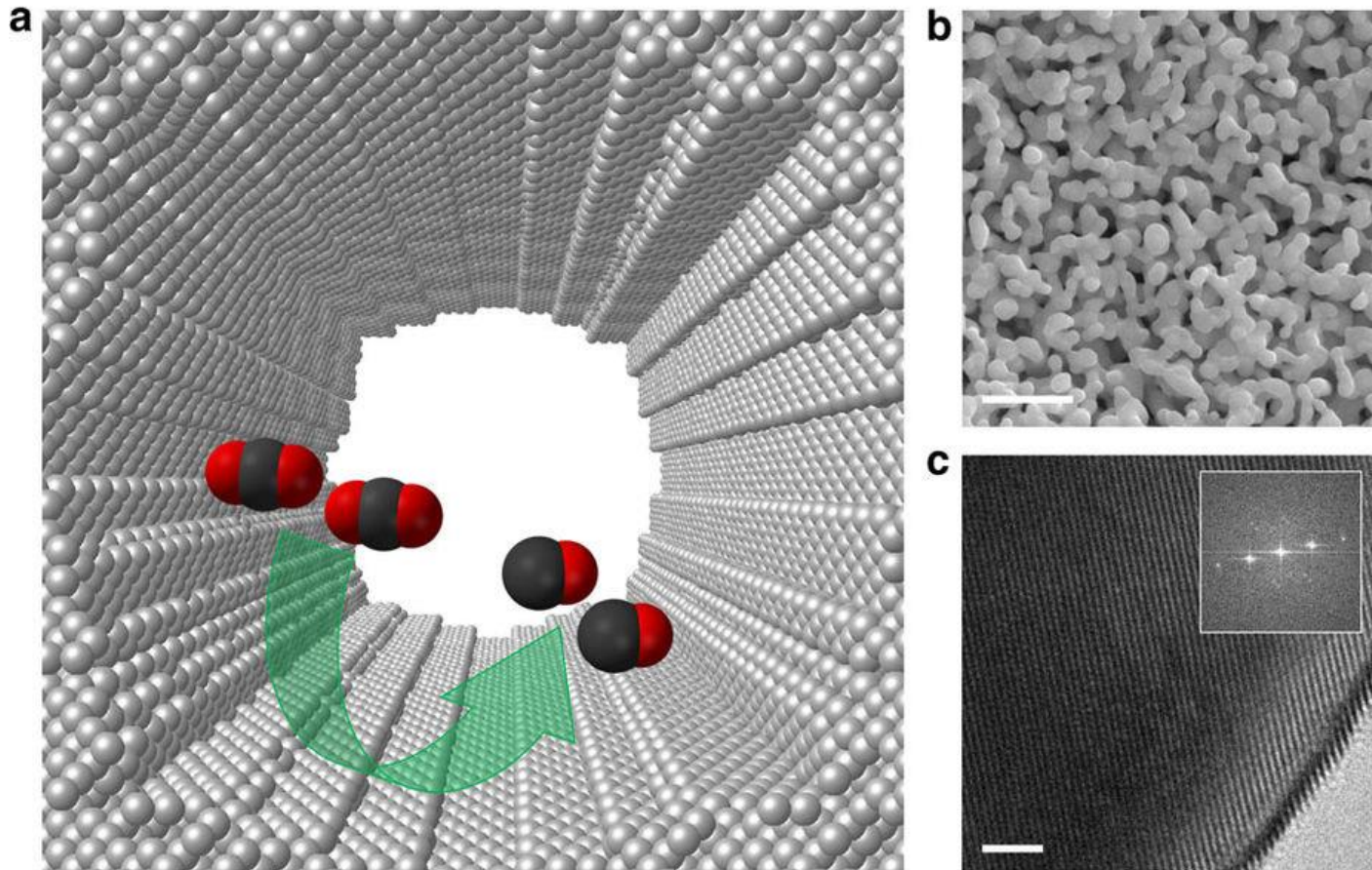
mater... and its absence in science and art

chillida

chillida
2

Nano and Catalysis

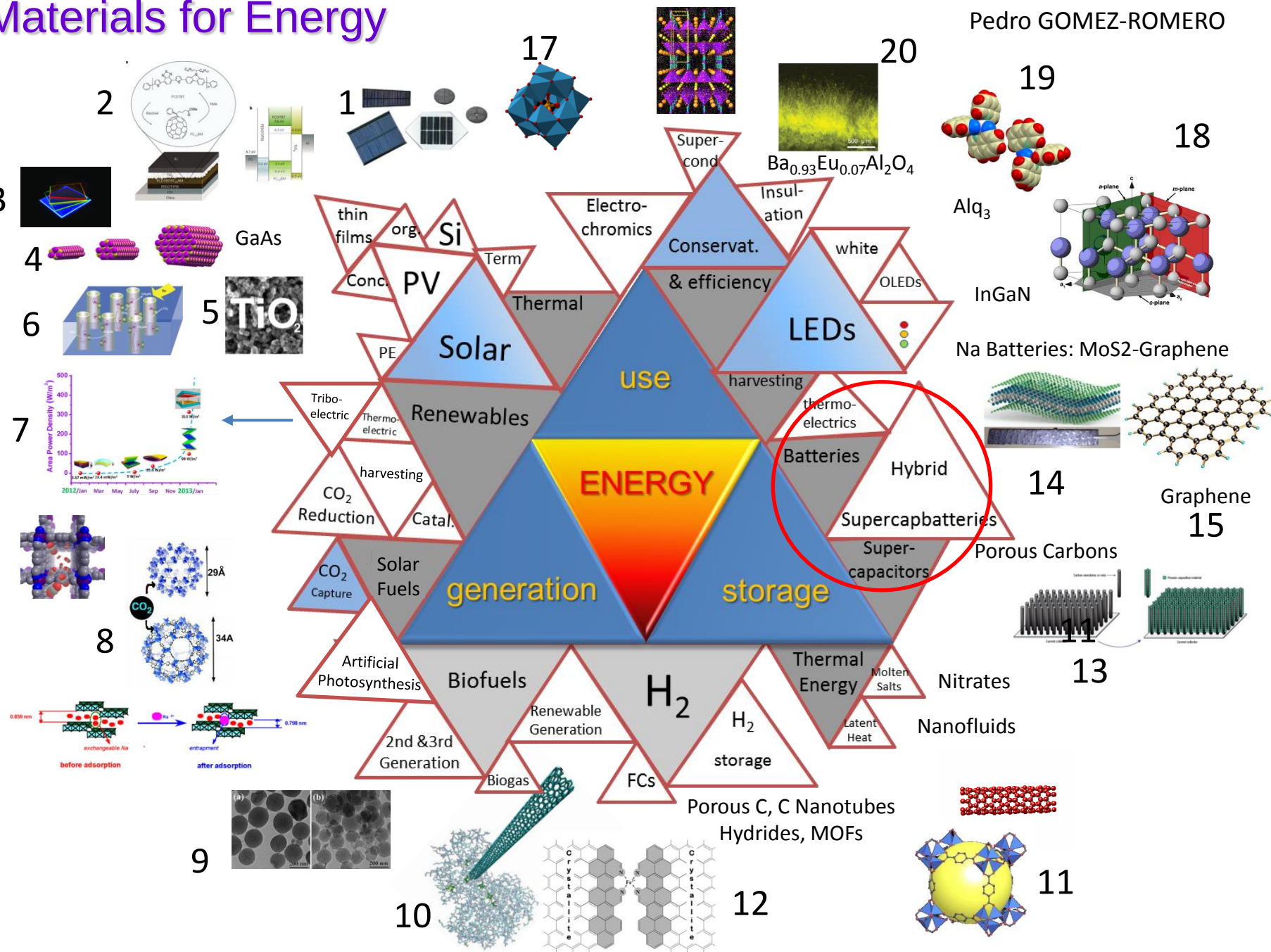
CO₂ Electroreduction to CO by nanopores in de-alloyed Ag from AgAl



activity three orders of magnitude higher than that of the polycrystalline counterpart at a moderate overpotential of <500 mV.

Qi Lu et al. *Nature Communications* **5**: 3242 (2014)

Pedro GOMEZ-ROMERO



Energy storage in transition



Towards flexible solid-state supercapacitors for smart and wearable electronics
D. P. Dubal,* N.R. Chodankar, D-H. Kim and P. Gomez-Romero*
Chemical Society Reviews, **2018**, 47(6), 2065-2129

Symmetric Graphene supercap printed on paper

+ KI

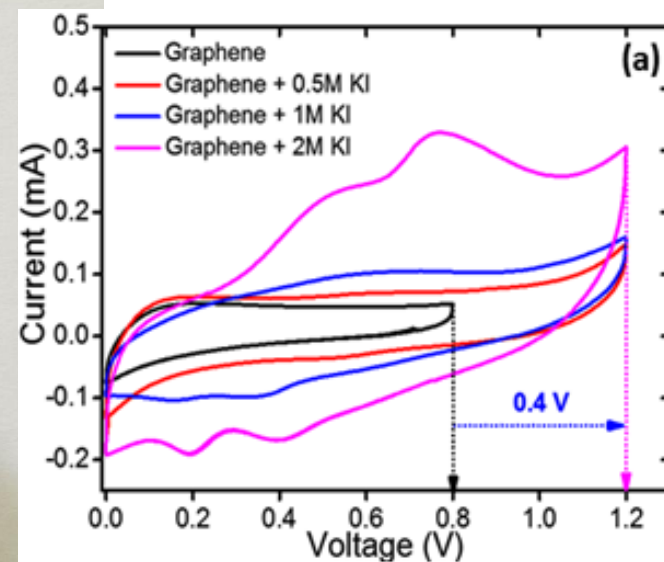
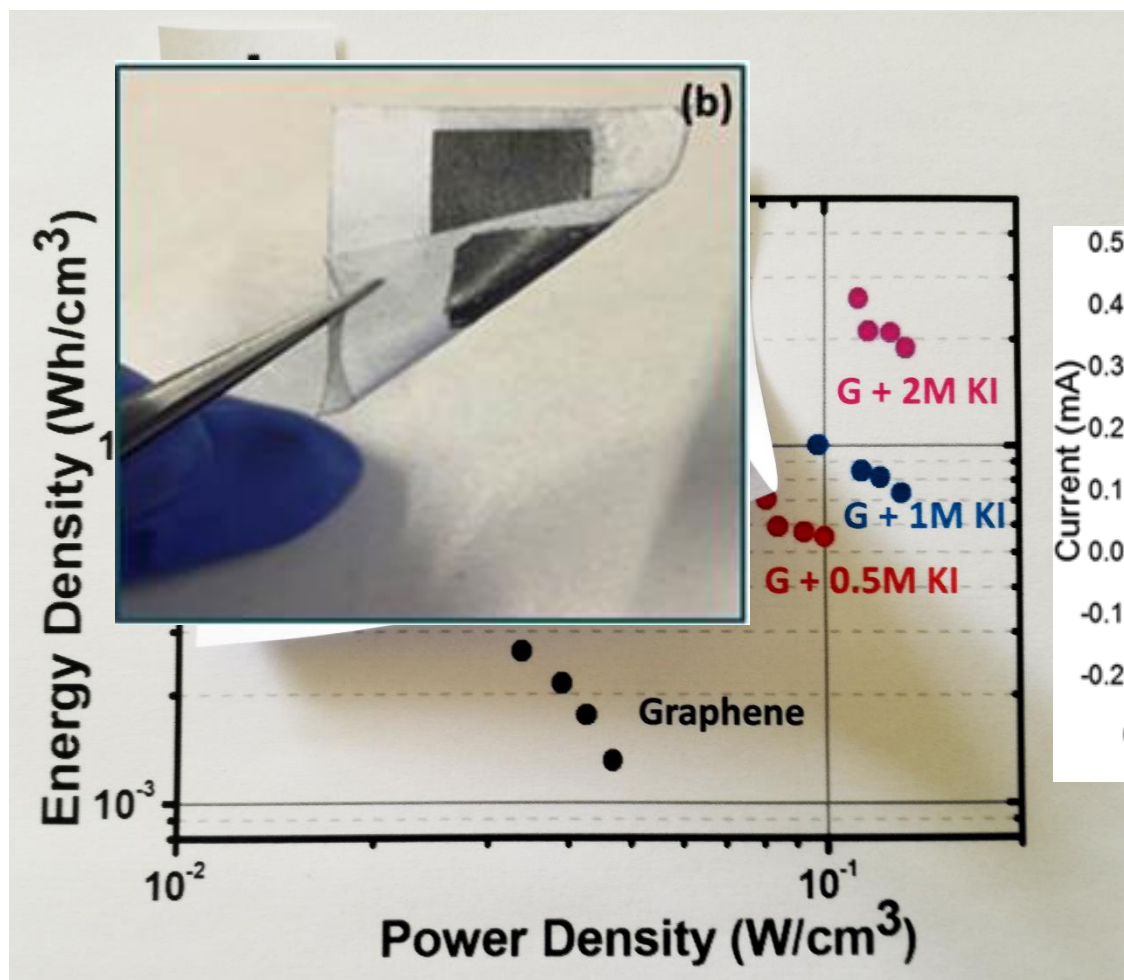


ICN2^R

Institut Català
de Nanociència
i Nanotecnologia



EXCELENCIA
SEVERO
OCHOA



1 M H_2SO_4 electrolyte

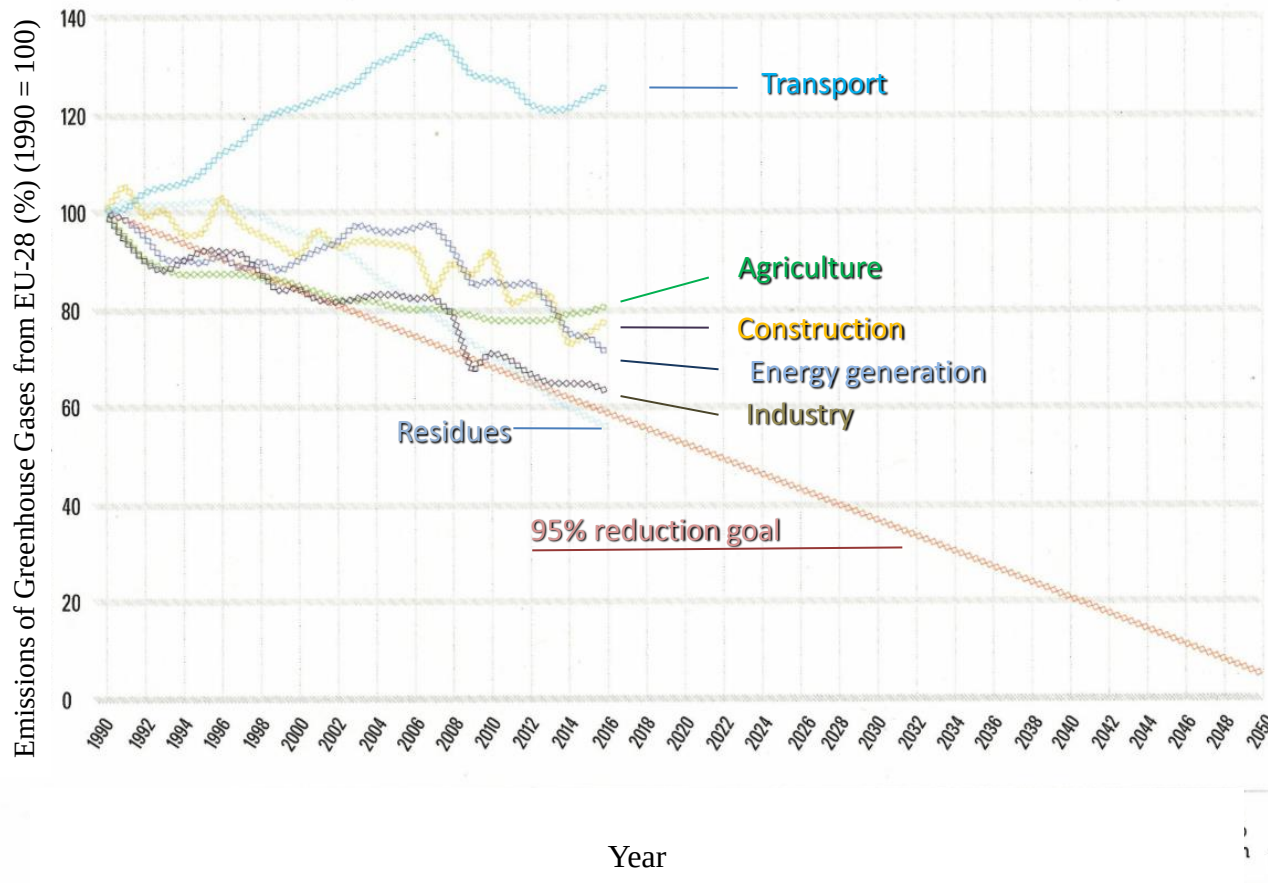
B Nagar, D. P. Dubal, L. Pires, A. Merkoçi and P. Gómez Romero*
ChemSusChem **2018**, 11(11), 1849-1856

Energy storage in transition



Towards flexible solid-state supercapacitors for smart and wearable electronics
D. P. Dubal,* N.R. Chodankar, D-H. Kim and P. Gomez-Romero*
Chemical Society Reviews, **2018**, 47(6), 2065-2129

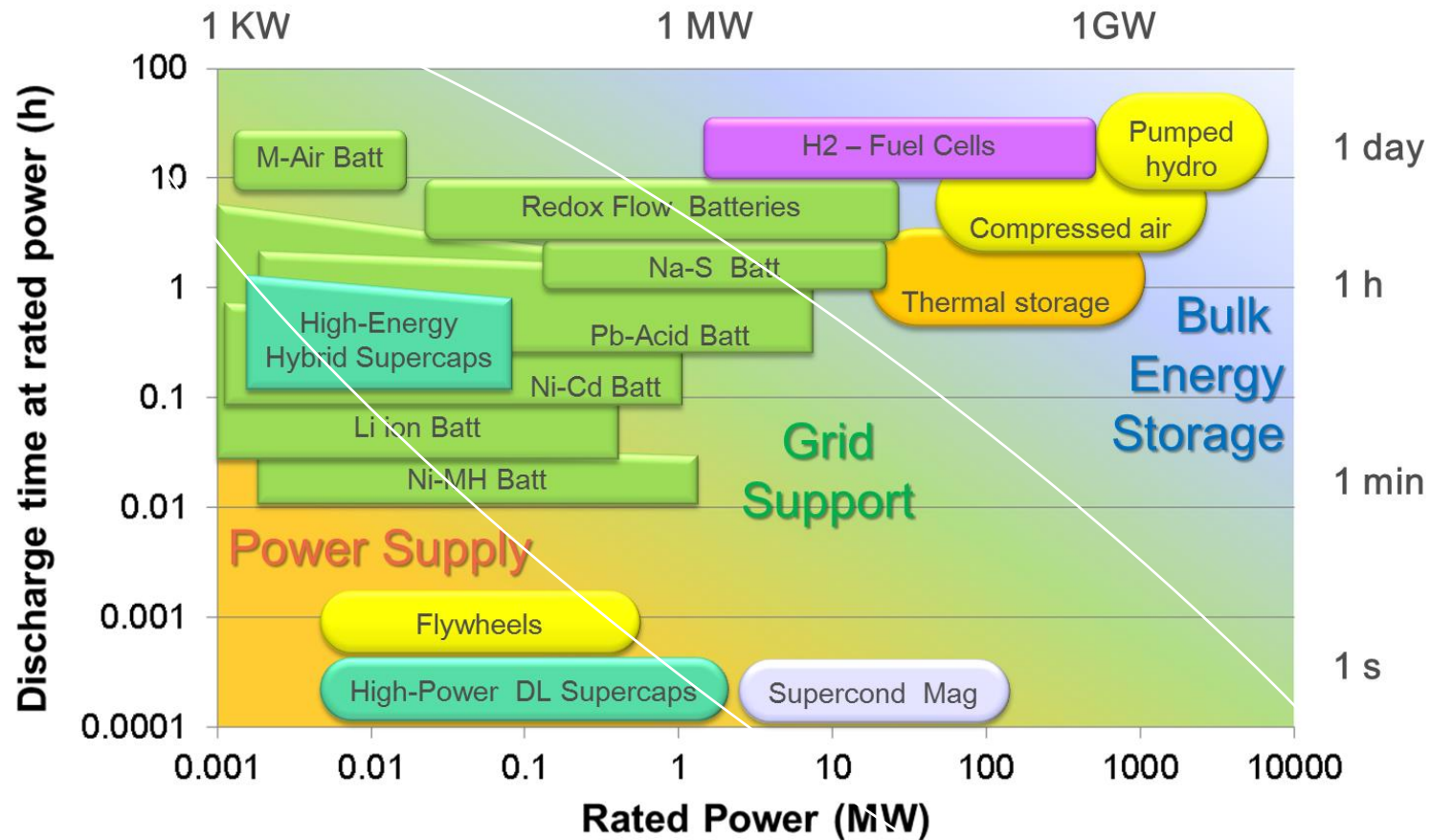
Which sector is not reducing emissions in Europe (yet)?



Source: Transport and Environment. From: Ballena Blanca, revista de medioambiente y economía, Nº 13, Dic 2017



Energy storage systems



P. Gómez-Romero, D. Muñoz-Rojas Intro Chapter 1 “Energy in Transition”
Materials for Sustainable Energy Applications (D. Muñoz-Rojas, X. Moya Eds.)
Pan Stanford Pub 2016

Instalación Hidroeólica de El Hierro



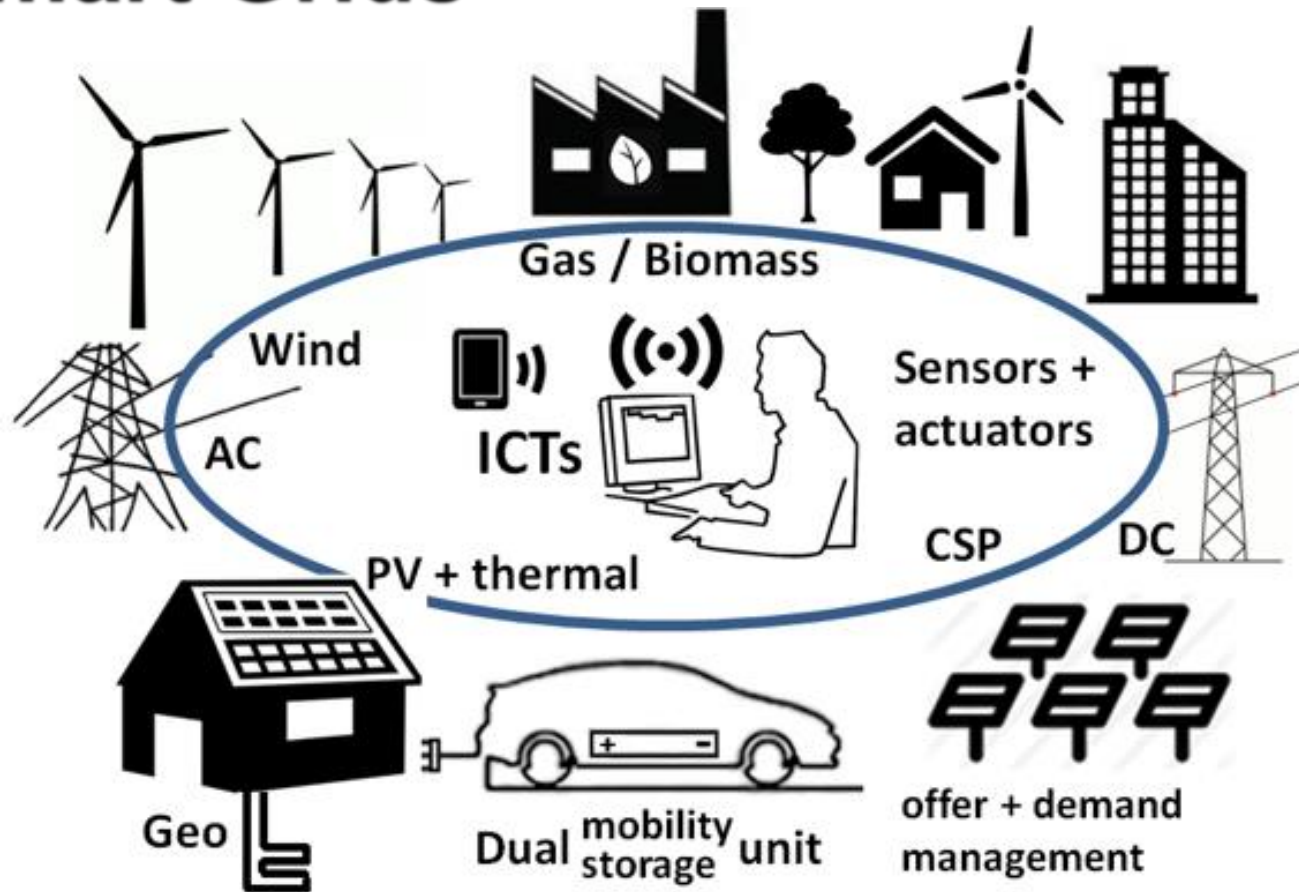
Isla de El Hierro
Canarias



Centralized AND Distributed Energy

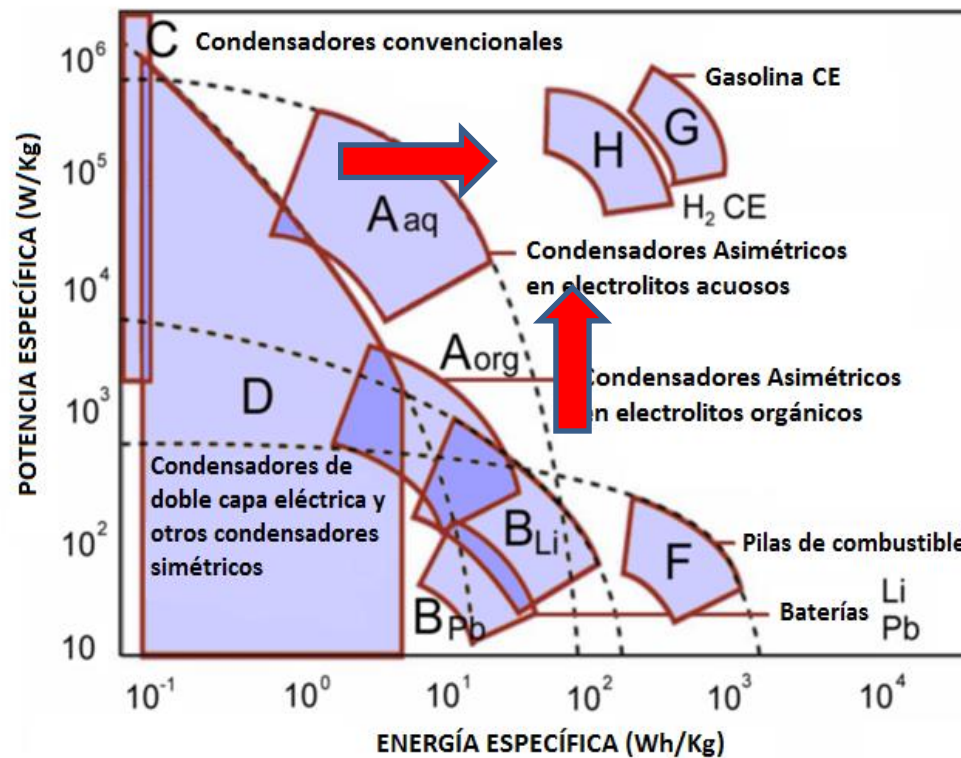
Smart Grids

Internet of Things

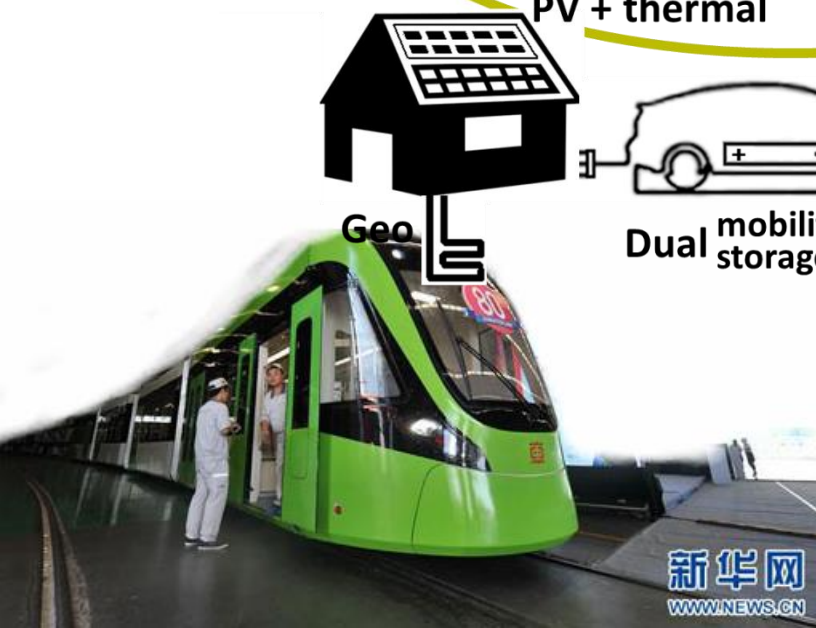
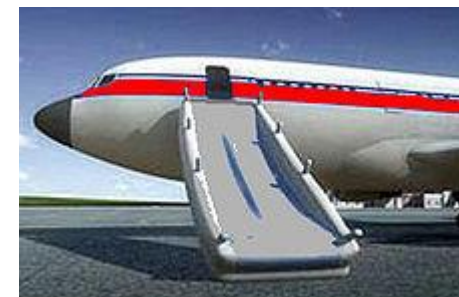
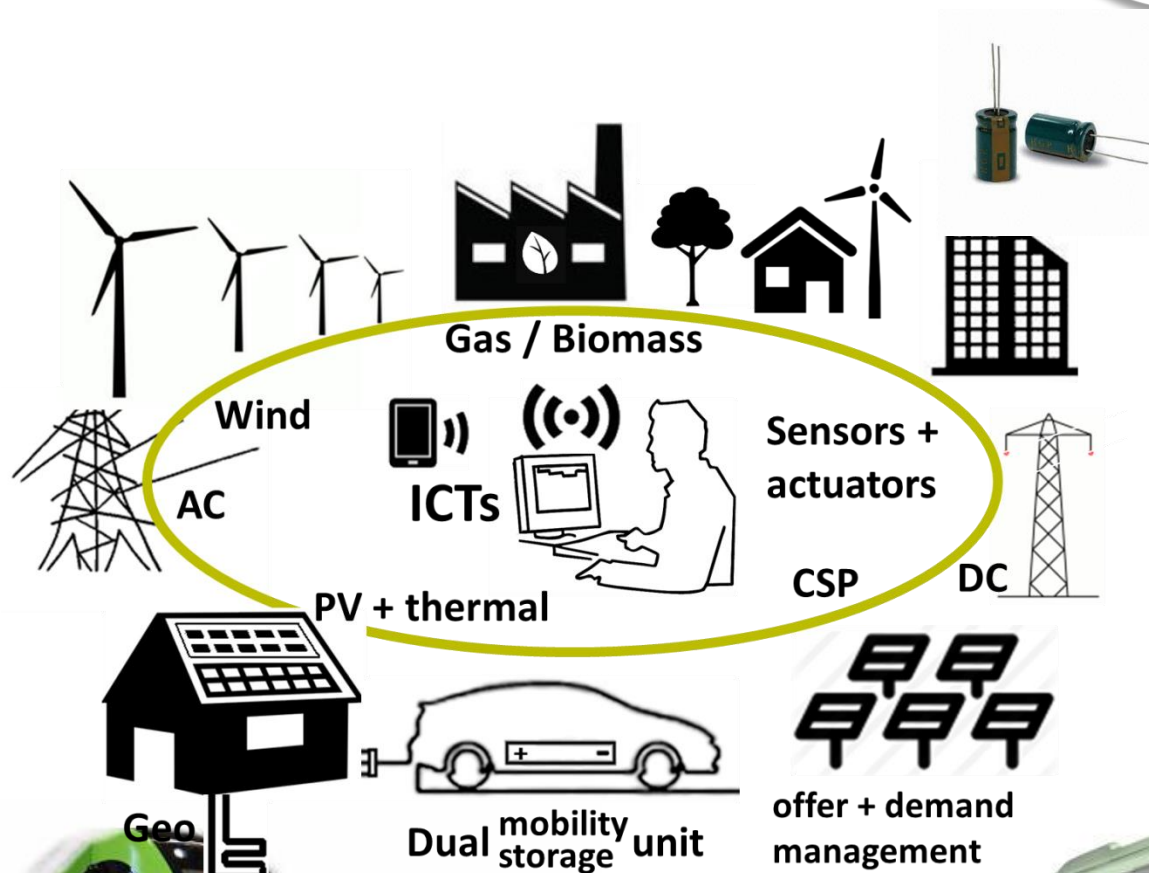


Energy vs. Power

Batteries vs. Supercapacitors



Supercapacitors applications



Hybrid approaches

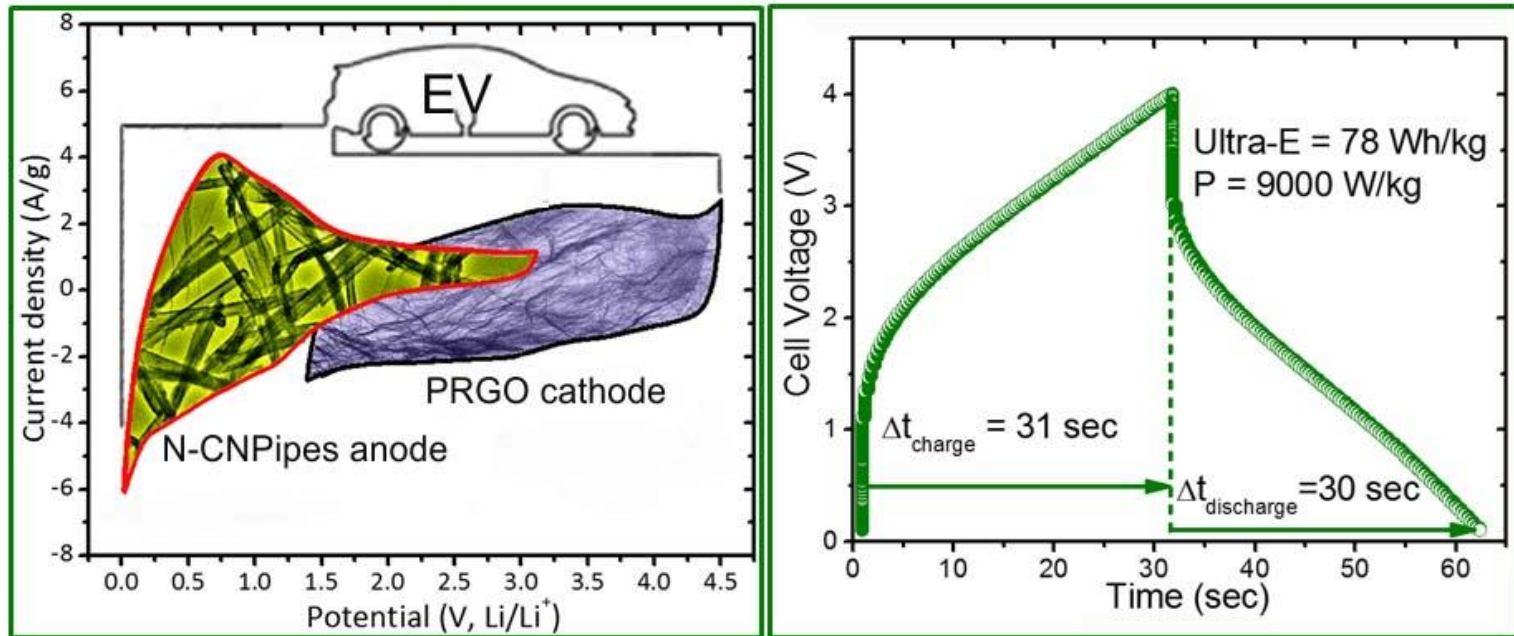
Hybrid devices

Electrode - Electrode
Electrode - Electrolyte

Hybrid materials

Hybrid Device

Nanocarbon Li-Ion Capacitor



Nanocarbon Li-Ion Capacitor:

Battery-like negative electrode + supercap-type positive electrode

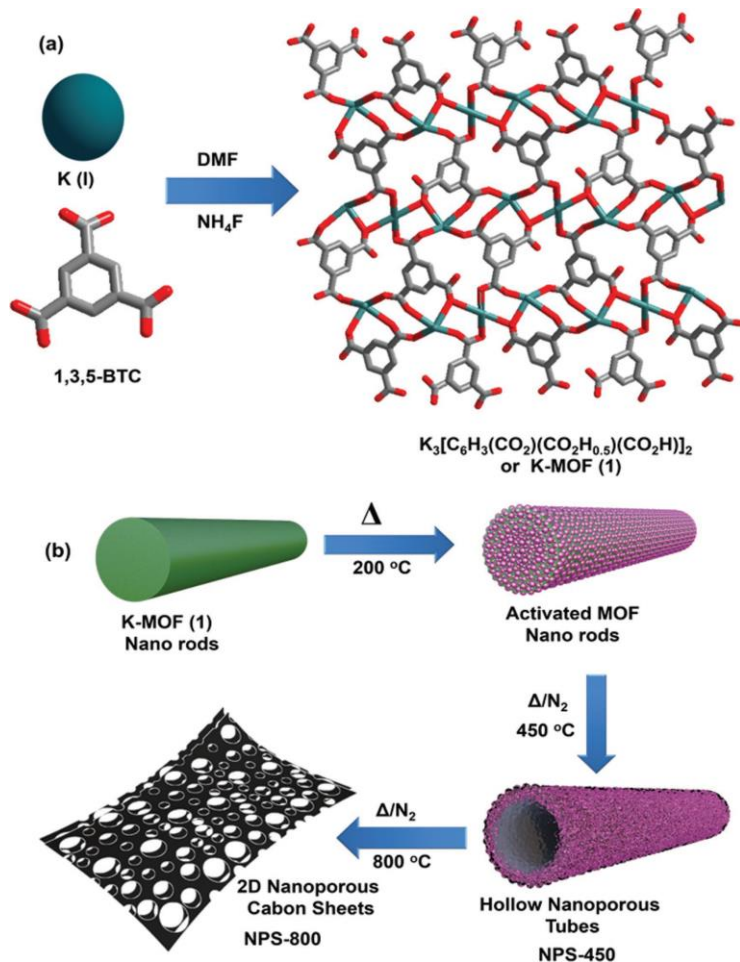
High energy and high power (fast charge) **SIMULTANEOUSLY**

All Nanocarbon Li-Ion Capacitor with High Energy and High Power Densities.

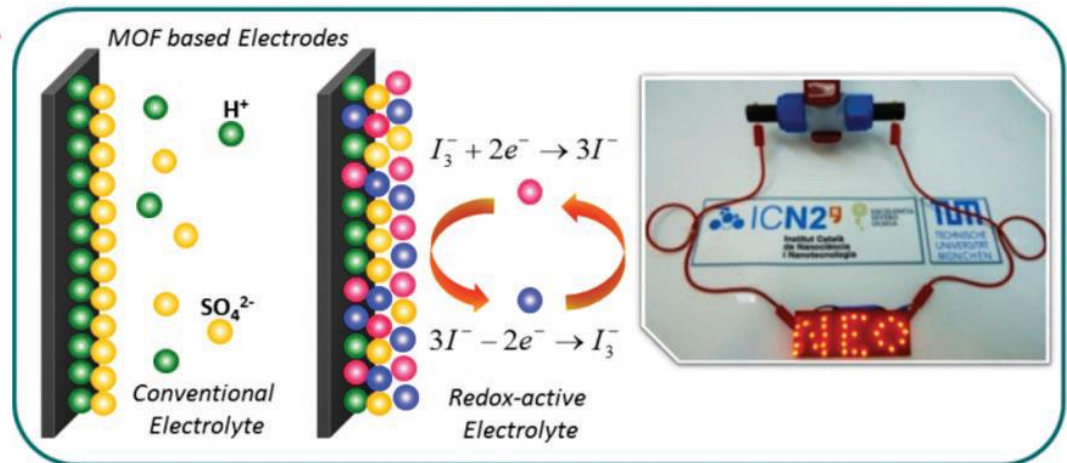
Deepak P. Dubal,* Pedro Gomez-Romero** *Materials Today Energy* **2018**, 8, 109-117

Electrode-Electrolyte Hybridization

2D Nanoporous Carbon Sheets



Capacitive electrodes +
Redox electrolytes



1200 m²/g



ICN2^R

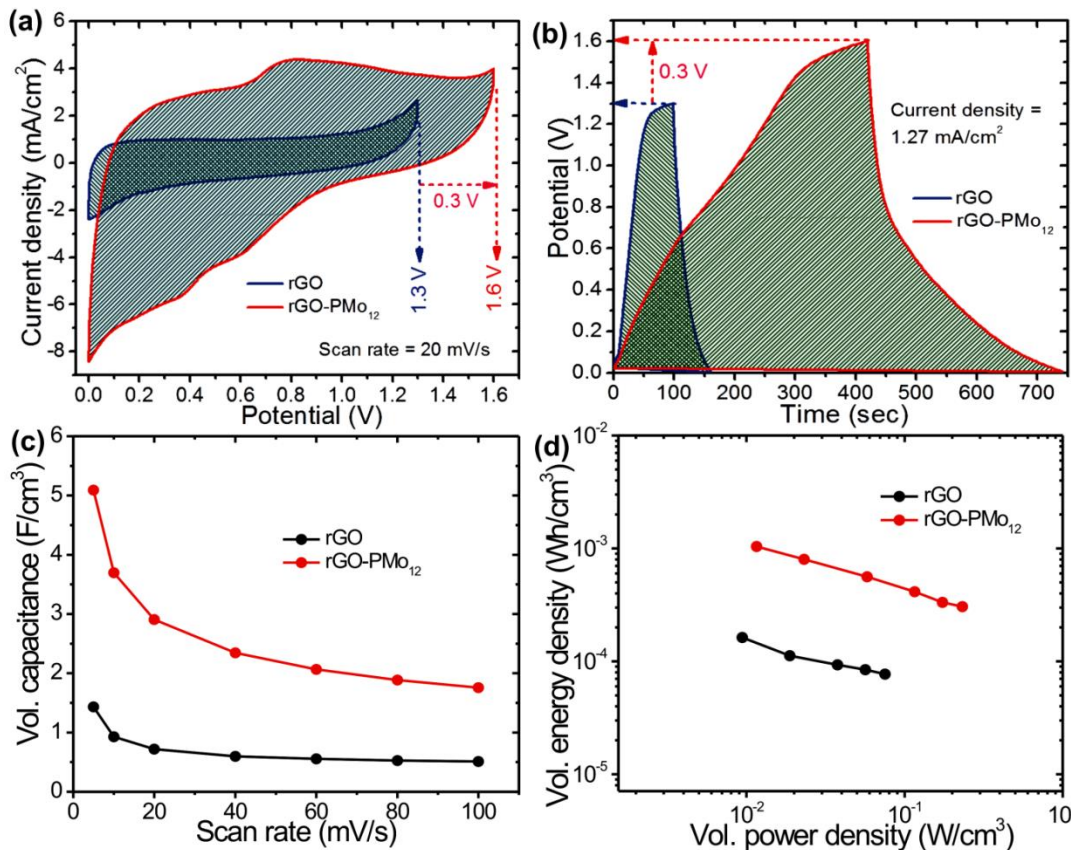
**Institut Català
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i Nanotecnologia**



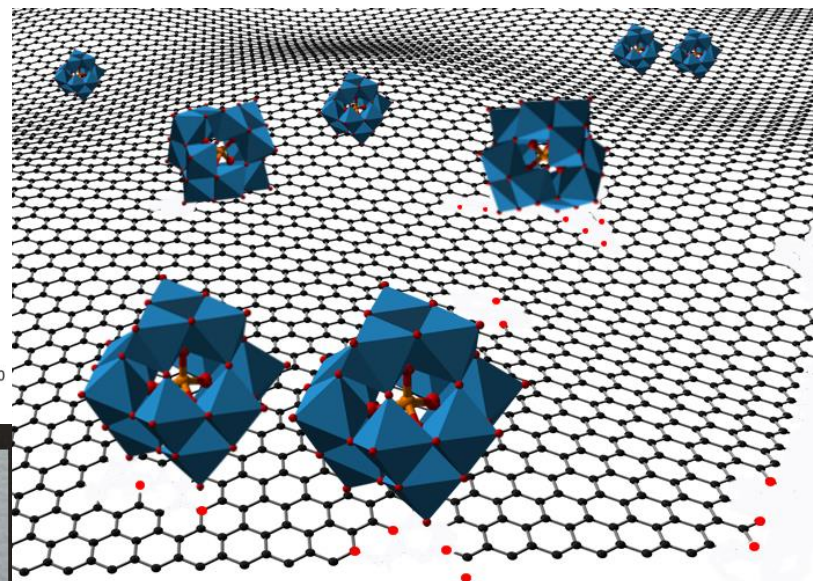
**EXCELENCIA
SEVERO
OCHOA**

Electrode Material Hybridization

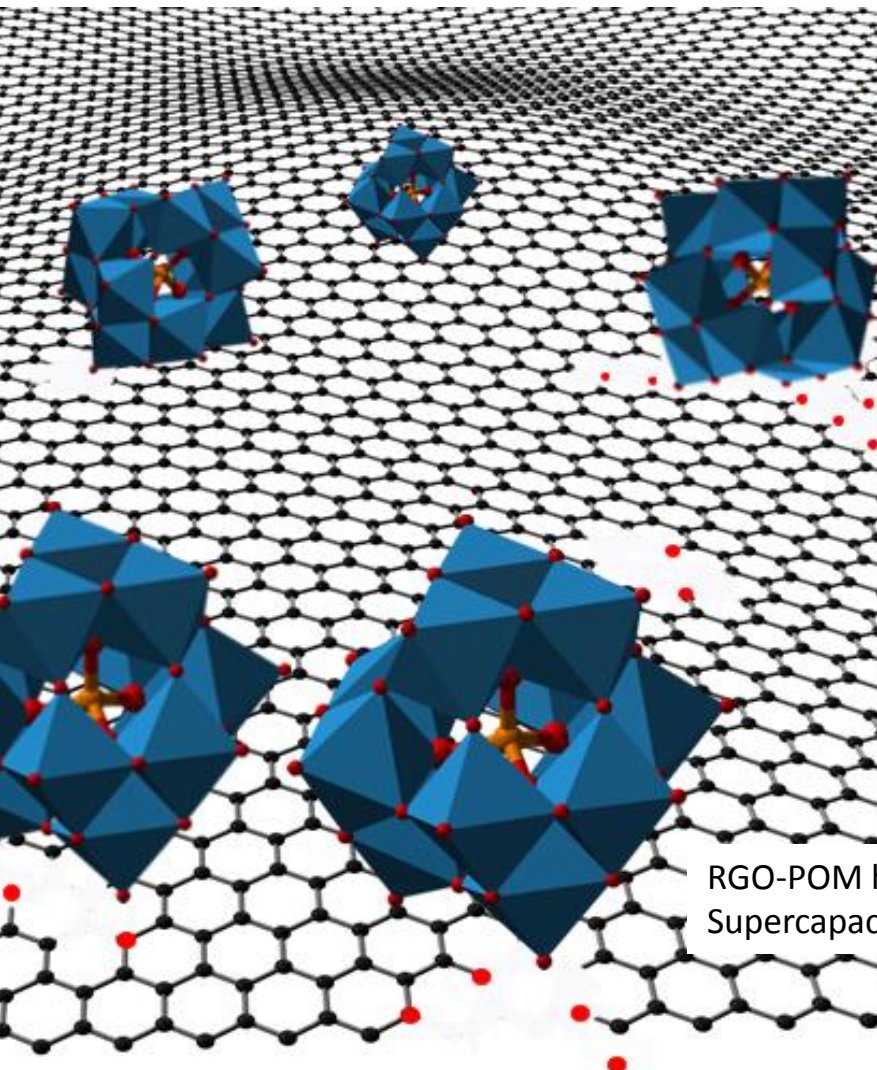
Hybrid Graphene Polyoxometalate Electrodes



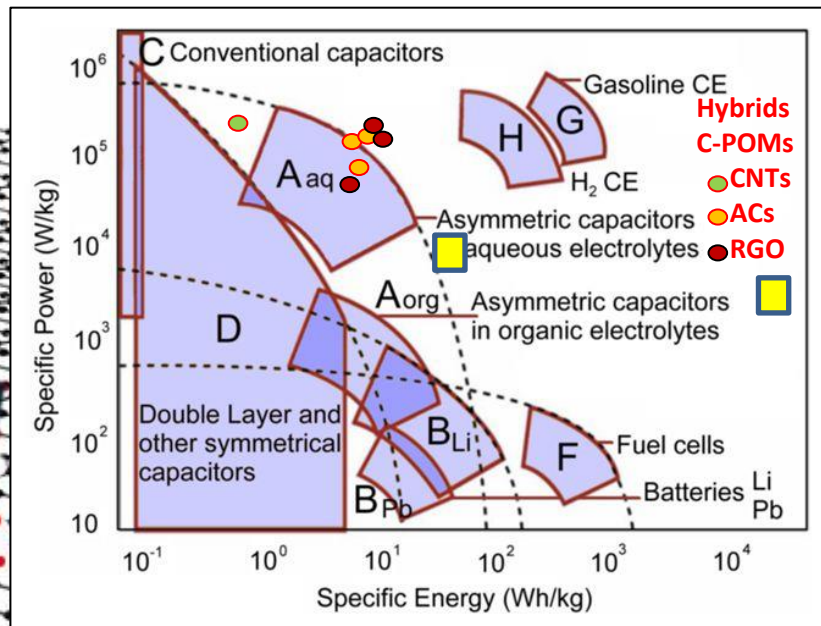
Hybrid Energy Storage: Hybrid electrode materials + Electroactive electrolyte rGO / H₃PMo₁₂O₄₀ / HQ



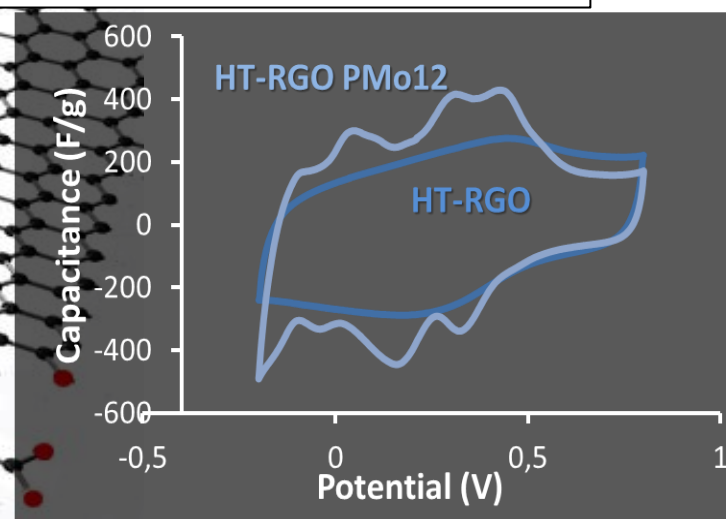
31 LEDs powered with a single rGO-PMo₁₂ symmetric cell with 0.2 M HQ doped polymer gel electrolyte.
30 s charge 2 min lit



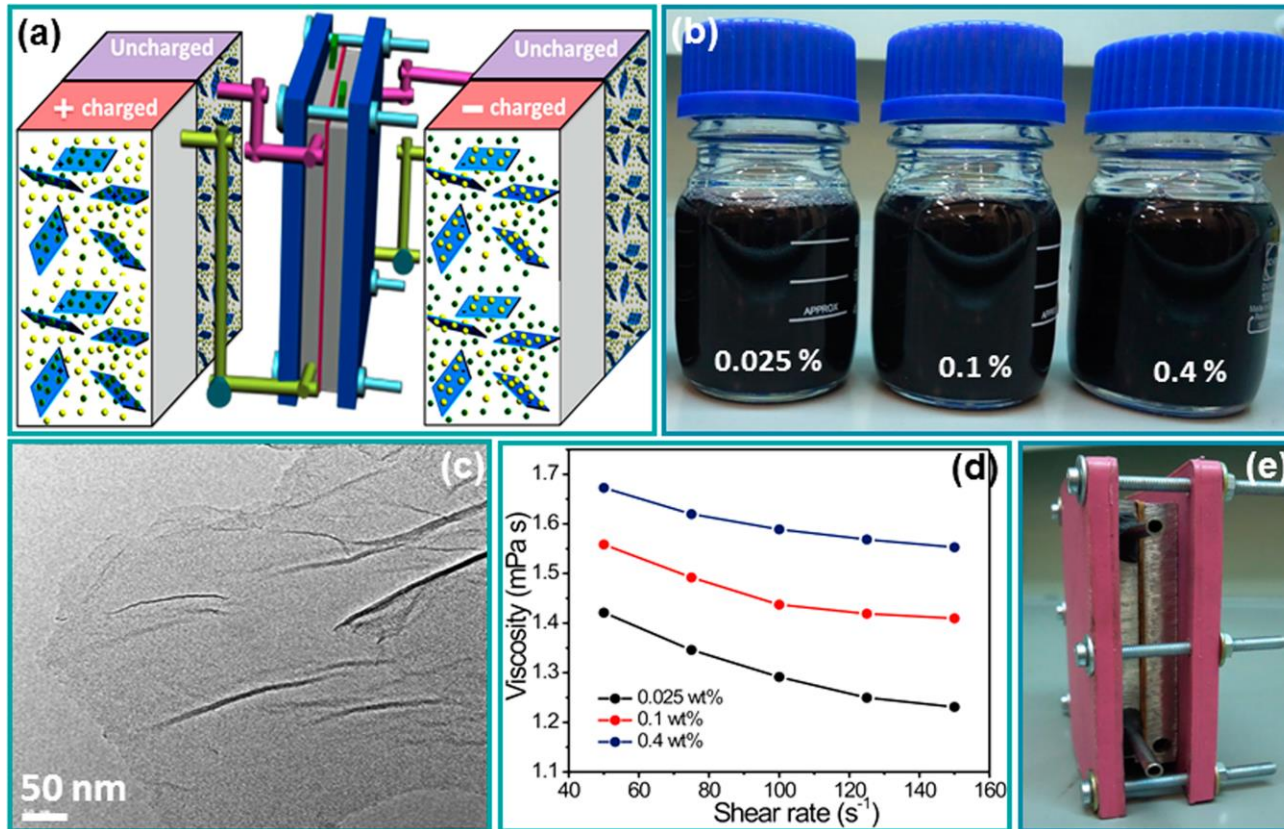
RGO-POM hybrid
Supercapacitor electrode



rGO Ni(OH)₂



Electroactive Graphene Nanofluids for New Flow Cell Concepts.

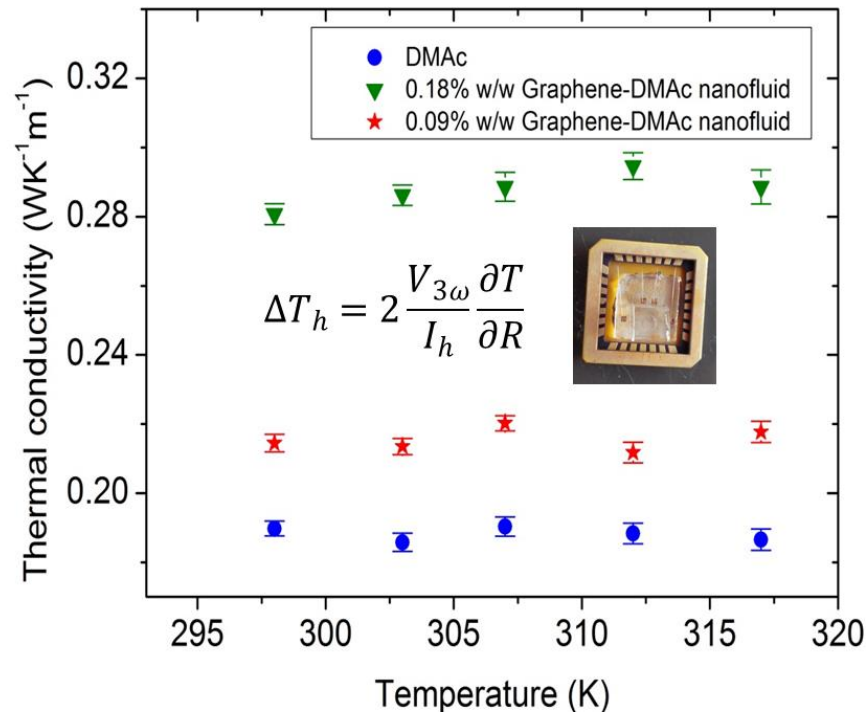


D. P. Dubal, D. Gomez, P. Gómez-Romero, Patent ES1641.1064. "Electroactive nanofluids on graphene-based materials for energy storage in flow cells." 20-05-2015

Electroactive Graphene Nanofluids for Fast Energy Storage.

D.P. Dubal and P. Gomez-Romero 2D-Materials **2016**, 3, 031004

Graphene Nanofluids as Heat Transfer Fluids



3-omega measurements.
Novel approach for thermal conductivity in liquids (colaboration with Prof. C. Sotomayor's and Prof. P Ordejón's groups at ICN2)

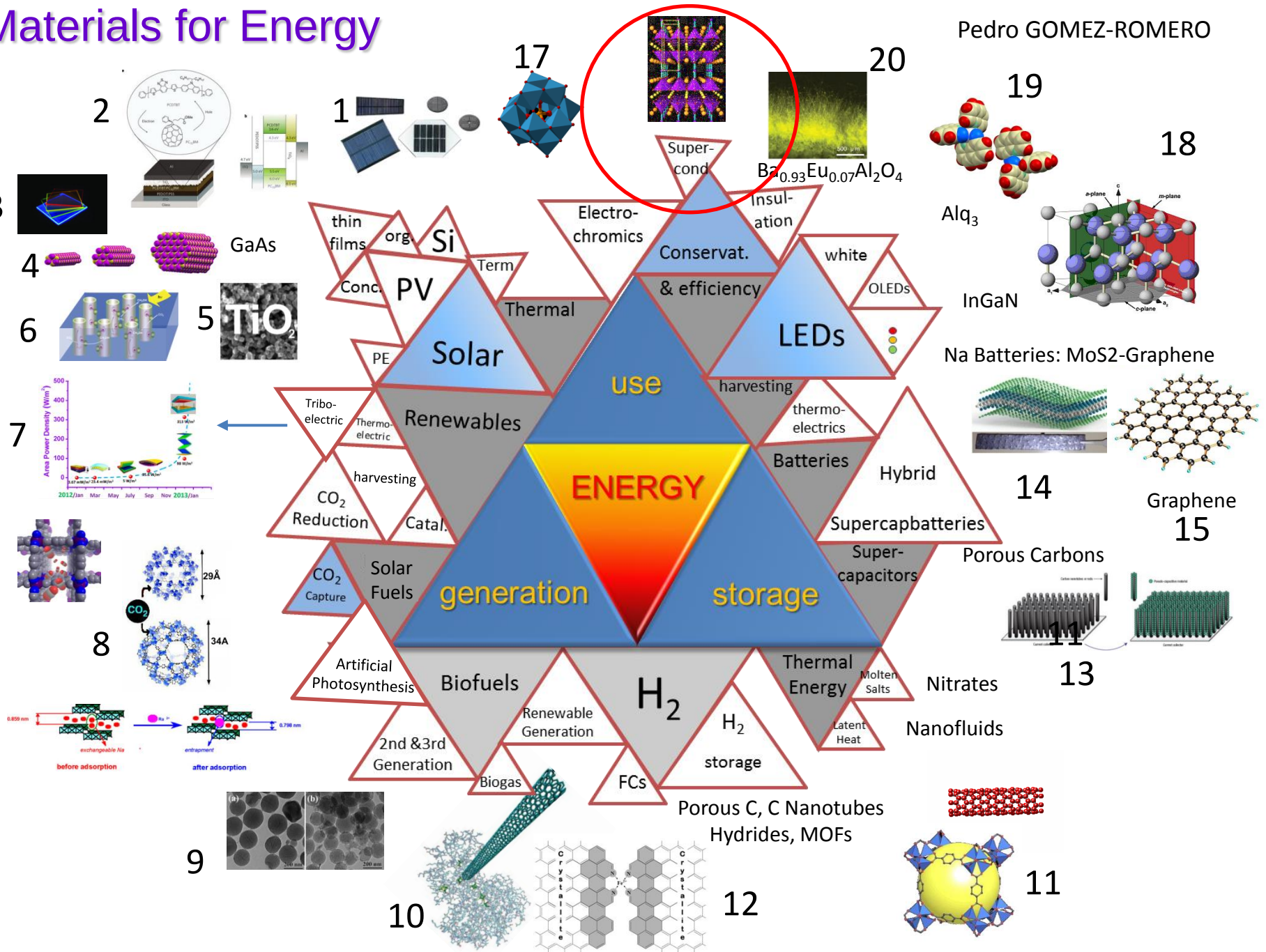
Less than 1% graphene in a perfectly stable solution (left) leads to a 50% improvement

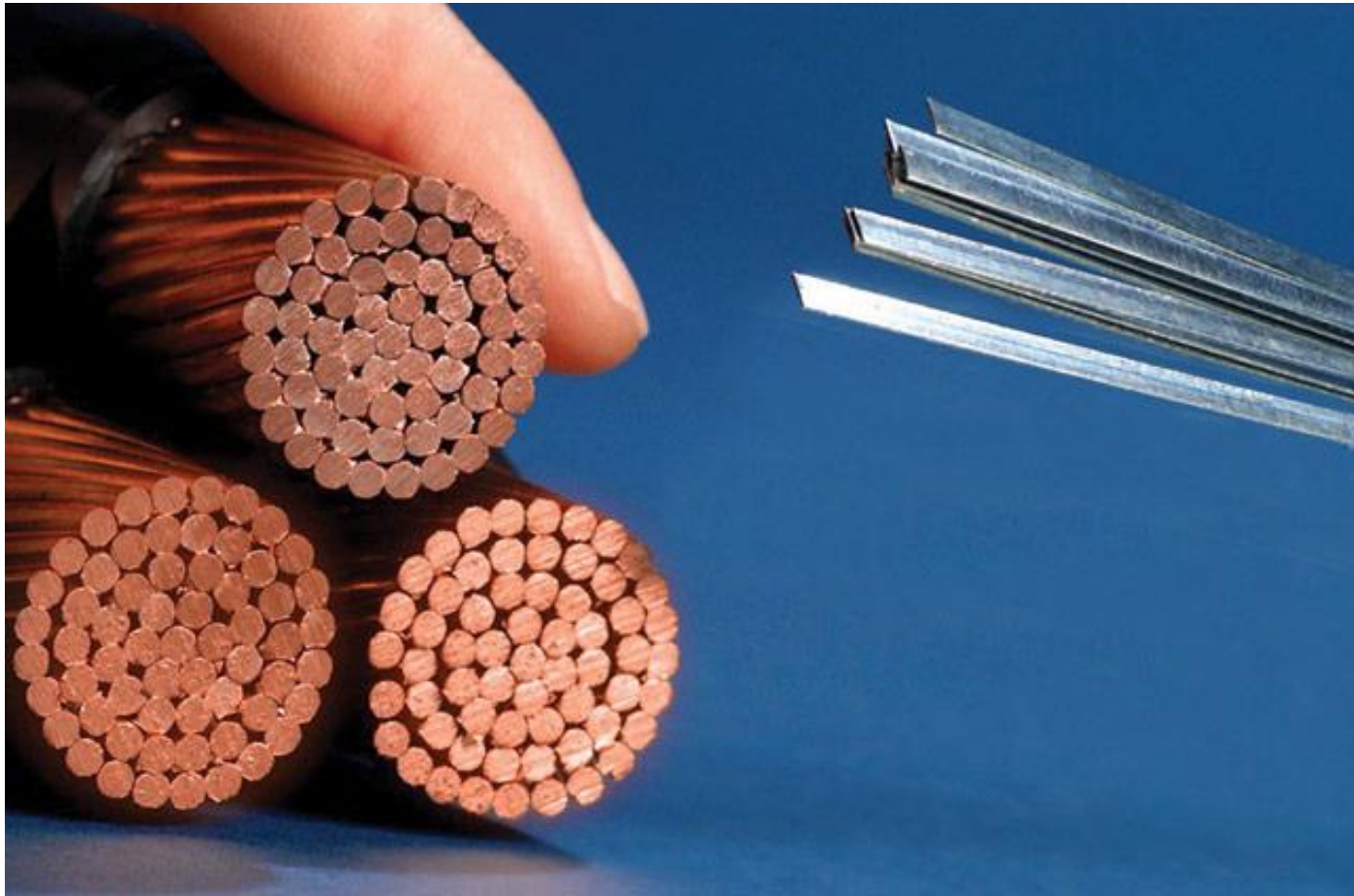
Graphene acting as an accelerating centre for thermal waves in organic nanofluids

M. R. Rodríguez-Laguna, A. Castro-Alvarez, M. Sledzinska, J. Maire, F. Costanzo, B. Ensing, P. Ordejón, C. M. Sotomayor-Torres, P. Gómez-Romero and E. Chávez-Ángel *Nanoscale* **2018**, 10, 15402-15409

Materials for Energy

Pedro GOMEZ-ROMERO



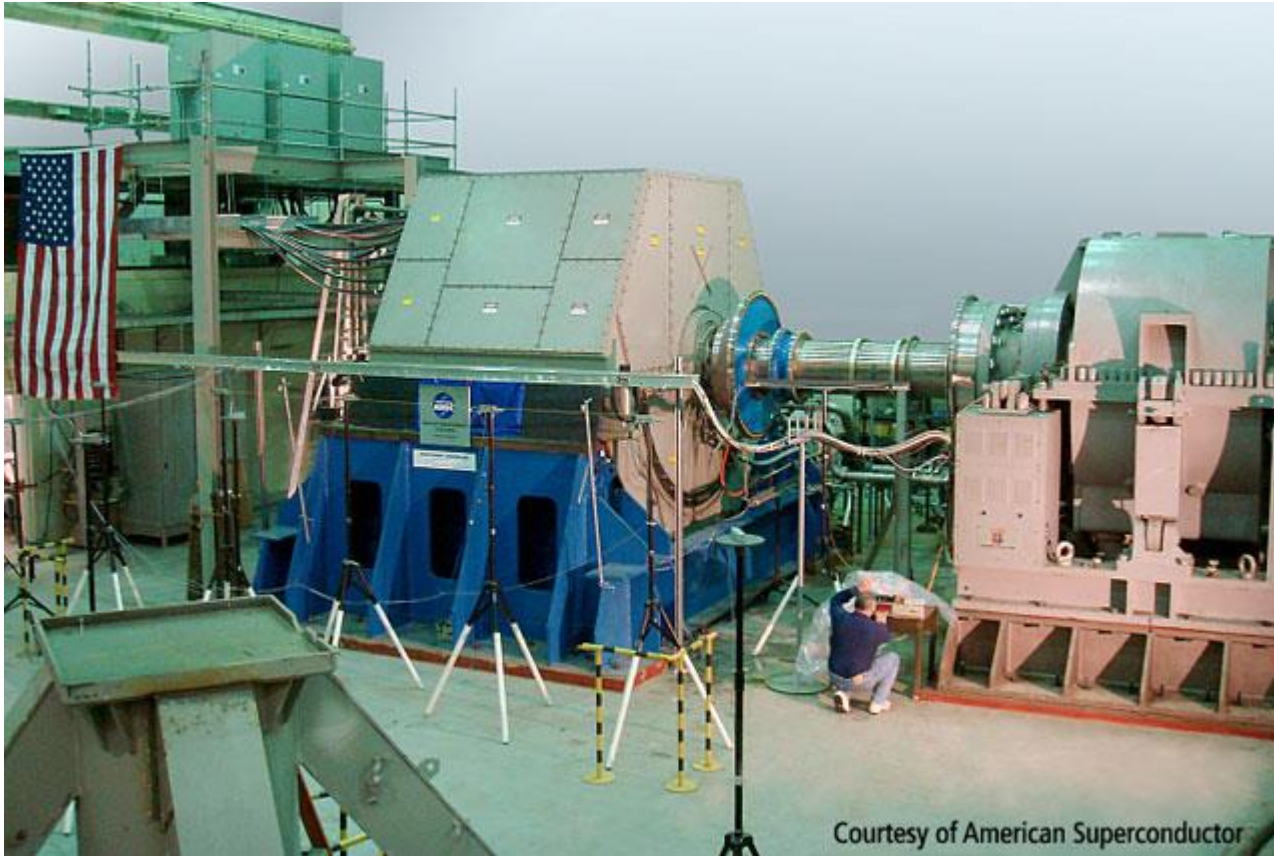


HTSC wires conduct more than 100 times the current of equivalent sized copper wire.

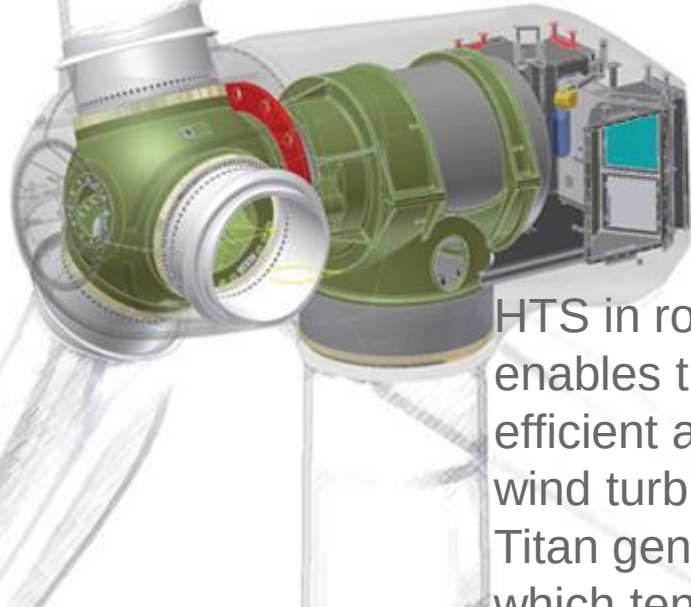
As a result, the few Amperium wires pictured on the right are able to carry as much power as all of the copper on the left.
(Courtesy of American Superconductor)



2008. world's first HTS power transmission cable system. Long Island April 2008.
At full capacity, the 138 kV system is capable of carrying 574 megawatts of power



2009. World's first 36.5 MW (49,000 hp) superconductor ship propulsion motor by AMSC and Northrop Grumman for the U.S. Navy
the motor is 1/3 the size and weight of a conventional 36.5 MW



HTS in rotors rather than copper enables the generator to be much smaller, lighter, more efficient and less expensive than conventional large-scale wind turbine generators.

Titan generators are direct drive. No need for a gearbox, which tends to be the most maintenance intensive wind turbine component. AMSC expects to license SeaTitan wind turbines and Titan generators to multiple manufacturers around the world by the end of 2010

Energy Critical Elements



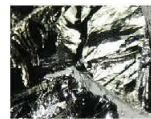
52
Te
Tellurium
127.60



TELLURIUM—brittle, silvery-white metallic element used in solar panels



32
Ge
Germanium
72.61



GERMANIUM—hard, grayish-white element with metallic luster; used in solar panels



78
Pt
Platinum
195.078



PLATINUM—silvery-white, lustrous, ductile and malleable; used in pollution control devices for cars, and in fuel cells



60
Nd
Neodymium
144.24



NEODYMIUM—bright, silvery rare-earth metal element; used in wind turbines and hybrid cars



75
Re
Rhenium
186.207



RHENIUM—silvery-white metal with one of the highest melting points of all elements; used to make advanced turbines and jet engine parts



65
Tb
Terbium
158.92534



TERBIUM—a soft, silvery-white rare earth metal; used along with its fellow rare earth europium in compact fluorescent light bulbs to provide an acceptable color balance

Chemistry. Nanocrystal nucleation and growth



Fundamental research at P. Alivisatos' laboratory. CdSe-CdS core shell NP



NanoSys Quantum Dot Technology from P. Alivisatos CdSe-CdS core shell nanoparticles

Getting away from *Toxic Tech*



**CADMIUM-FREE
QUANTUM DOTS**

Quantum dots exhibit excellent photoluminescence and electroluminescence properties such as narrow emission bandwidth and high brightness. The toxicity of cadmium makes Cd-free quantum dots more desirable for consumer applications.

Use Cd-free Quantum Dots for these applications:

- LEDs
- Displays
- Solid-state lighting
- Photovoltaics
- Transistors, etc.

Learn more about quantum dot properties and applications, including our complete product offering, at aldrich.com/quantumdots



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85851
09-2015

SIGMA-ALDRICH®

ALIEN, our future material(s)

Abundant

Lower costs

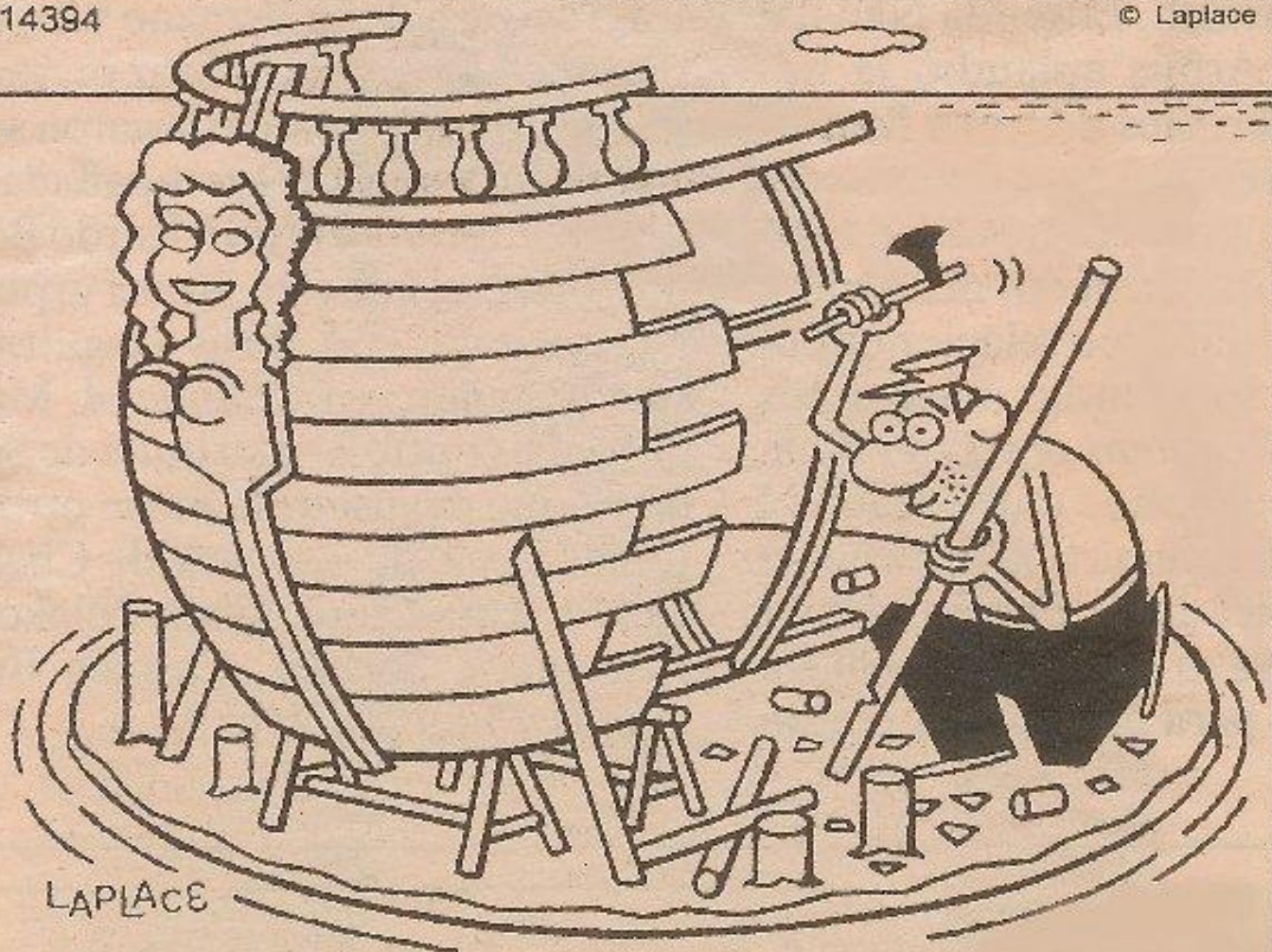
Improved performance

EEnvironmentally friendly

New props tricks and apps

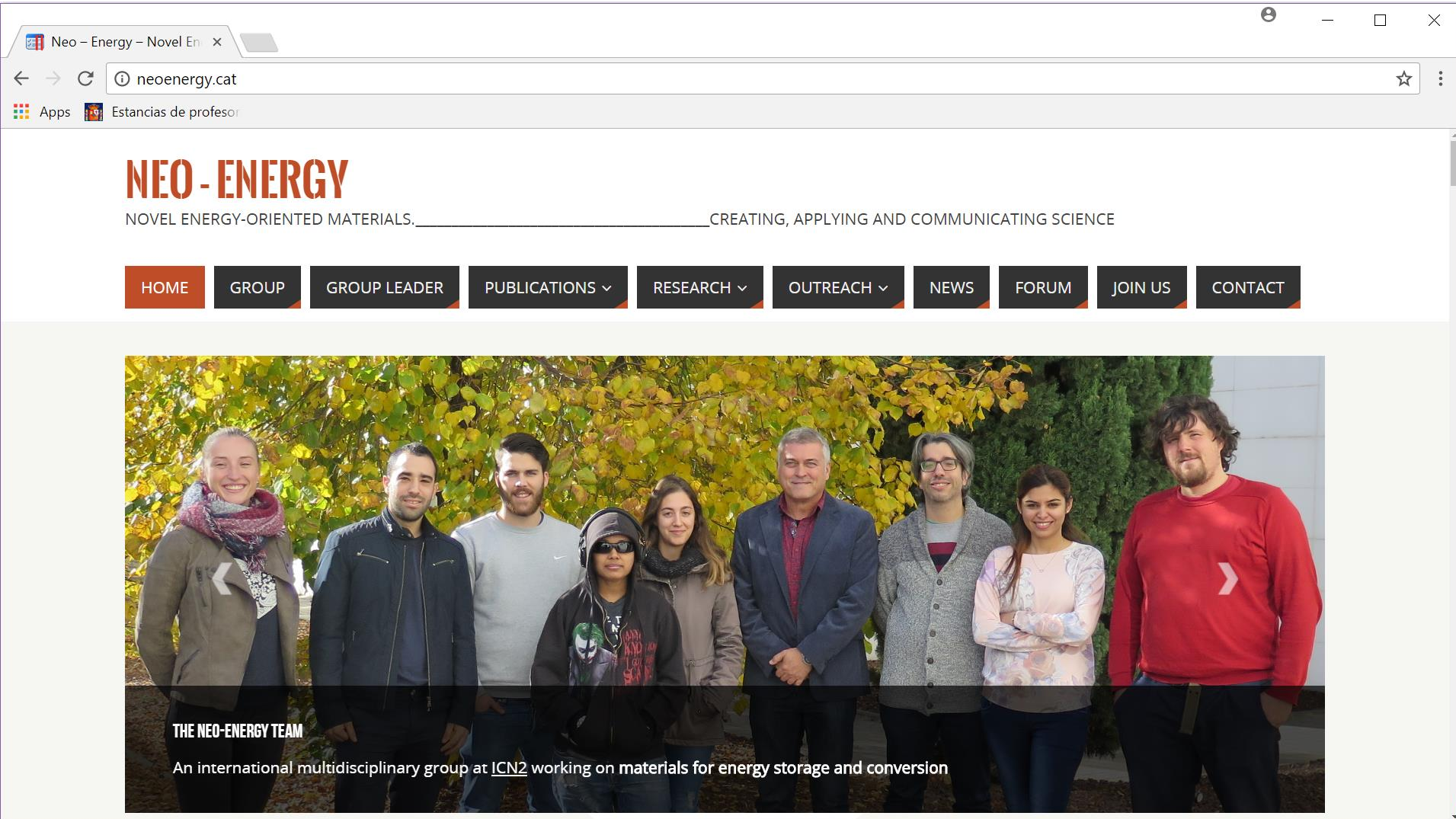
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LAPLACE

The NEO-Energy Group



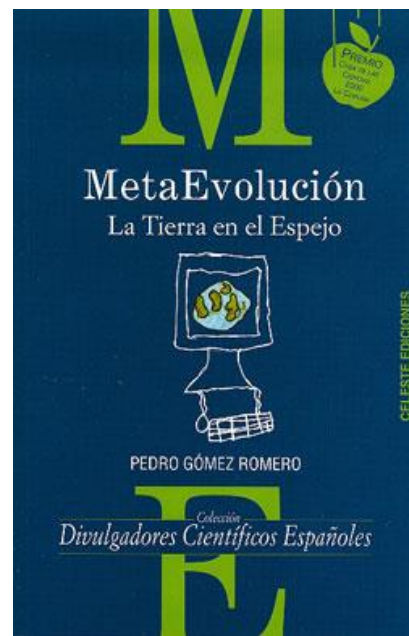
www.neoenergy.cat



Gracias

QUESTIONS

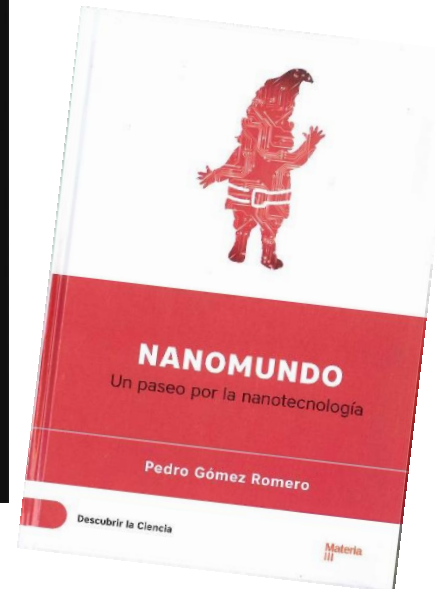
Premio Casa de las Ciencias
de Divulgación 2000
(Celeste, 2001).



Premio Europeo de
Divulgación Universidad de
Valencia (Ed. Bromera, 2016)



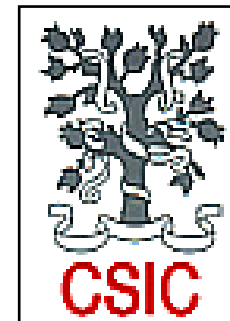
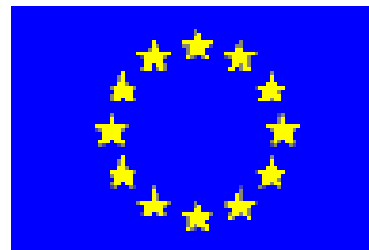
Premio Internacional de
Ensayo "Esteban de
Terreros"
(Editorial Síntesis, 2007)



Editorial Materia/El País
(Nov 2016)

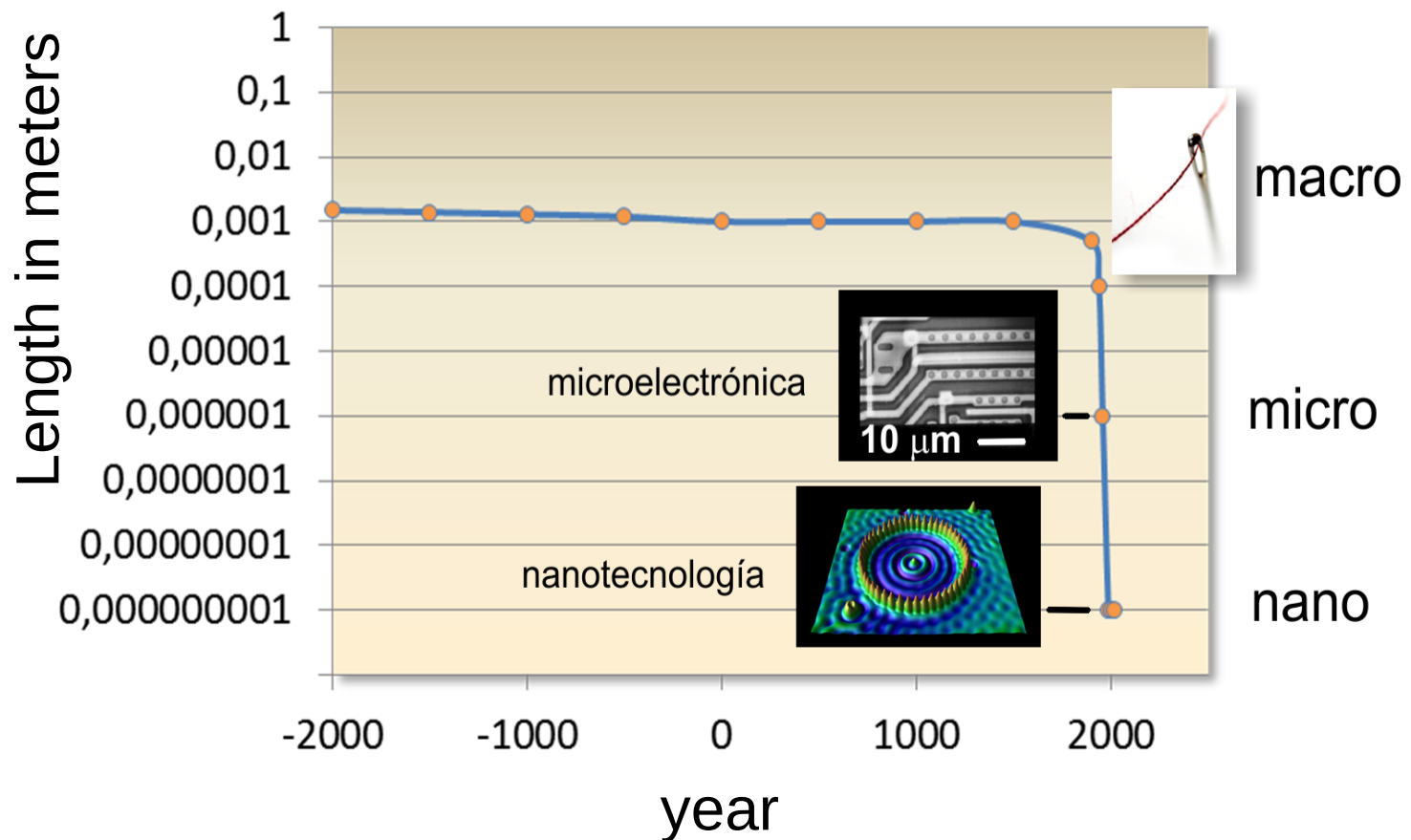
NEO-Energy Lab

Prof. Pedro Gómez-Romero



Nano-Science and Nano-Technology. Much more than just fashion

Length of the smallest object made by humans

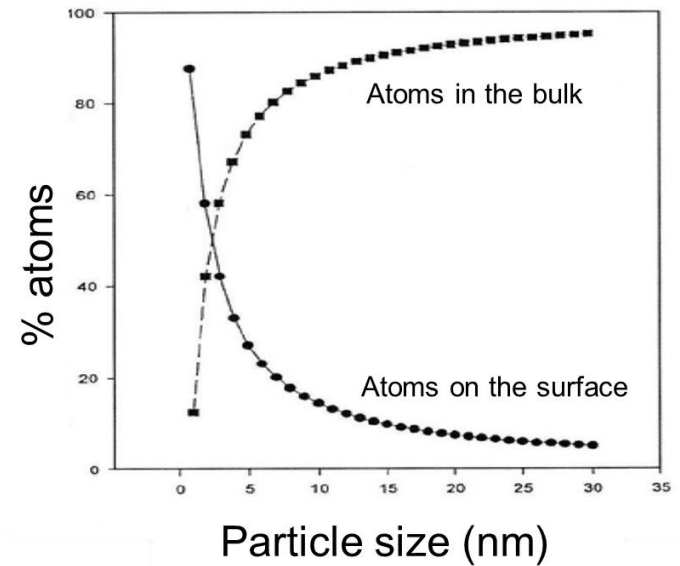
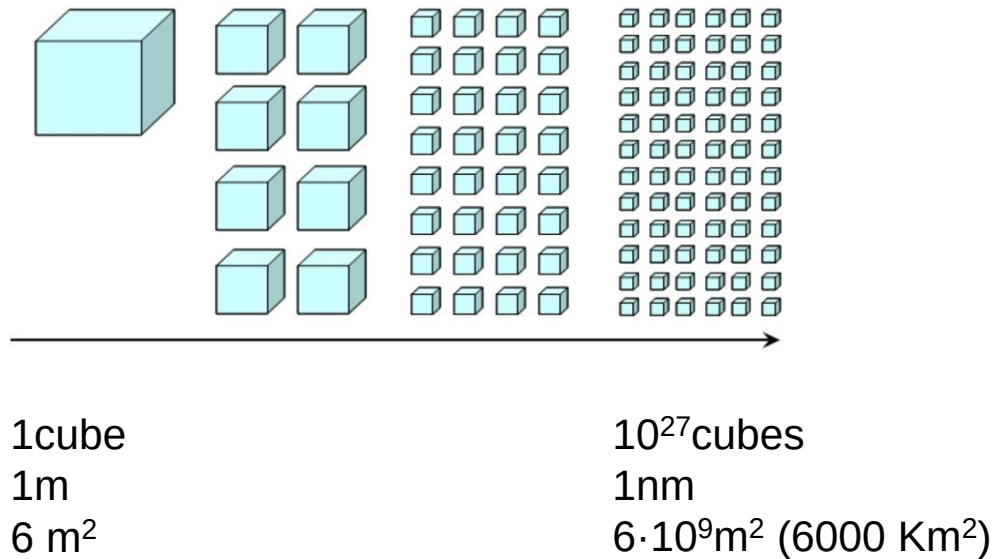


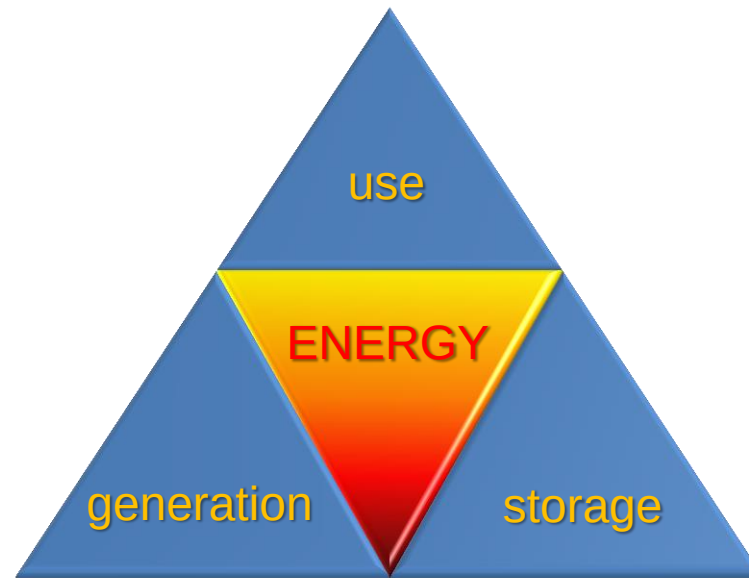
From Materials to Nanomaterials

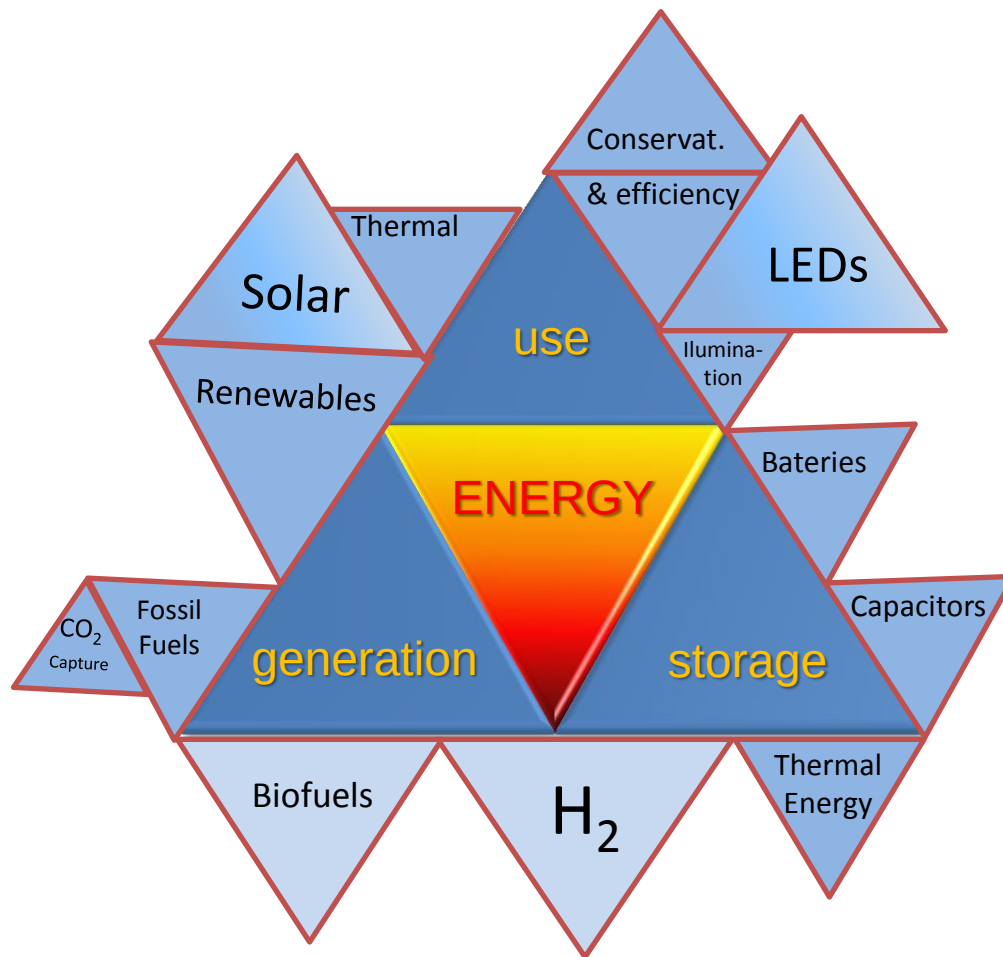
Extrinsic changes in properties

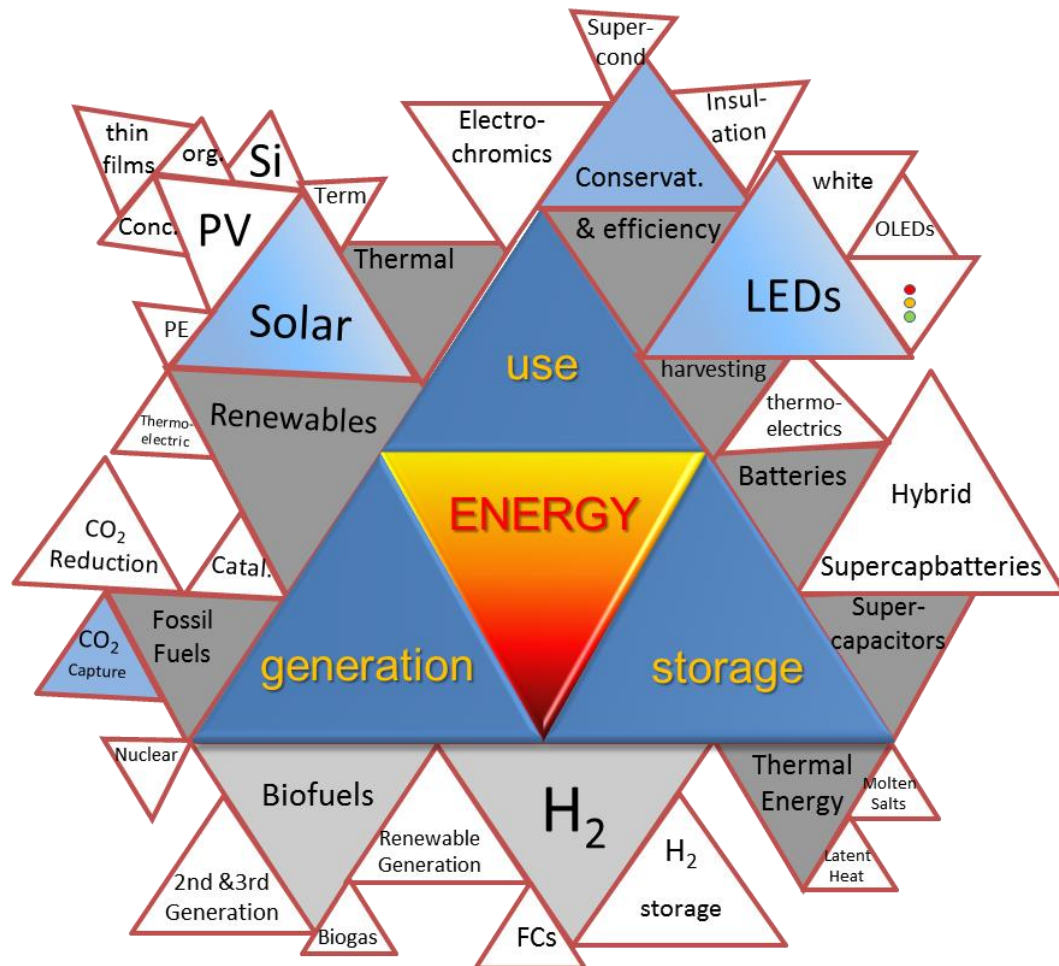
From Materials to Nanomaterials

Extrinsic changes in properties



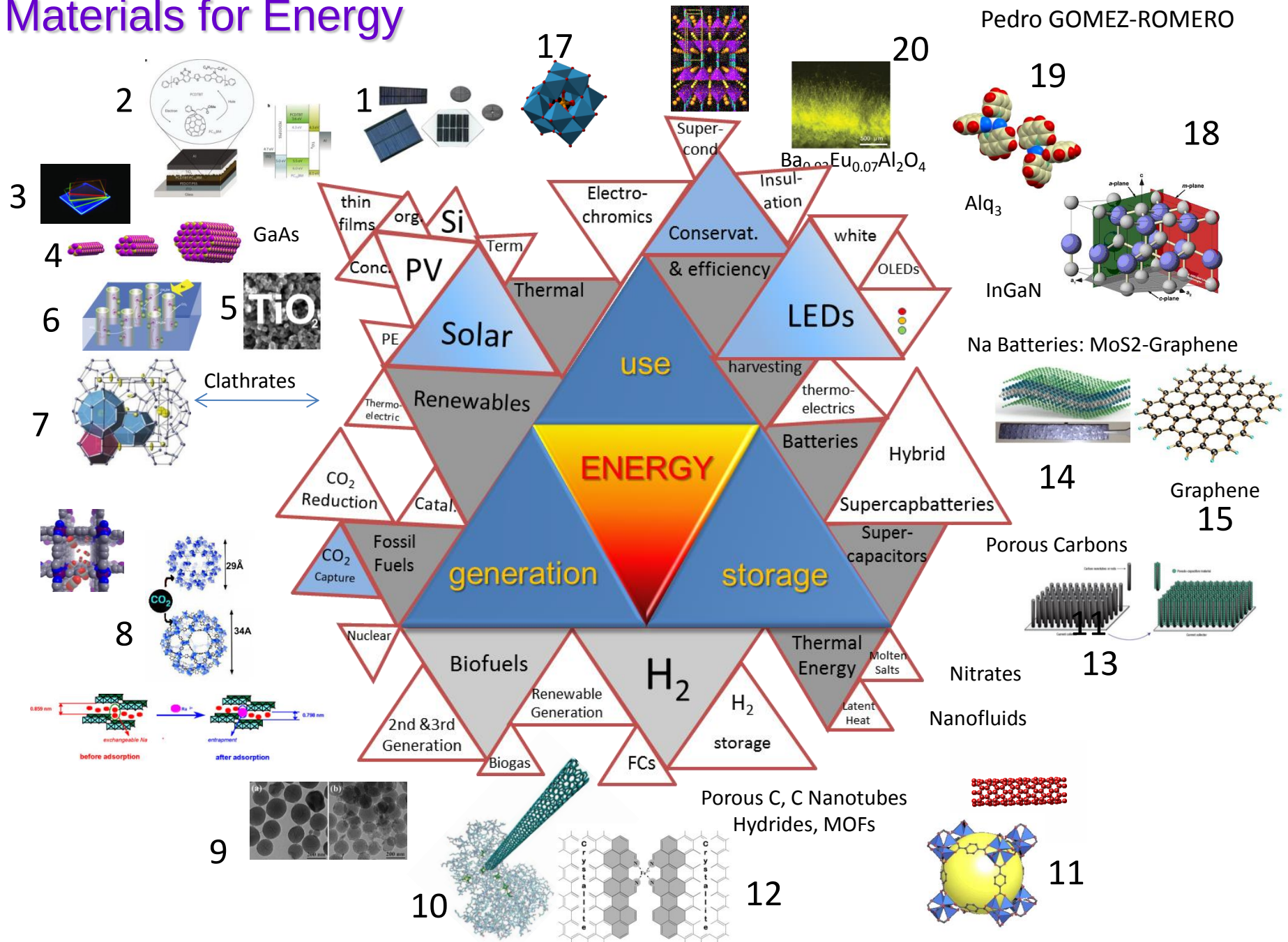






Materials for Energy

Pedro GOMEZ-ROMERO



1. CUESTIÓN DE ENERGÍA

Una introducción fundamental a la cuestión de la energía.



Termodinámica: Energía frente a Entropía



Metabolismo biológico

Biosíntesis
Crecimiento celular
Respiración
Movimiento
Comunicación
Defensa

Energía somática

2,000 Kcal/persona·día



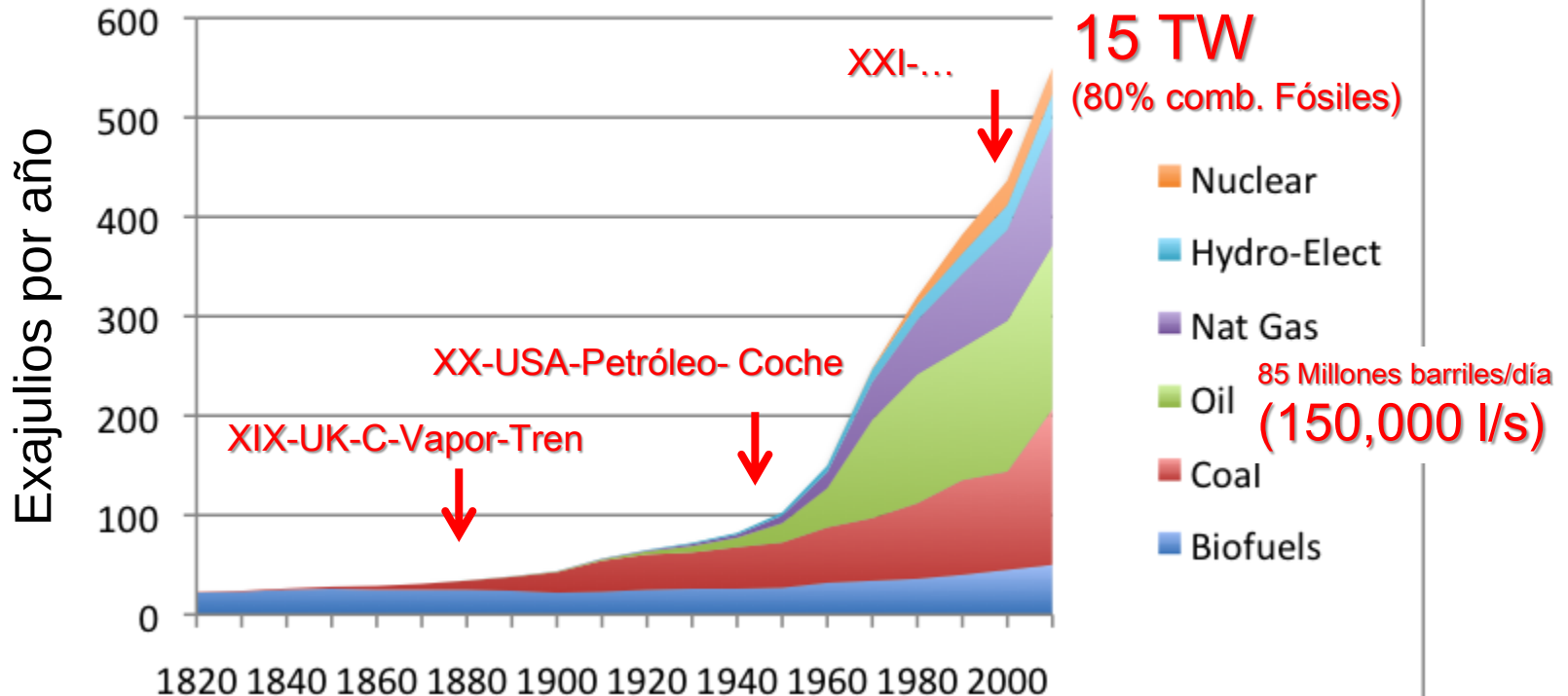
Metabolismo social

Extracción de recursos
Fabricación / manufactura
Construcción
Transporte
Comunicación
Defensa

Energía exosomática

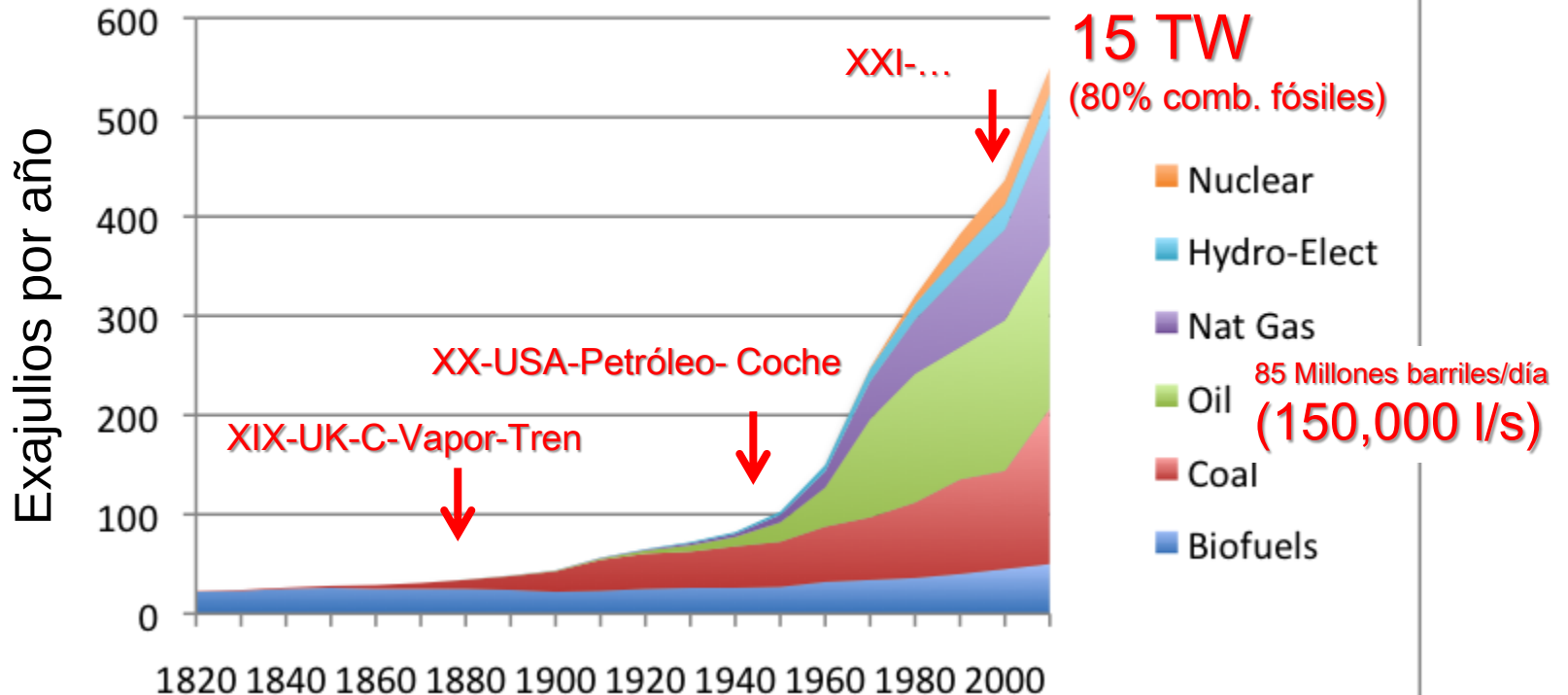
200,000 Kcal/persona·día

Consumo global de energía



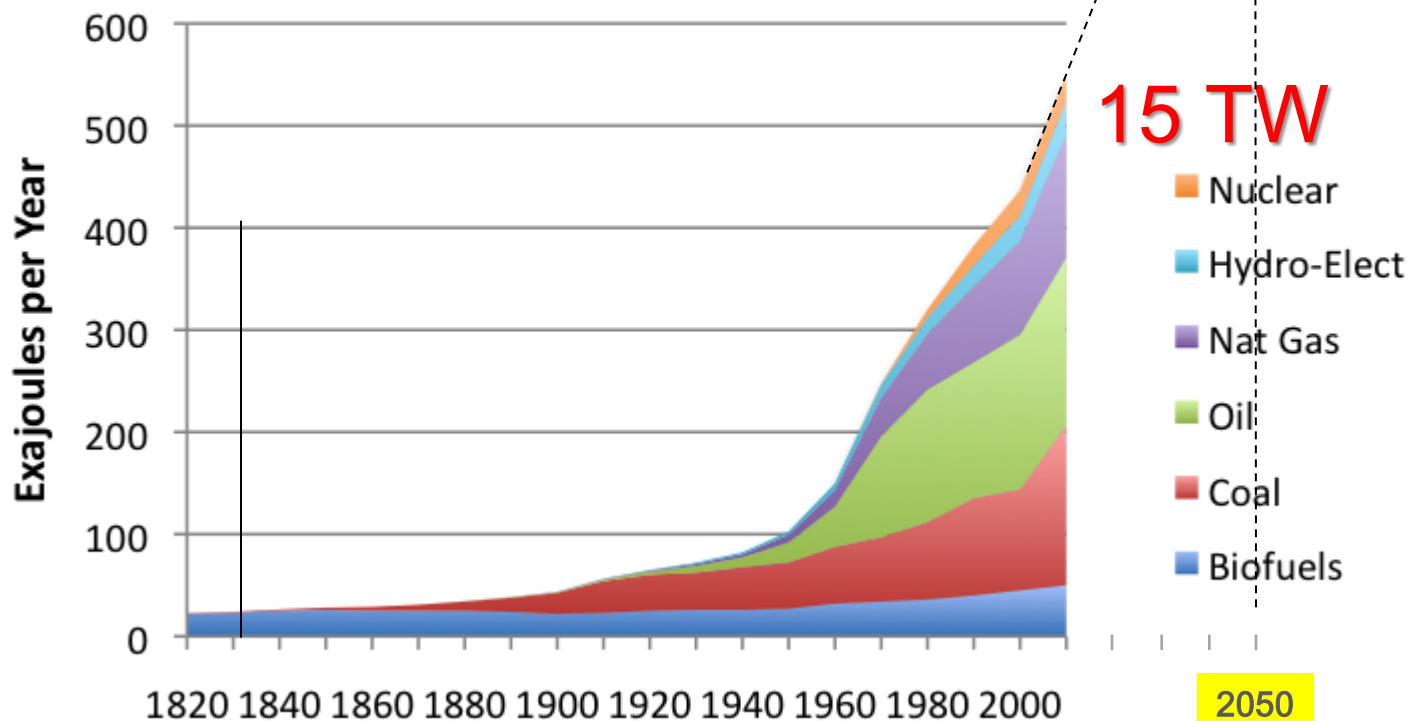
Based on Vaclav Smil estimates from Energy Transitions: History, Requirements and Prospects together with BP Statistical Data for 1965 and subsequent

Consumo global de energía



28 TW

World Energy Consumption



15 TW

2050

De recolectores a agricultores.

Una perspectiva histórica

Adios a los combustibles “primarios”

elige un vector energético

Vector

Bio: Biocombustibles

Biodiesel (20% bio 80% diesel), (oleaginosas)

Bioetanol (con Gasolina)(carbohidratos/azúcares)

Elect: Baterías recargables para Vehículos Eléctricos

Quim: Pilas de Hidrógeno (Pilas de Combustible)

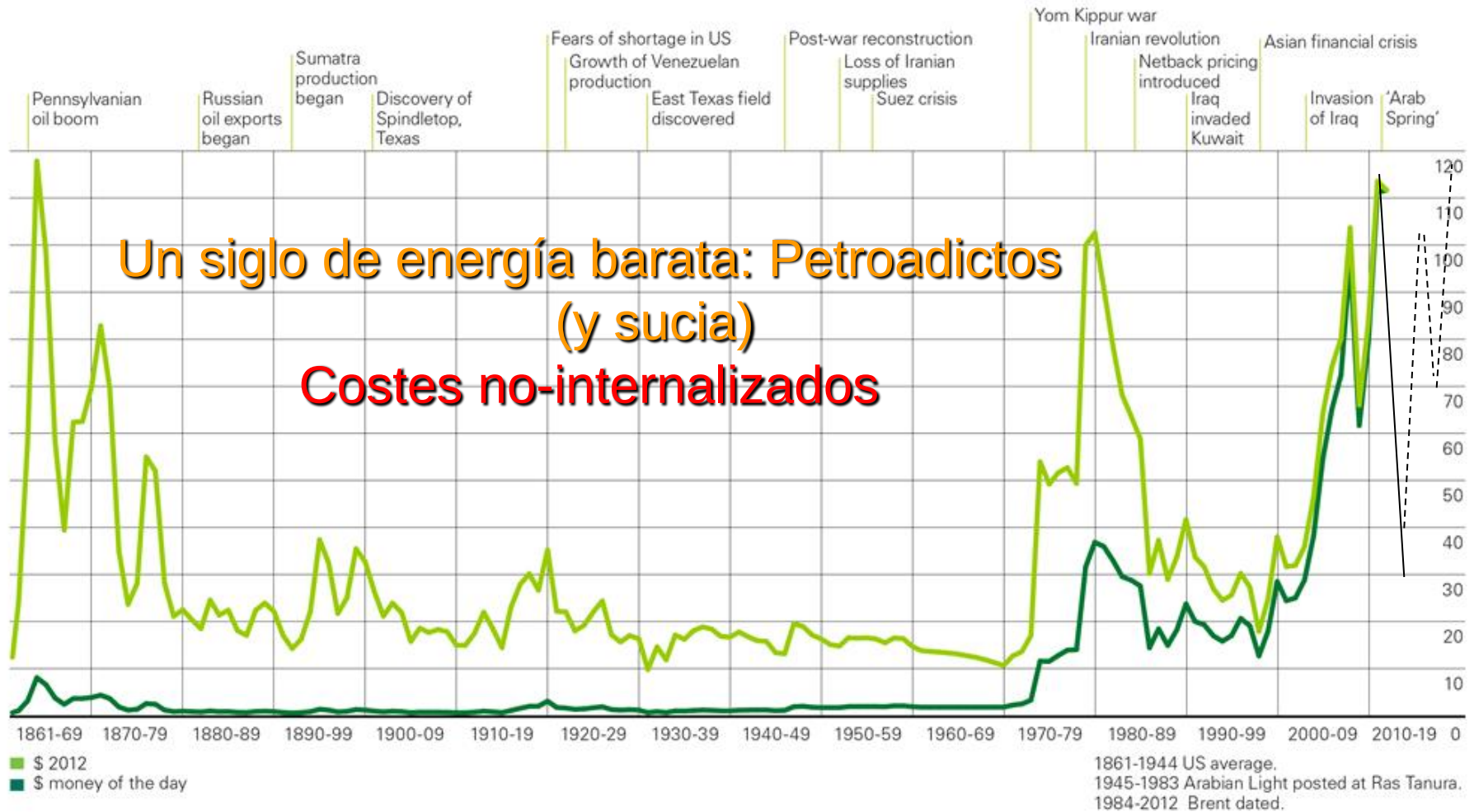
H₂... El combustible del futuro?

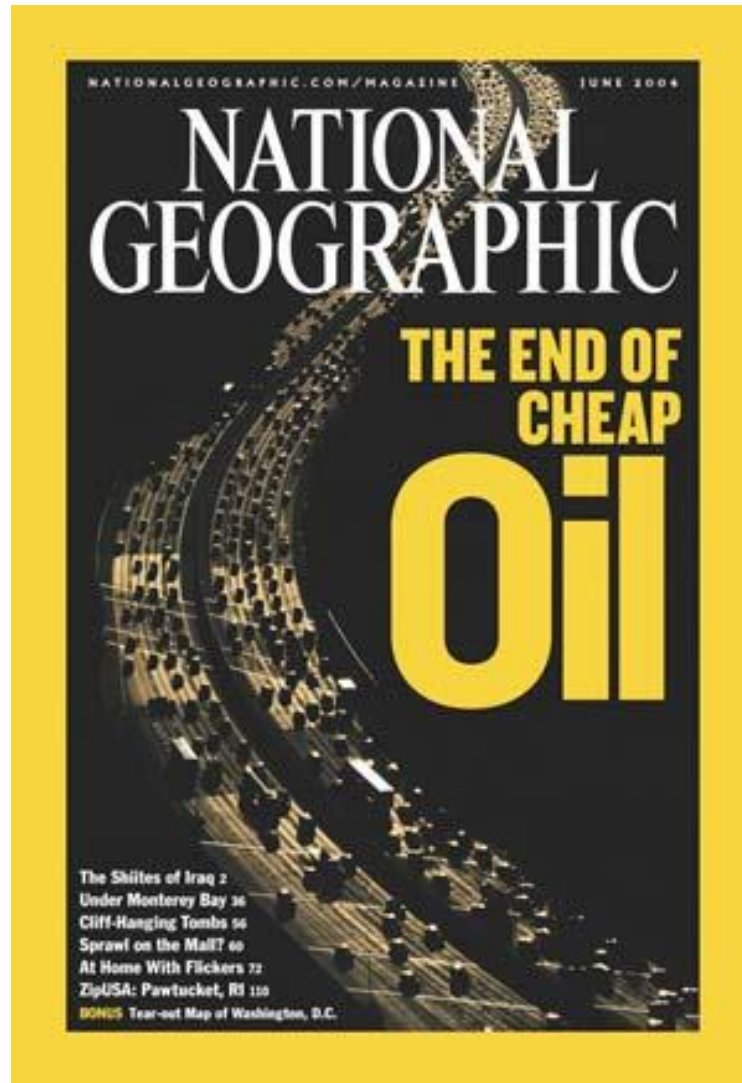


Crude oil prices 1861-2012

Nuevos vectores: ¿Competitivos?

US dollars per barrel, world events





Junio 2004, Volumen 205, Numero 6





¿La solución al negro panorama de la energía?

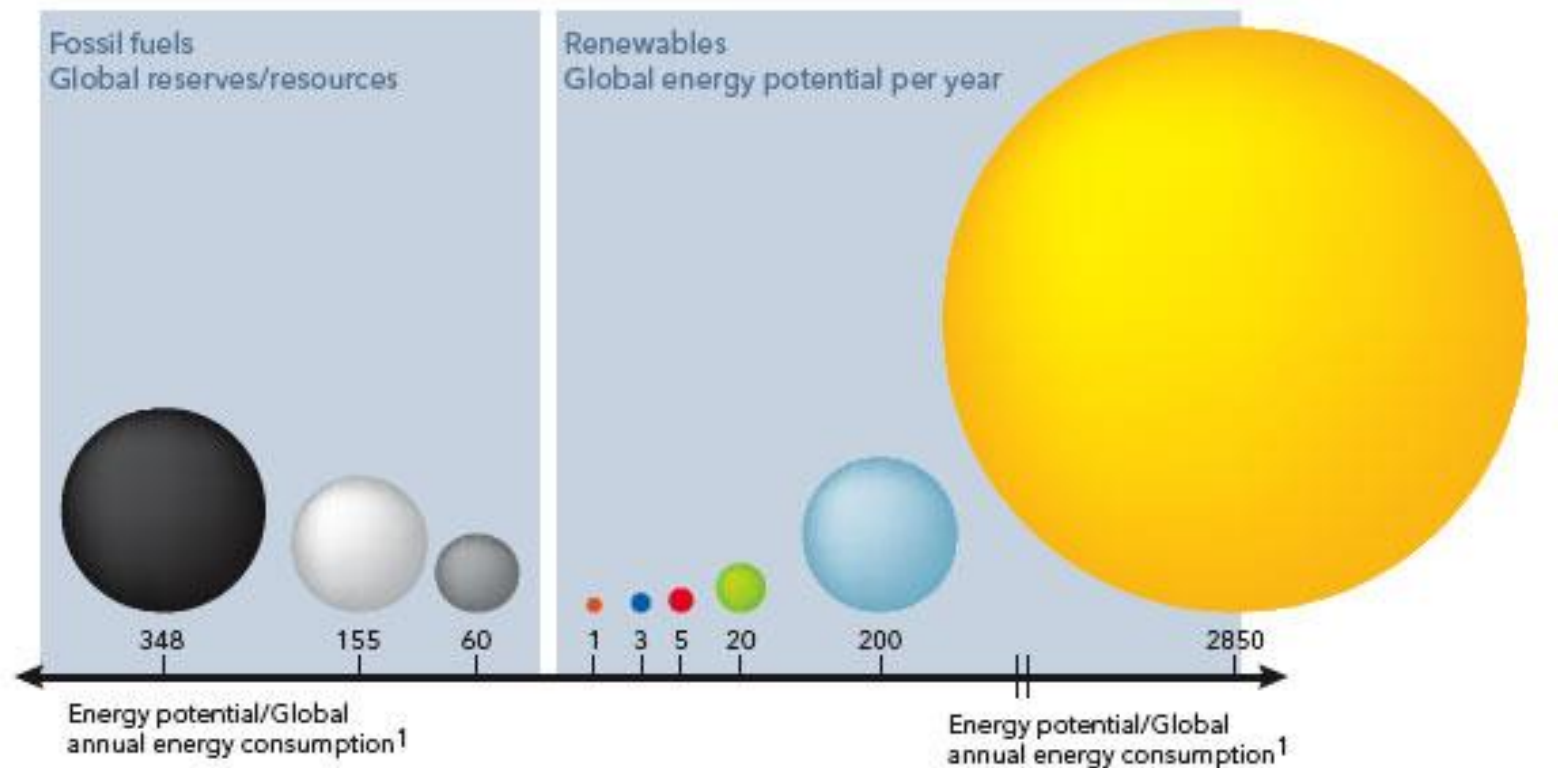


¿La solución al negro panorama de la energía?

fuentes
RENOVABLES
de energía primaria



Fuentes primarias: ¿Sol arcaico o moderno?



	Energy potential Reserves/Resources ²	Thereof conven- tionally utilisable ²		Energy potential (amount of energy p. a.) ²	technologically utilis- able (state of the art) ²
Coal	~ 135.000 EJ		Solar radiation	~ 1.111.500 EJ	~ 1.482 EJ
Natural gas	~ 60.400 EJ	~ 12.000 EJ	Wind energy	~ 78.000 EJ	~ 195 EJ
Crude oil	~ 23.000 EJ	~ 9.800 EJ	Biomass	~ 7.800 EJ	~ 156 EJ
			Geothermal	~ 1.950 EJ	~ 390 EJ
			Hydro/tide power	~ 1.170 EJ	~ 78 EJ
Global energy demand 2006:	~ 470 EJ				

1 EJ = 1 exajoule or 10^{18} joules or ~163 million barrels of oil.

Data source: German Federal Institute for Geosciences and Natural Resources

El nuevo paisaje de la energía



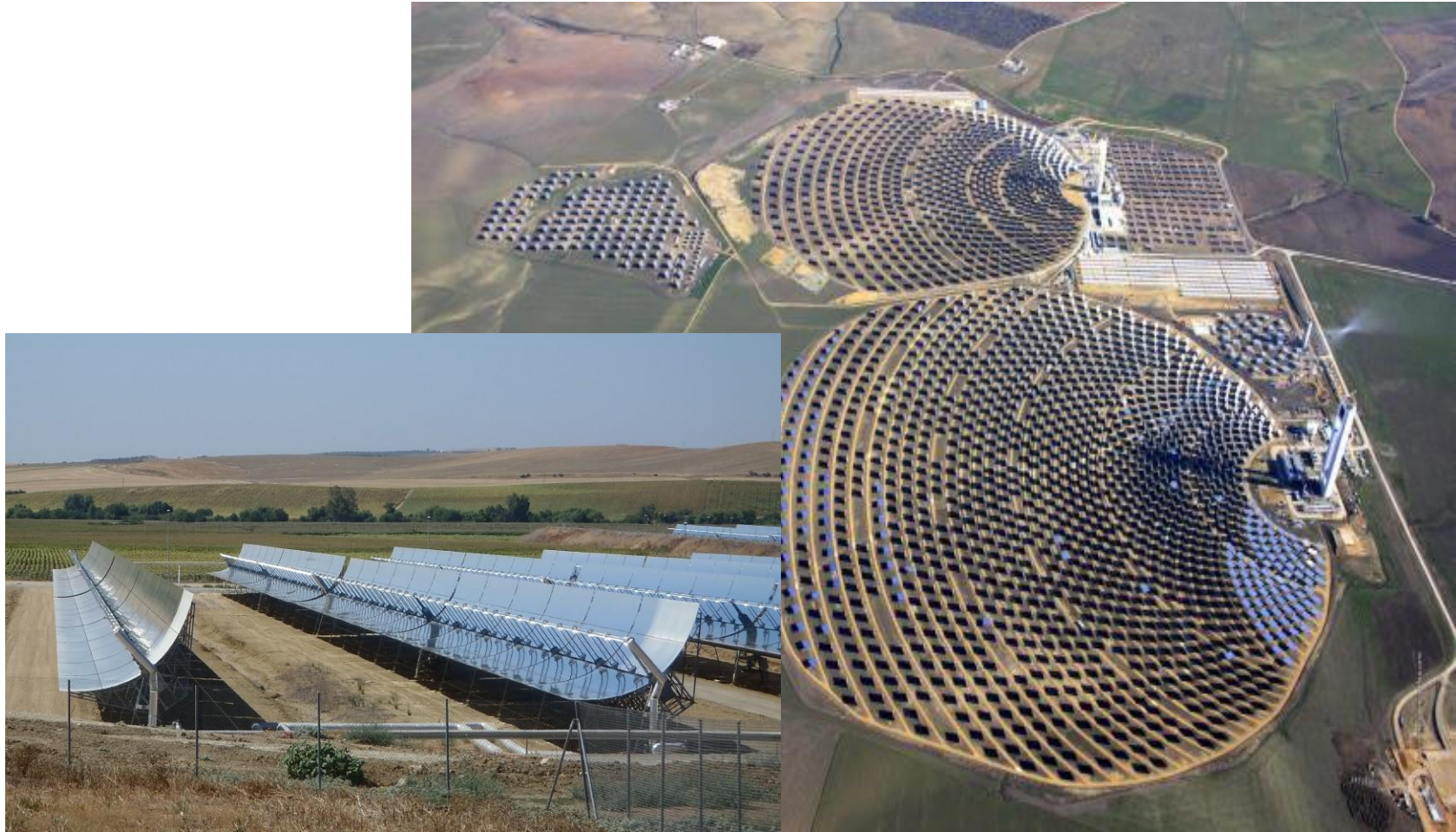
Wind power at Sisante

La Mancha
Spain



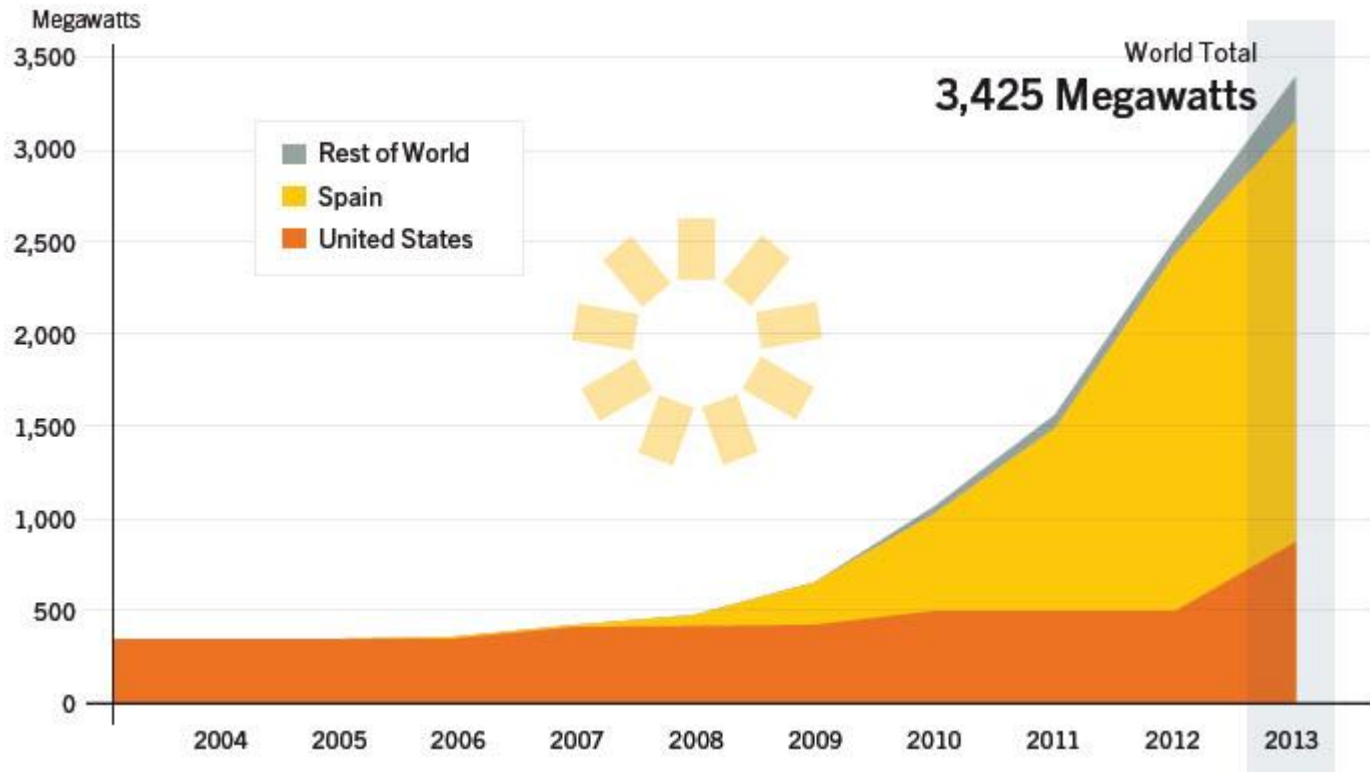
Solar PV crop in Castilla La Mancha (Solaer)

El nuevo paisaje de la energía

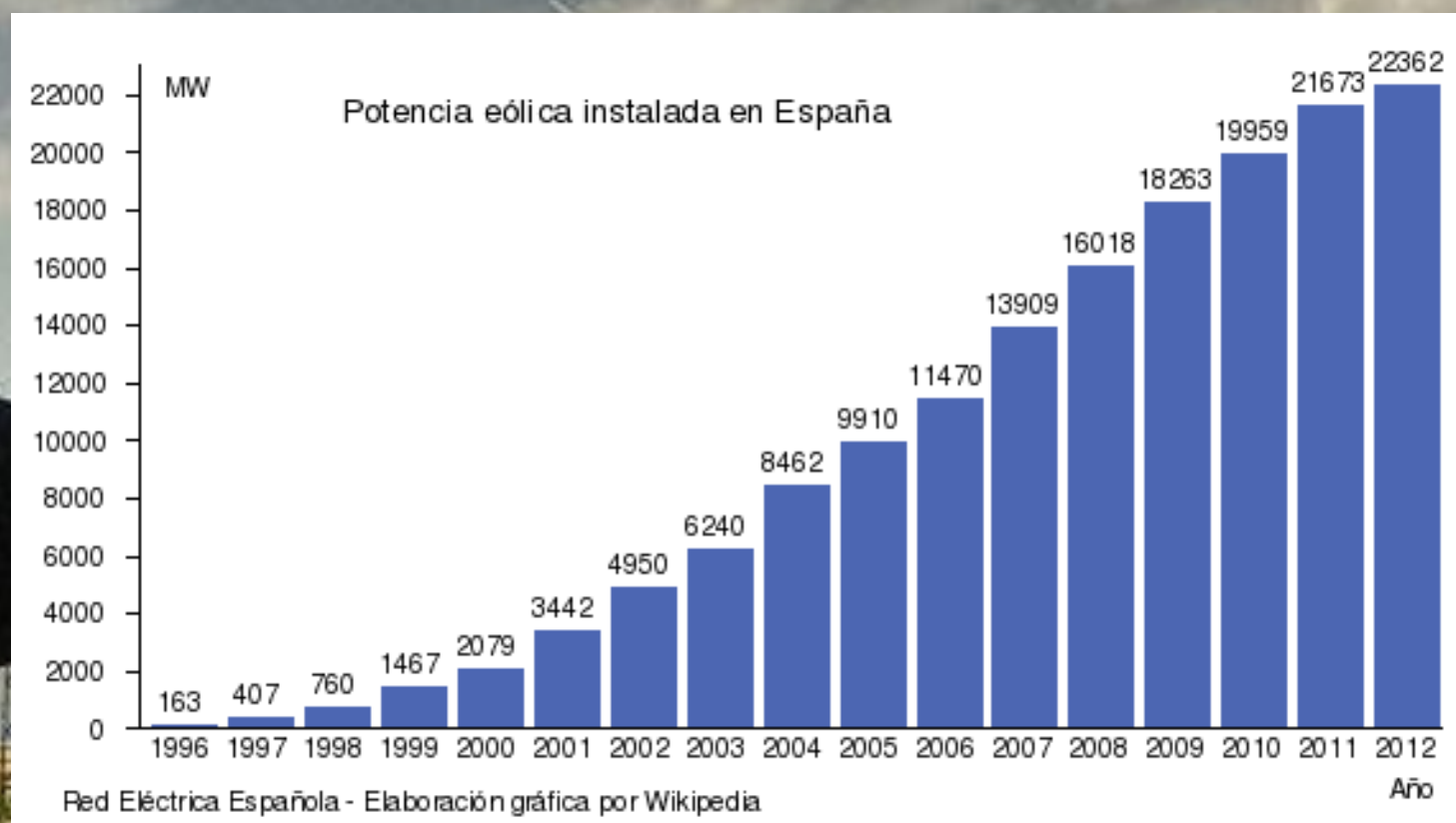


Pioneer thermosolar power plant. Sanlúcar (Sevilla) Spain (Abengoa)

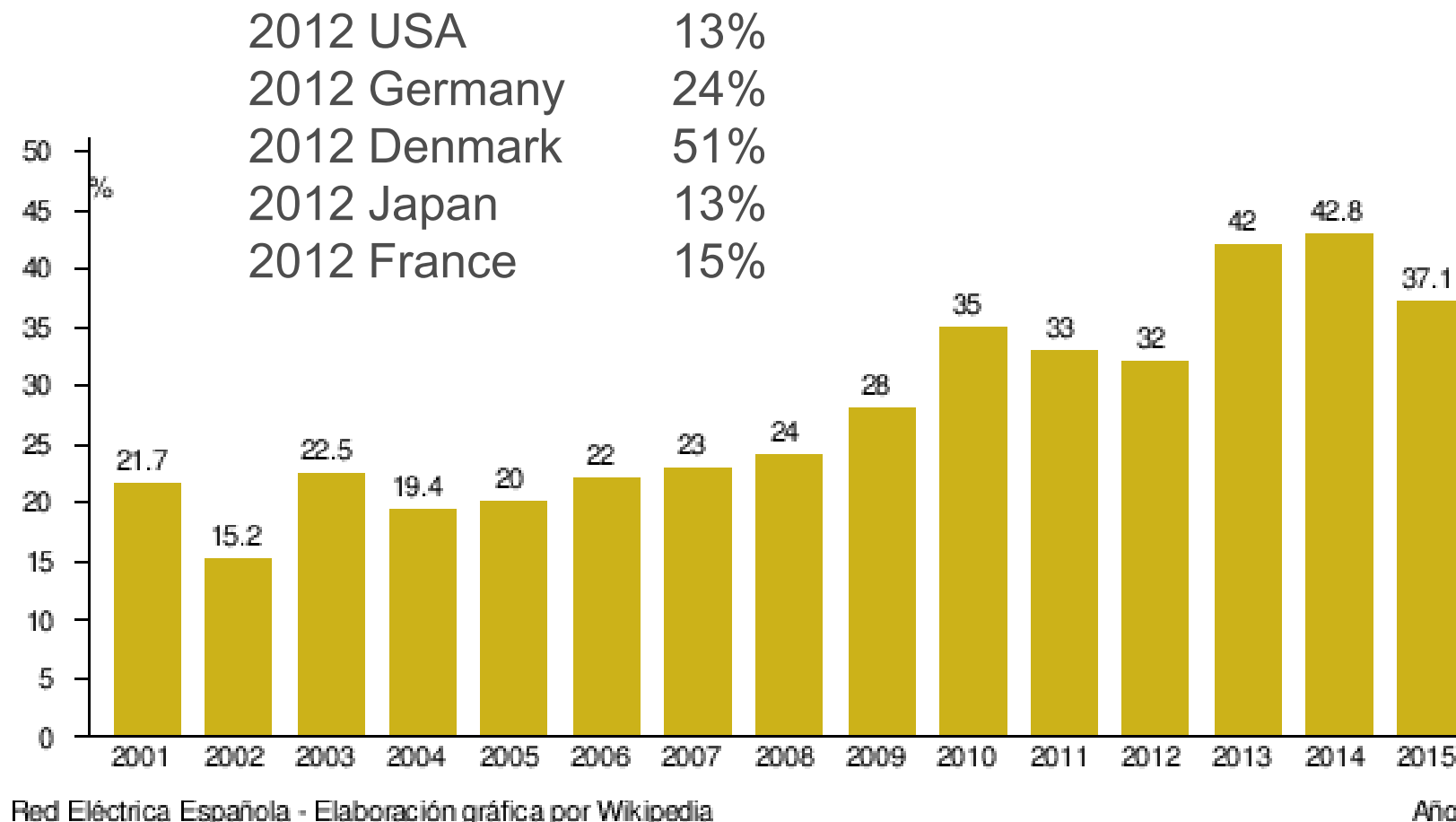
España, potencia mundial en energía solar térmica de concentración (CSP).



El nuevo paisaje de la energía

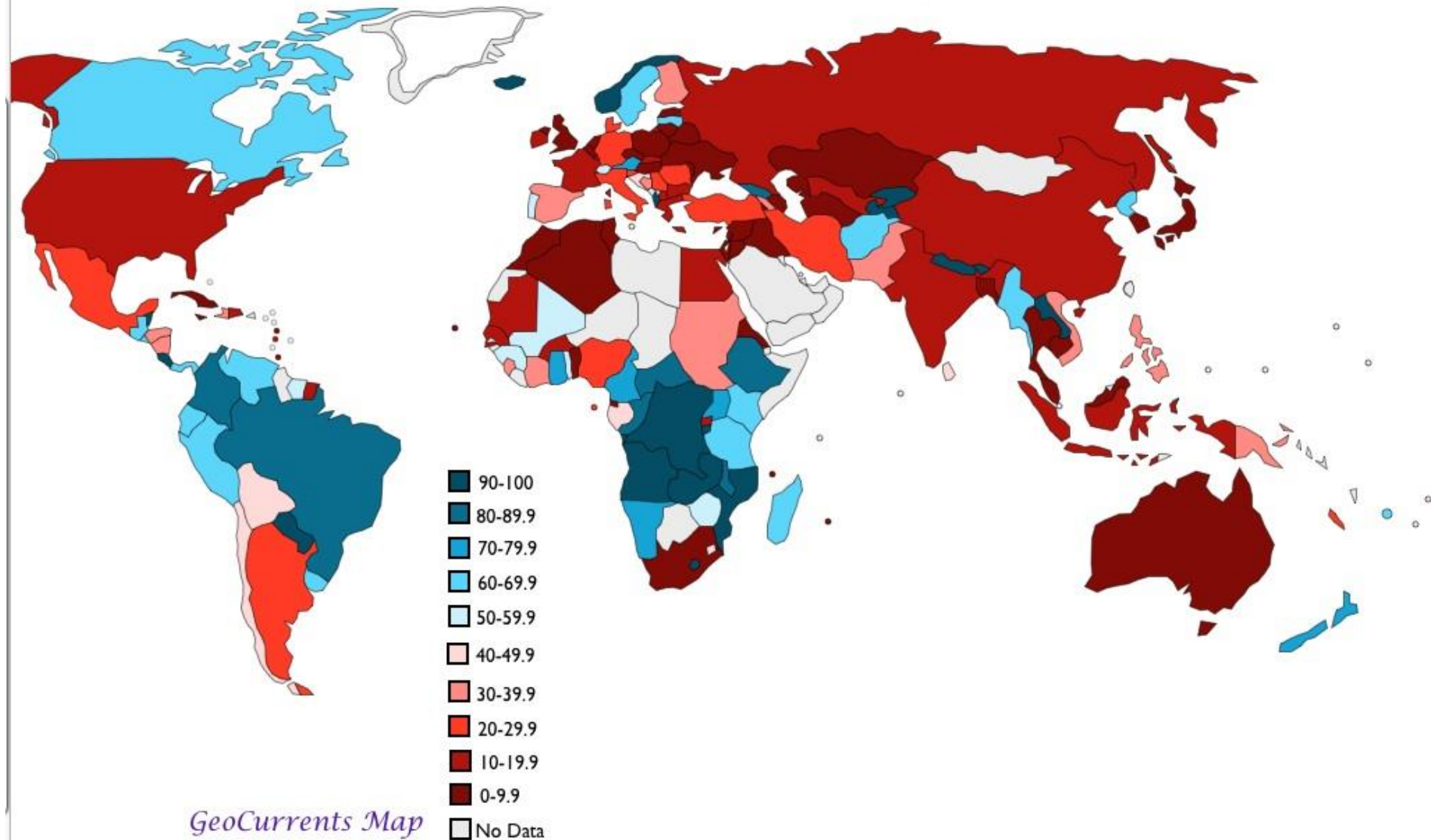


Renewable electricity in Spain (%)



Percentage of Electricity Generation from Renewable Sources

(Hydro, Geothermal, Solar, Biomass, Wind)



Data Source: http://en.wikipedia.org/wiki/List_of_countries_by_electricity_production_from_renewable_sources

¿La solución al negro panorama de la energía?

Combustibles
más LIMPIOS
vectores



De recolectores a agricultores

Adios a los combustibles “primarios”
elige un vector energético

Vector

Bio: Biocombustibles
Biodiesel (20% bio 80% diesel), (oleaginosas)
Bioetanol (con Gasolina)(carbohidratos/azúcares)

Elect: Baterías recargables para Vehículos Eléctricos

Quim: Pilas de Hidrógeno (Pilas de Combustible)
H₂... El combustible del futuro?



Biocombustibles



Biocombustibles

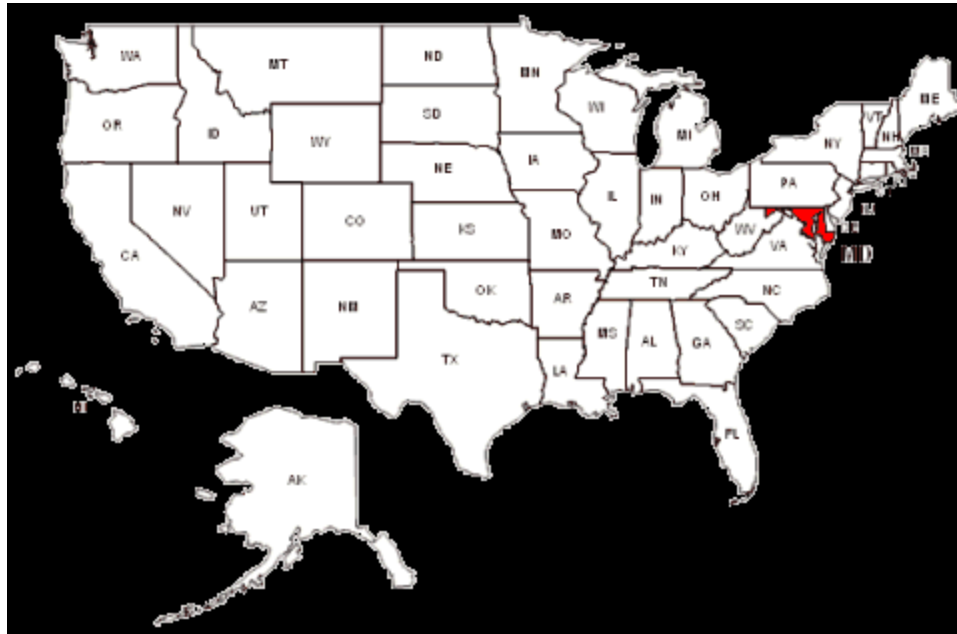






Biocombustibles 2G

DOE: aprox. 40000 Km²
Suficientes para sustituir
Todo el petróleo consumido
en USA por algas



Biocombustibles 3G

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Quim: Pilas de Hidrógeno (Pilas de Combustible)

H₂... El combustible del futuro?



H₂ ¿El combustible del futuro?



Honda FCX 2005
Hydrogen car

Revoluciones pendientes:

- Generación H₂ renovable
- Almacenamiento eficaz
- Pilas de Combustible mejoradas



13 August 2004

Tipos de aplicaciones

Generación local estacionaria



Off-grid and emergency power
PEM FC

Transporte



Honda FCX 2005
Coche de hidrógeno

Miniaturización



Pilas de combustible para
ordenadores portátiles
20h de autonomía.

De recolectores a agricultores

Adios a los combustibles “primarios”
elige un vector energético

Vector

Bio: Biocombustibles

Biodiesel (20% bio 80% diesel), (oleaginosas)

Bioetanol (con Gasolina)(carbohidratos/azúcares)

Elect: Baterías recargables para Vehículos Eléctricos

Quim: Pilas de Hidrógeno (Pilas de Combustible)

H₂... El combustible del futuro?



Energía Centralizada... o Distribuída

Largest Stationary Battery System.
Power Back-up, Fairbanks (AK, USA)



46 MW

1370 NiCd cells

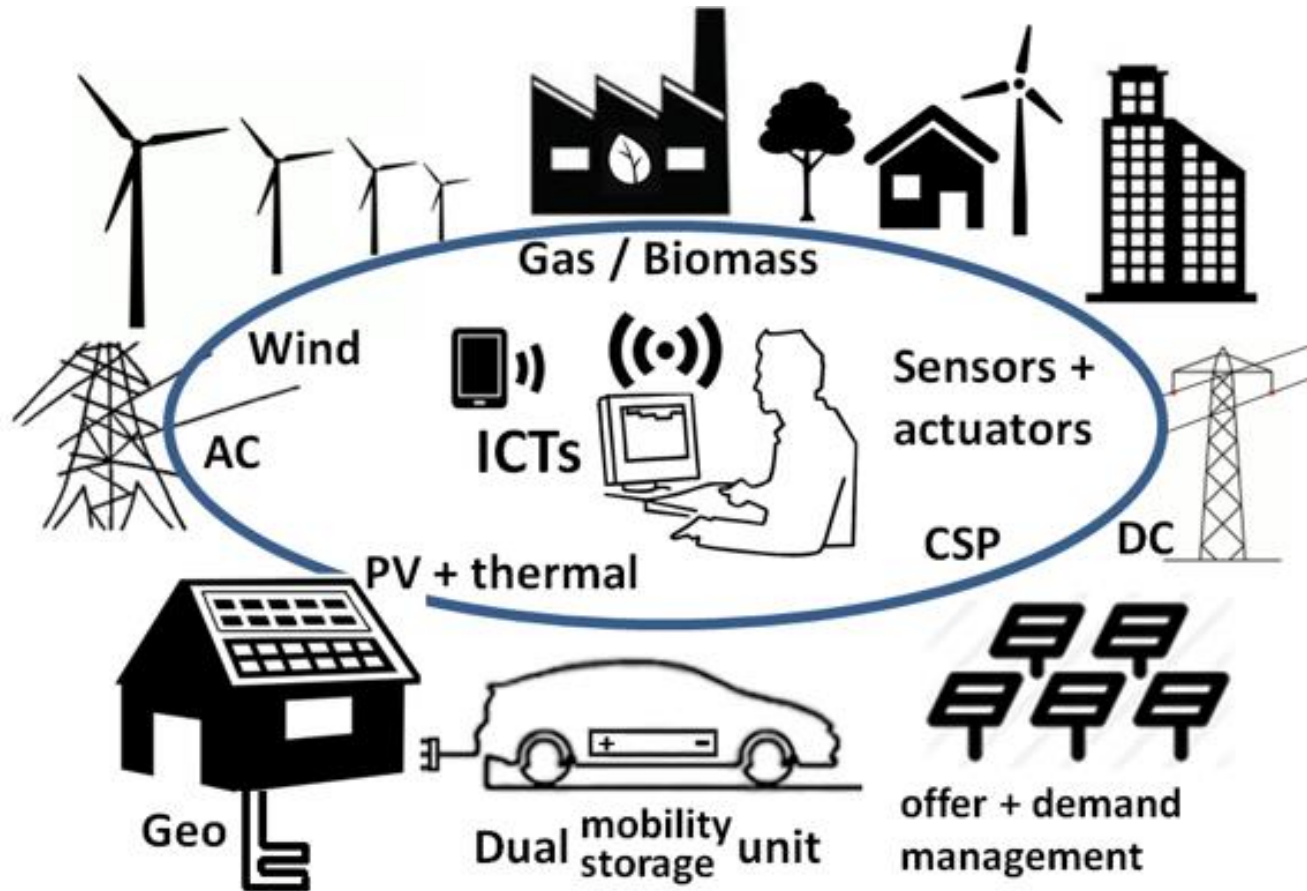
2000 m²

1300 T

7 min elec. Fairbanks

Energía Centralizada... o Distribuída

Smart Grids



¿puede haber algo más sostenible?



Claude Guillot, (1673-1722) **The Two Coaches** (ca. 1710) Musée Louvre, Paris

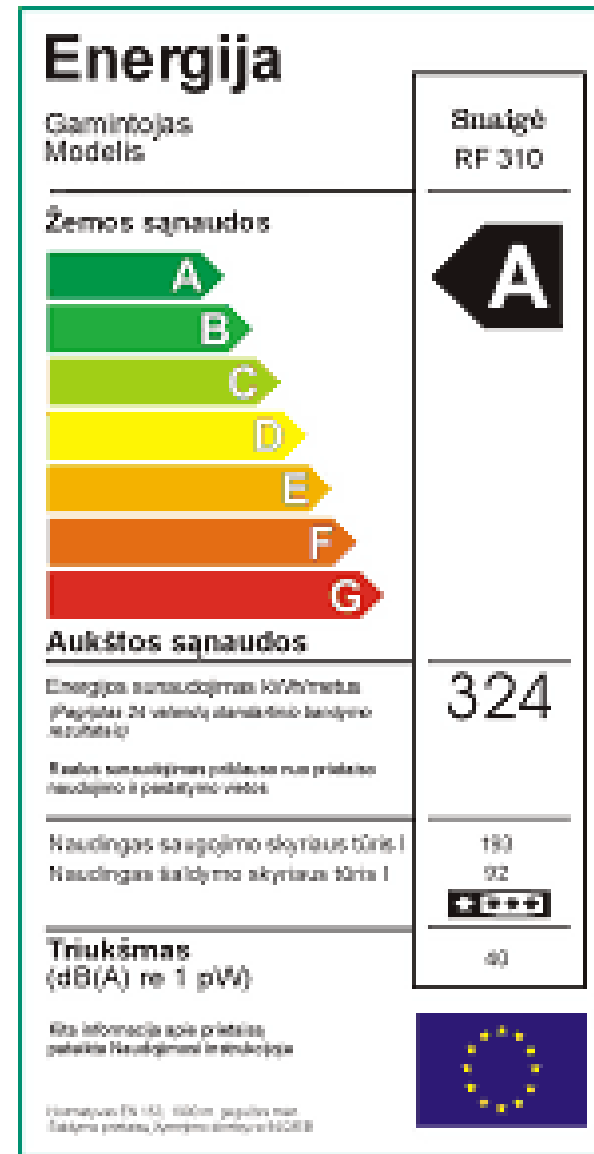
¿La solución al negro panorama de la energía?



**AHORRO
Y
EFICIENCIA ENERGÉTICOS**

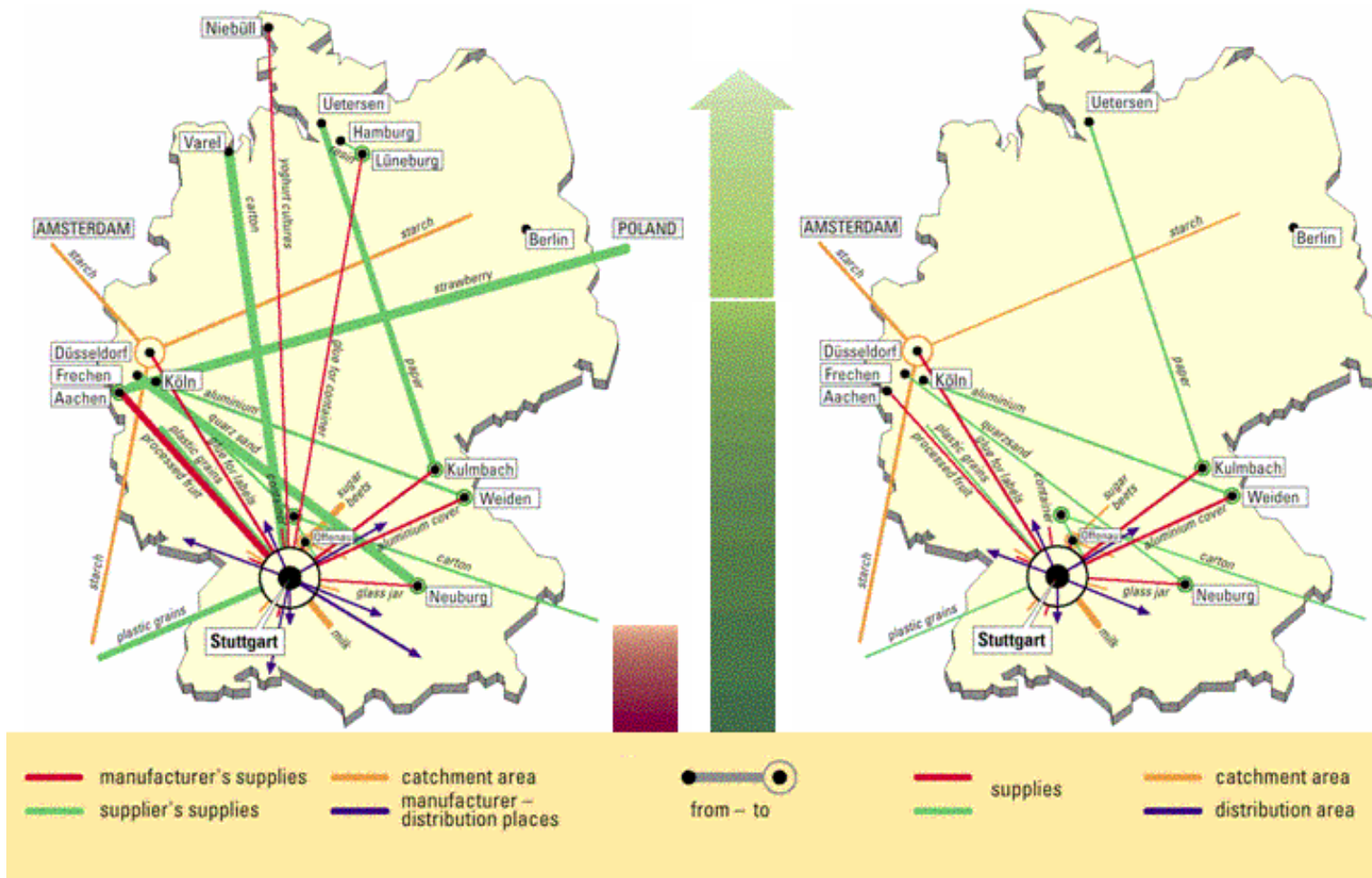
Eficiencia y ahorro energéticos

Frigo tipo A vs. tipo D
Se amortiza en ≈ 3 años



Eficiencia energética / Cadena de suministro

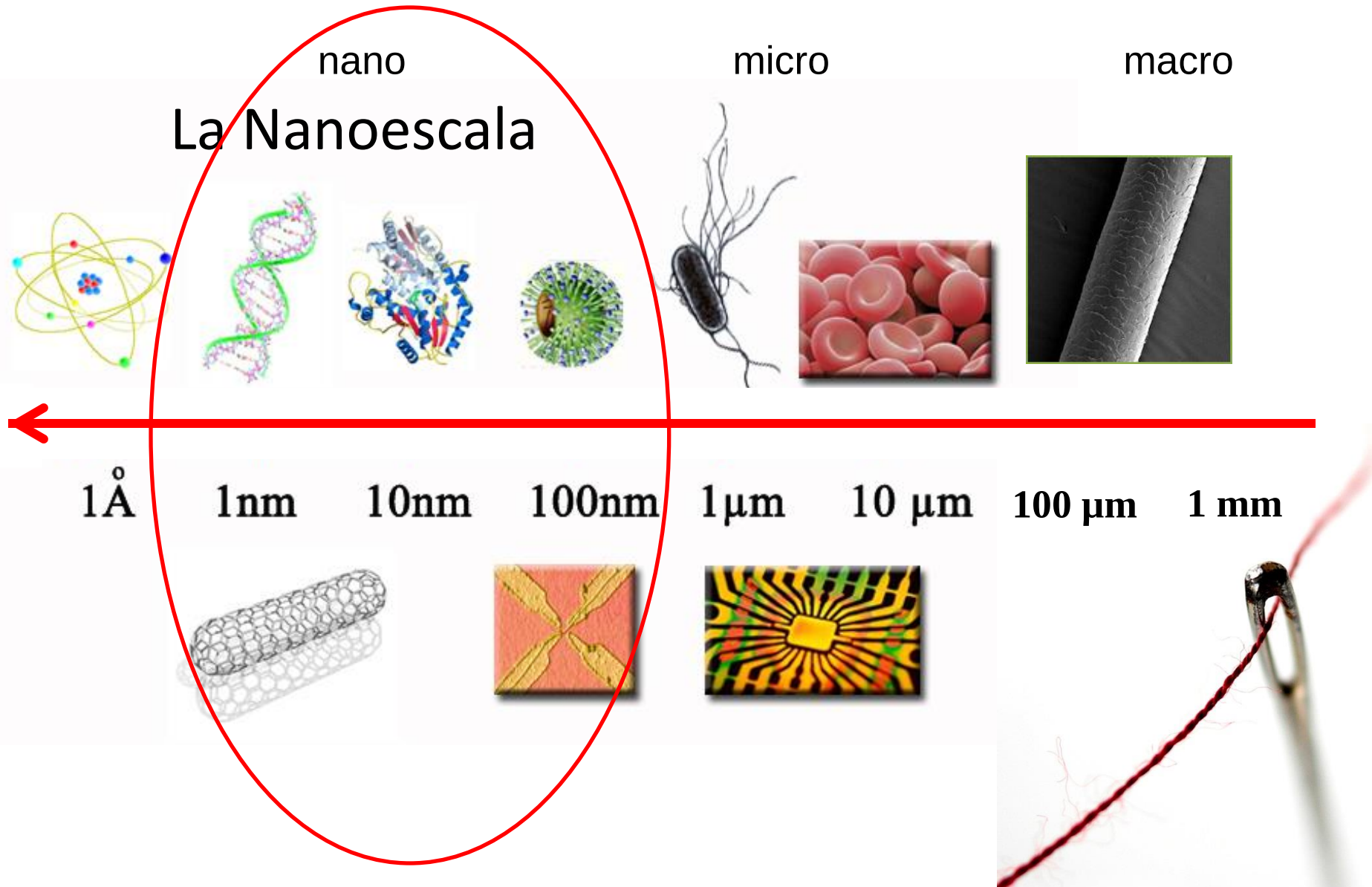
Strawberry yoghurt



Fuente: E.U. von Weizsäcker. "China: Building a Resource-Efficient Society" Beijing, 25 June, 2005

**la nanotecnología
como motor de cambio
del modelo energético**

2. NANOCIENCIA Y NANOTECNOLOGÍA



2. NANOCIENCIA Y NANOTECNOLOGÍA

Temas de investigación transversales en Nanociencia

- Catálisis por parte de materiales en la nanoescala
- Uso de interfaces para manipular energy carriers
- Relaciones entre estructura y función en la nanoescala
- Ensamblado y arquitectura de estructuras en la nanoescala
- Teoría, modelado y simulación de la nanociencia de la energía
- Métodos de síntesis escalables

El principio de la nanotecnología moderna

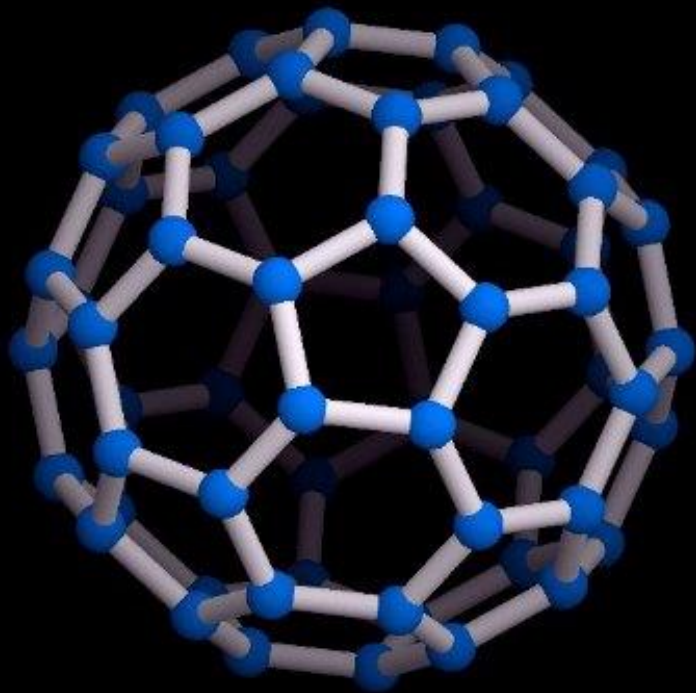
O

Cómo ganar el premio Nobel con un balón de fútbol

Kroto, Curl and Smalley
Fullereno, 1985. Premio Nobel de Química 1996



Por su descubrimiento de los fullerenos

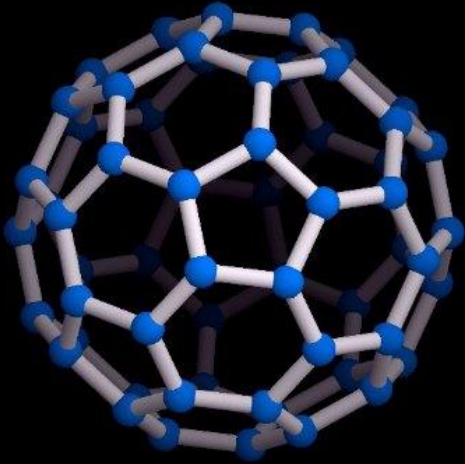


nanobalón

$1\text{nm} = 0.000000001\text{ m}$

Nano mundo

1nm = 0.000000001 m



nanobalón

1nm = 0.000000001 m

mundo humano

1 m



balón

0.22 m

Giga mundo

1Gm = 1,000,000,000 m

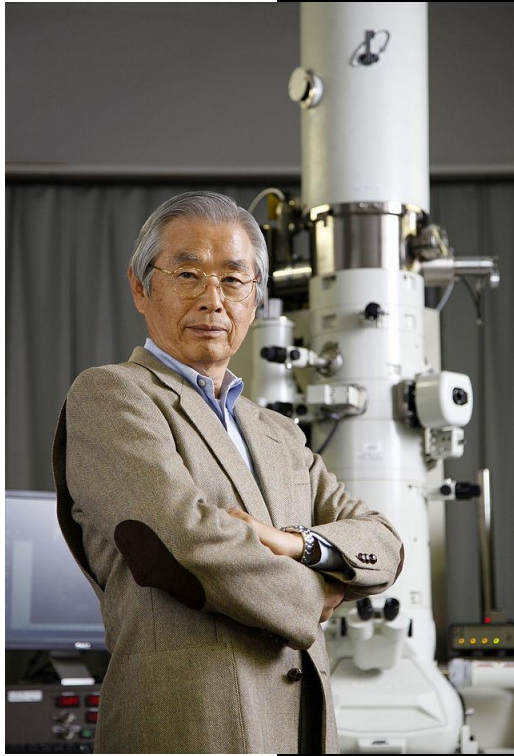
4 x



4 x Tierra

4 x 12,700,000 m

diámetros



Sumio Iijima
1991



Carbon Nanotubes



University of Manchester 2010 Nobel Prize in Physics

"for groundbreaking experiments regarding the two-dimensional material graphene".

We have been able to prepare graphitic sheets of thicknesses down to a few atomic layers (including single-layer graphene), to fabricate devices from them, and to study their electronic properties. Despite being atomically thin, the films remain of high quality, so that 2D electronic transport is ballistic at submicrometer distances. No other film of similar thickness is known to be even poorly metallic or continuous under ambient conditions. Using FLG, we demonstrate a metallic field-effect transistor in which the conducting channel can be switched between 2D electron and hole gases by changing the gate voltage.

Our graphene films were prepared by mechanical exfoliation (repeated peeling) of small mesas of highly oriented pyrolytic graphite (15). This approach was found to be highly reliable and allowed us to prepare FLG films up to 10 μm in size. Thicker films ($d \geq 3$ nm) were up to 100 μm across and visible by the naked eye. Figure 1 shows examples of the prepared films, including single-layer graphene [see also (15)]. To study their electronic properties, we processed the films into multiterminal Hall bar devices placed on top of an oxidized Si substrate so that a gate voltage V_g could be applied. We have studied more than 60 devices with $d < 10$ nm. We focus on the electronic properties of our thinnest (FLG) devices, which contained just one, two, or three atomic layers (15). All FLG devices exhibited essentially identical electronic properties characteristic for a 2D semimetal, which differed from a more complex (2D plus 3D) behavior observed for thicker, multilayer graphene (15) as well as from the properties of 3D graphite.

In FLG, the typical dependence of its sheet resistivity ρ on gate voltage V_g (Fig. 2) exhibits a sharp peak to a value of several kilohms and decays to ~ 100 ohms at high V_g (note that 2D resistivity is given in units of ohms rather than ohms $\times$ cm as in the 3D case). Its conductivity $\sigma = 1/\rho$ increases linearly with V_g on both sides of the resistivity peak (Fig. 2B). At the same V_g where ρ has its peak, the Hall coefficient R_H exhibits a sharp reversal of its sign (Fig. 2C). The observed behavior resembles the ambipolar field effect in semiconductors, but there is no zero-conductance region associated with the Fermi level being pinned inside the band gap.

Our measurements can be explained quantitatively by a model of a 2D metal with a small overlap $\delta\epsilon$ between conduction and valence bands (15). The gate voltage induces a surface charge density $n = \epsilon_0 \epsilon V_g / te$ and, accordingly, shifts the position of the Fermi energy ϵ_F . Here, ϵ_0 and ϵ are the permittivities of free space and SiO_2 , respectively; e is the electron charge; and t is the thickness of our SiO_2 layer (300 nm). For

typical $V_g = 100$ V, the formula yields $n \approx 7.2 \times 10^{12} \text{ cm}^{-2}$. The electric field doping transforms the shallow-overlap semimetal into either completely electron or completely hole conductor through a mixed state where both electrons and holes are present (Fig. 2). The three regions of electric field doping are clearly seen on both experimental and theoretical curves. For the regions with only

electrons or holes left, R_H decreases with increasing carrier concentration in the usual way, as $1/ne$. The resistivity also follows the standard dependence $\rho^{-1} = \sigma = ne\mu$ (where μ is carrier mobility). In the mixed state, σ changes little with V_g , indicating the substitution of one type of carrier with another, while the Hall coefficient reverses its sign, reflecting the fact that R_H is proportional to

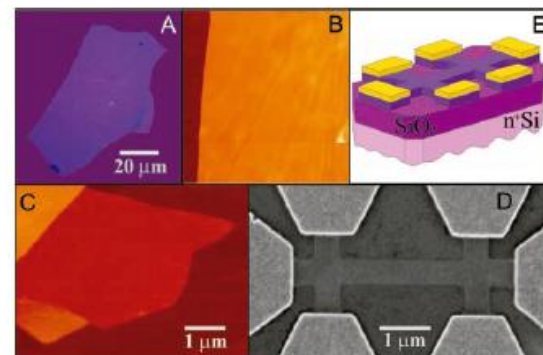


Fig. 1. Graphene films. (A) Photograph (in normal white light) of a relatively large multilayer graphene flake with thickness ~ 3 nm on top of an oxidized Si wafer. (B) Atomic force microscope (AFM) image of $2 \mu\text{m}$ by $2 \mu\text{m}$ area of this flake near its edge. Colors: dark brown, SiO_2 surface; orange, 3 nm height above the SiO_2 surface. (C) AFM image of single-layer graphene. Colors: dark brown, SiO_2 surface; brown-red (central area), 0.8 nm height; yellow-brown (bottom left), 1.2 nm; orange (top left), 2.5 nm. Notice the folded part of the film near the bottom, which exhibits a differential height of ~ 0.4 nm. For details of AFM imaging of single-layer graphene, see (15). (D) Scanning electron microscope image of one of our experimental devices prepared from FLG. (E) Schematic view of the device in (D).

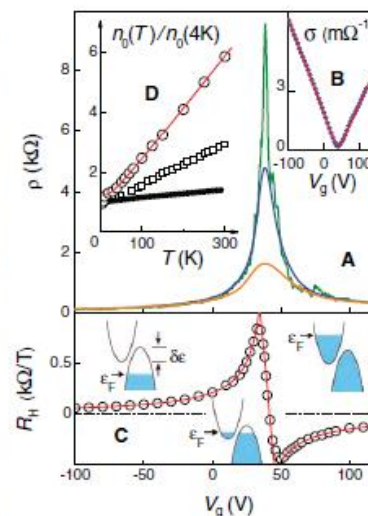
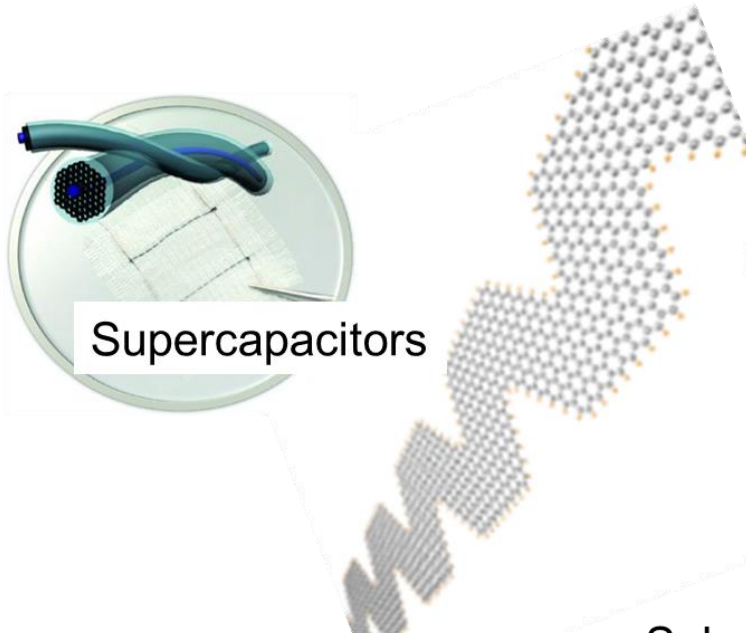
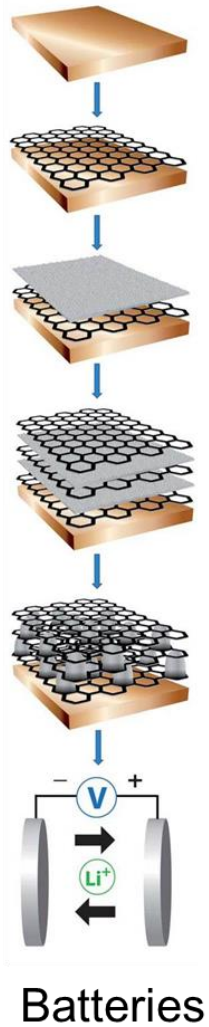
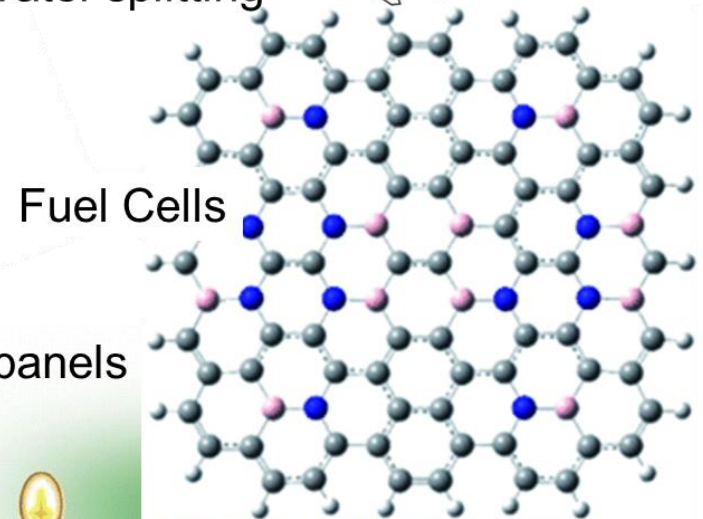
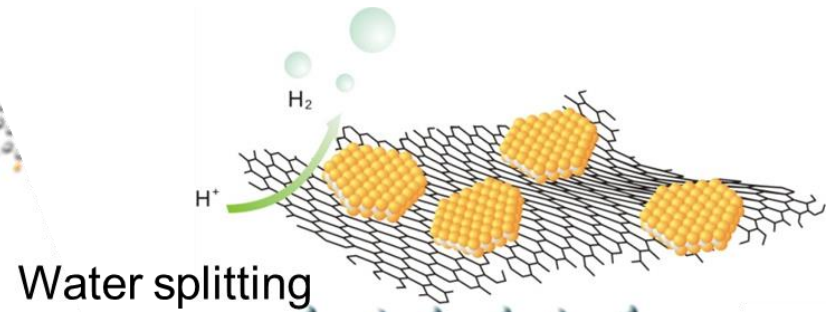
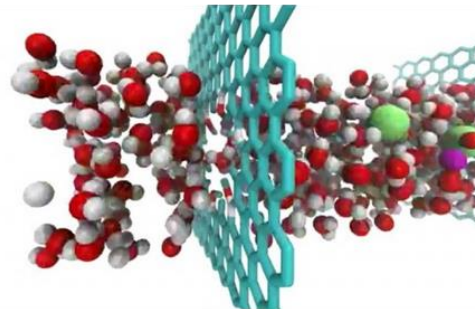


Fig. 2. Field effect in FLG. (A) Typical dependences of FLG's resistivity ρ on gate voltage for different temperatures ($T = 5, 70$, and 300 K for top to bottom curves, respectively). (B) Example of changes in the film's conductivity $\sigma = 1/\rho(V_g)$ obtained by inverting the 70 K curve (dots). (C) Hall coefficient R_H versus V_g for the same film; $T = 5$ K. (D) Temperature dependence of carrier concentration n_0 in the mixed state for the film in (A) (open circles), a thicker FLG film (squares), and multilayer graphene ($d \approx 5$ nm; solid circles). Red curves in (B) to (D) are the dependences calculated from our model of a 2D semimetal illustrated by insets in (C).

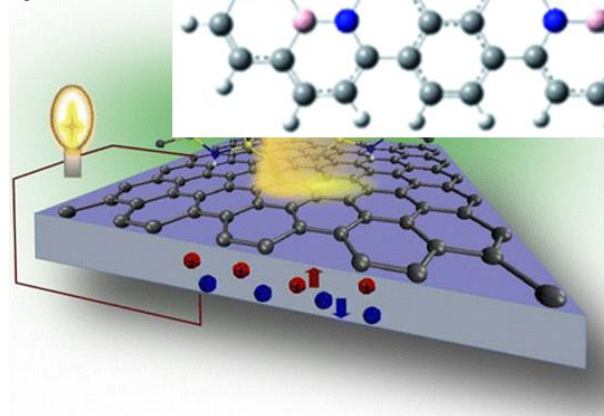
Grafeno para un futuro sostenible



Desalination.
Water purification



Solar panels



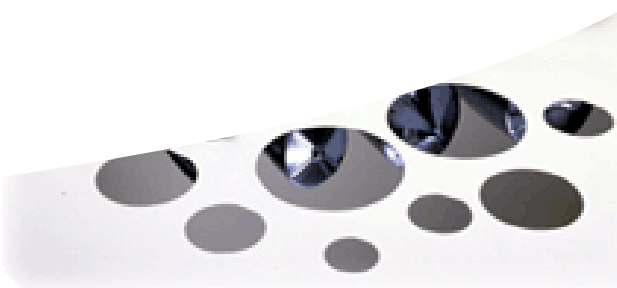
3. NANOMATERIALES PARA CONVERSIÓN DE ENERGÍA

Lo que en términos de ingeniería se conoce como generación:

- Conversión fotovoltaica
- Energía Solar Térmica
- Hidrógeno y pilas de combustible

Se busca: Silicio Solar barato

Silicio 99.999%



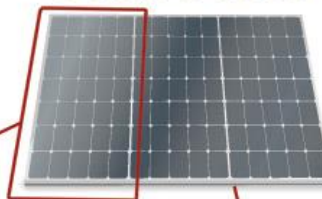
Solar Cell



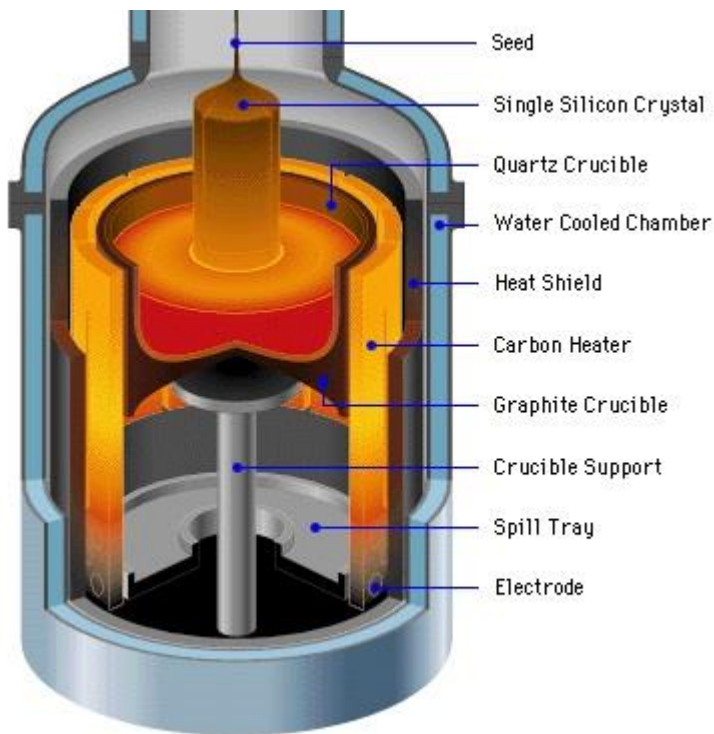
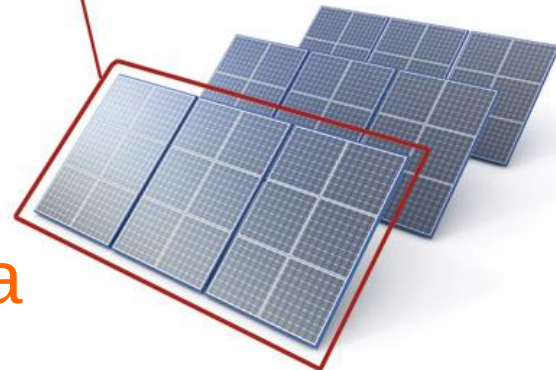
Solar Module



Solar Panel



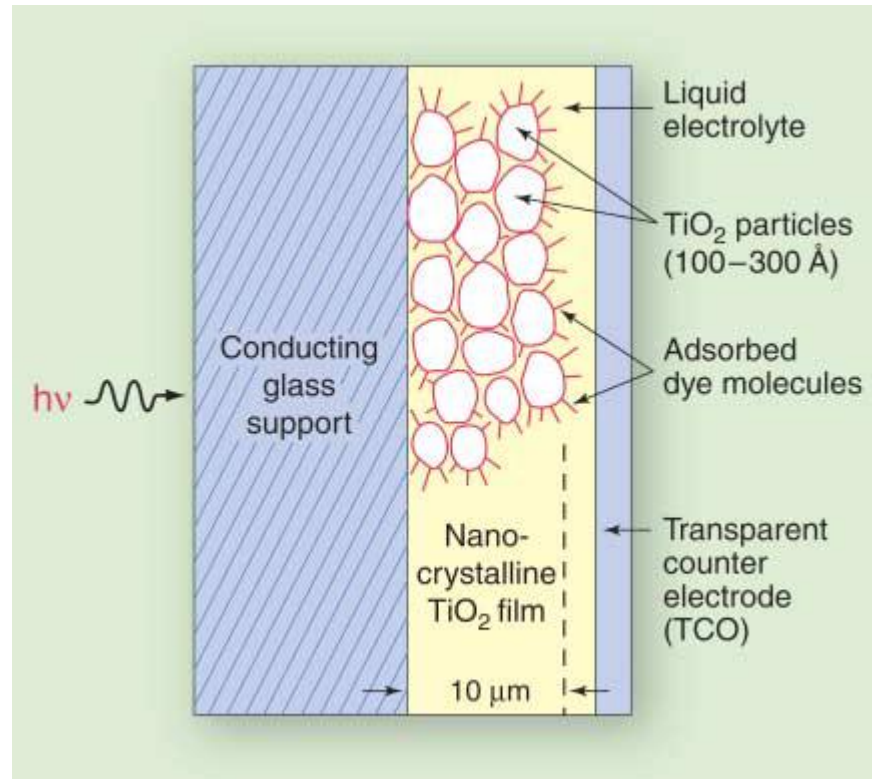
Solar Array



15% eficiencia

Nano y Celdas Solares

Dye-Sensitized Solar Cells (Grätzel)



Nathan S. Lewis, et al. Toward Cost-Effective Solar Energy Use. Science 315, 798 (2007)

Nano y Celdas Solares

Nanoestructura para “ortogonalizar” la absorción y la separación de carga

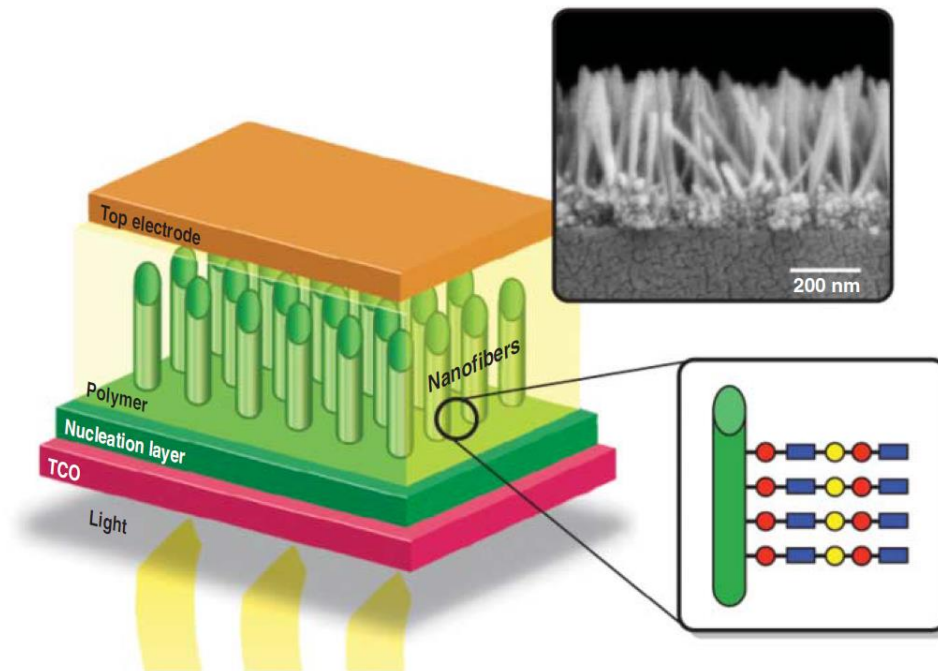
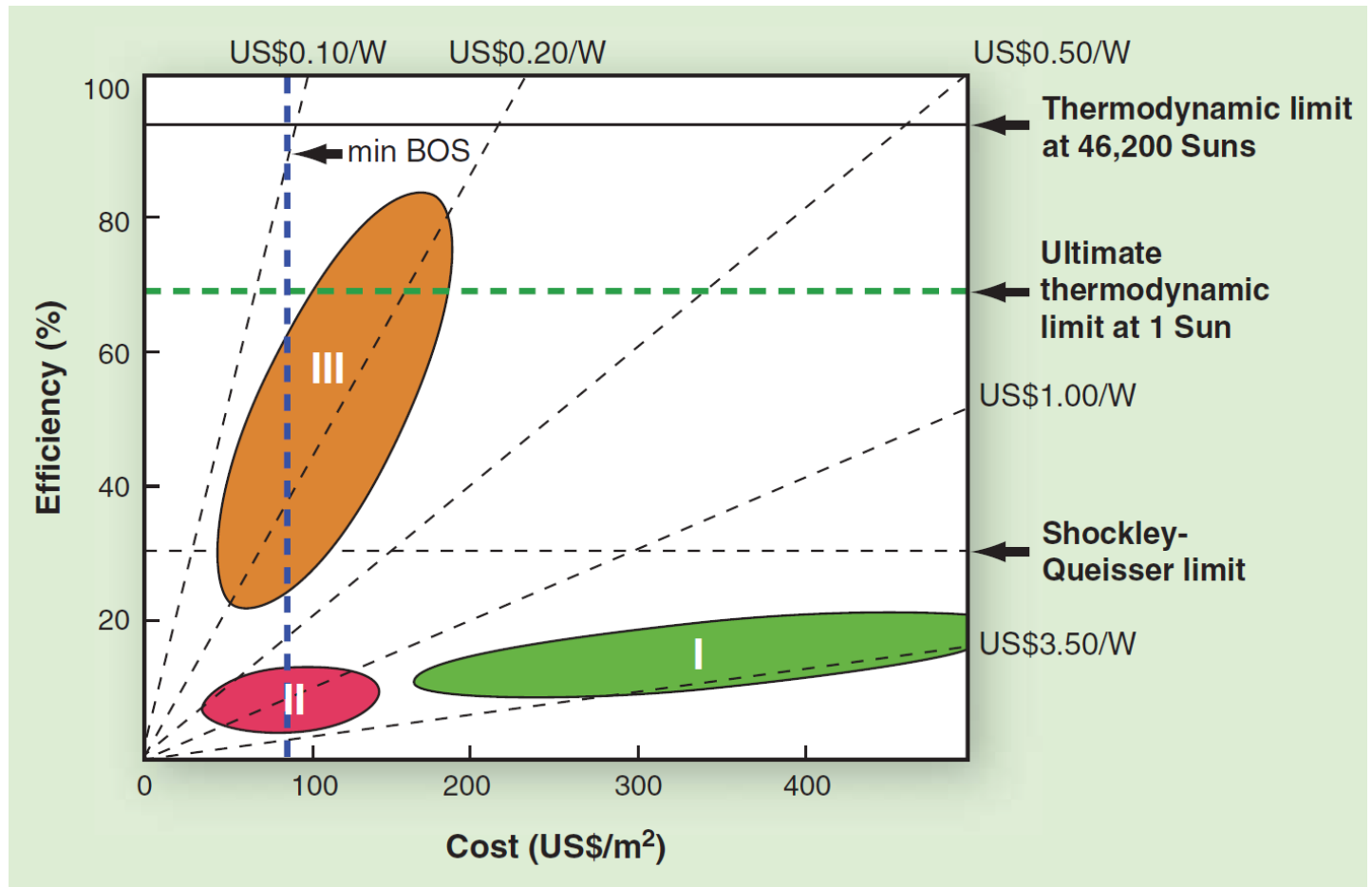


Fig. 3. Arrays of nanorods, illustrating an approach to orthogonalization of the directions of light absorption (down the length of the rods) and charge carrier collection (radially outward to the surface of the rods). [Adapted from (2)]

Nathan S. Lewis, et al. Toward Cost-Effective Solar Energy Use. Science 315, 798 (2007)

Nano y Celdas Solares

Costes y eficiencias



Nathan S. Lewis, et al. Toward Cost-Effective Solar Energy Use. Science 315, 798 (2007)

Nano y Celdas Solares

La importancia de ser nano

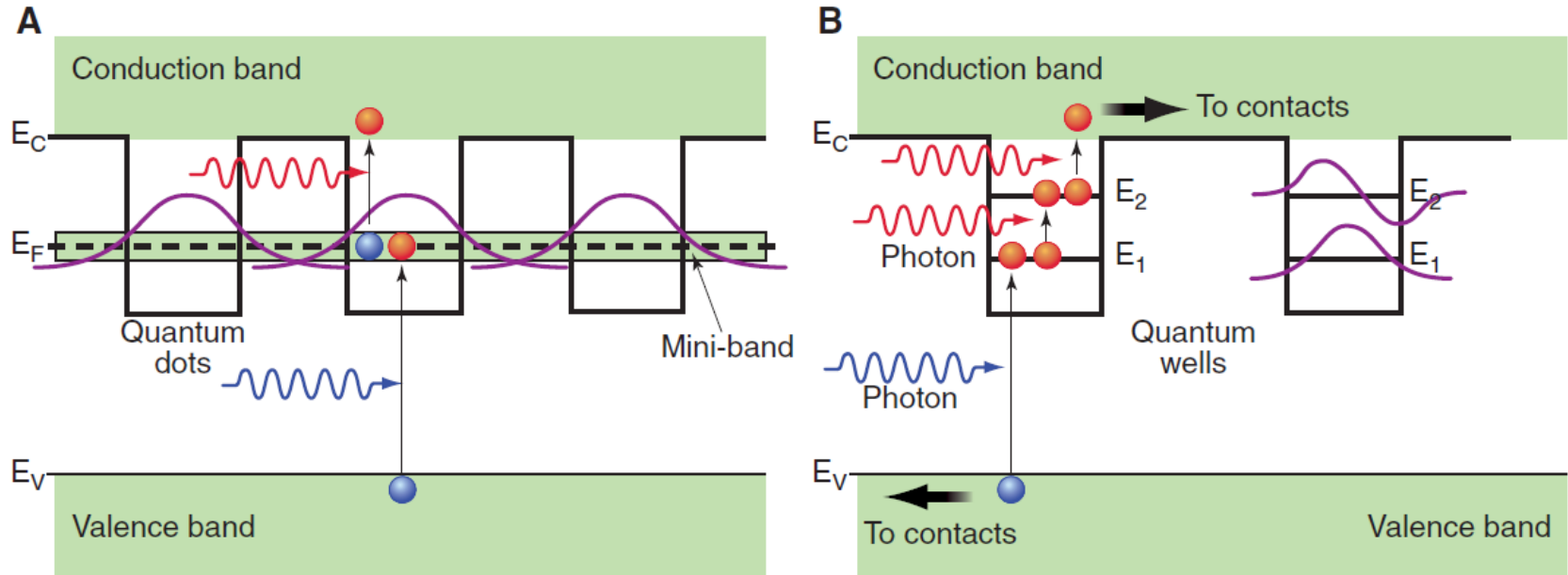


Fig. 2. Possible methods of circumventing the 31% efficiency limit for thermalized carriers in a single-band gap absorption threshold solar quantum conversion system. **(A)** Intermediate-band solar cell; **(B)** quantum-well solar cell. [Adapted from (2)]

Nathan S. Lewis, et al. Toward Cost-Effective Solar Energy Use. Science 315, 798 (2007)

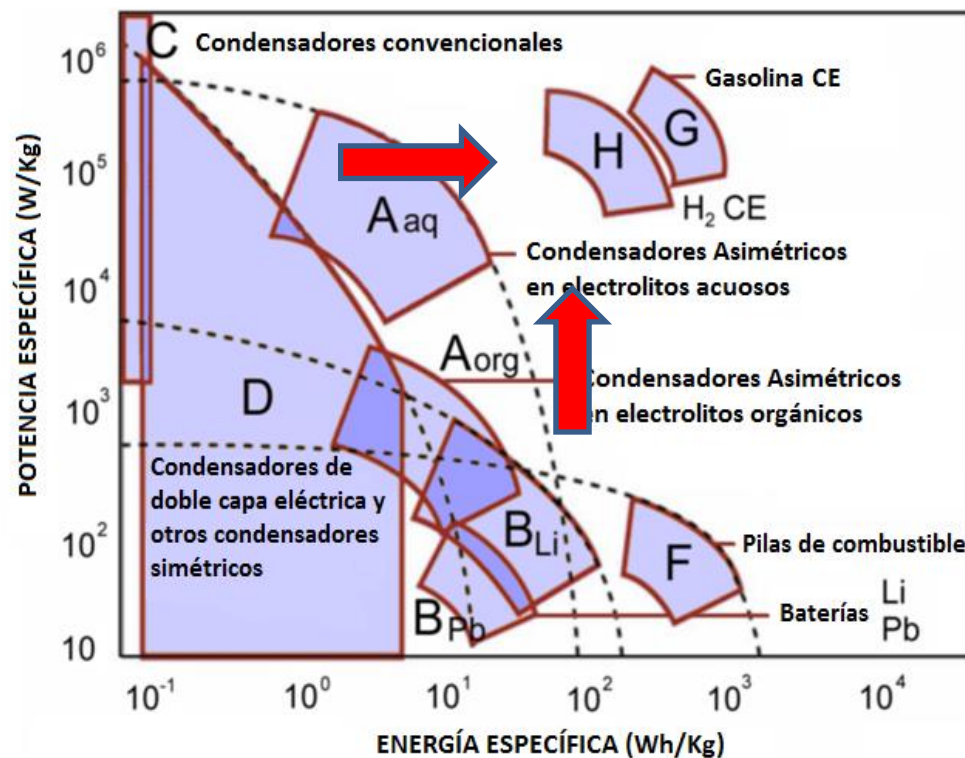
3. NANOMATERIALES PARA ALMACENAMIENTO DE ENERGÍA

Un campo tradicionalmente desatendido (excepto aplicaciones nicho) y actualmente en despegue:

- Baterías... de litio, de ¿sodio?
- Supercondensadores
- Dispositivos Híbridos (Supercapbatteries)

Diagrama de Ragone.

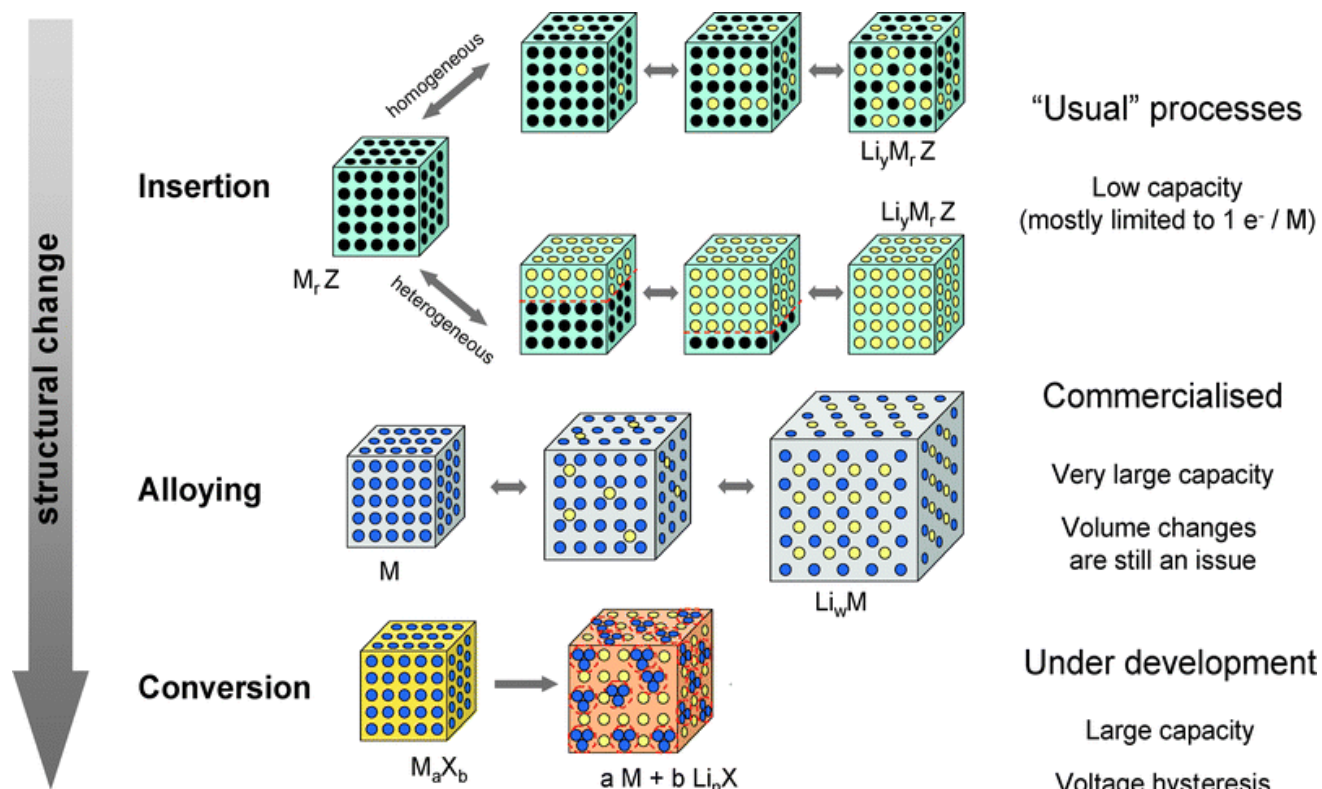
El tablero de ajedrez del almacenamiento de energía



Hybrid Energy Storage. The merging of battery and supercapacitor chemistries.
D. P. Dubal, O. Ayyad, V. Ruiz, and P. Gomez-Romero* Chemical Society Reviews, **2015**, 44, 1777.

Baterías

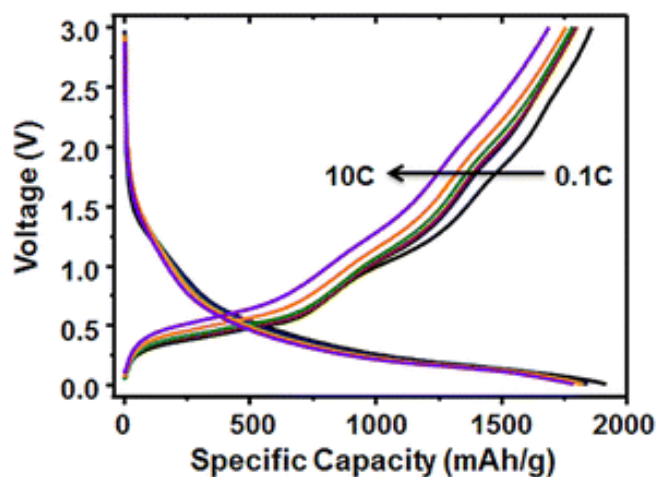
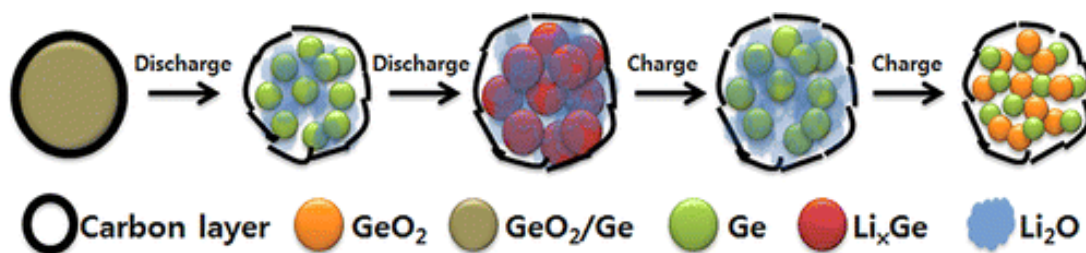
Tipos de materiales



Recent advances in rechargeable battery materials: a chemist's perspective
 M. Rosa Palacín Chem. Soc. Rev., 2009,38, 2565-2575

Nuevos materiales

Ejemplo de ánodo de conversión en baterías de Li



Prof Z. Guo y col. Self-Assembled Germanium/Carbon Nanostructures as High-Power Anode Material for the Lithium-Ion Battery. Angew. Chem. Int. Ed. 2012, 51(23), 5657

¿baterías más rápidas?

Electrodos nanoestructurados
Microestructura fractal

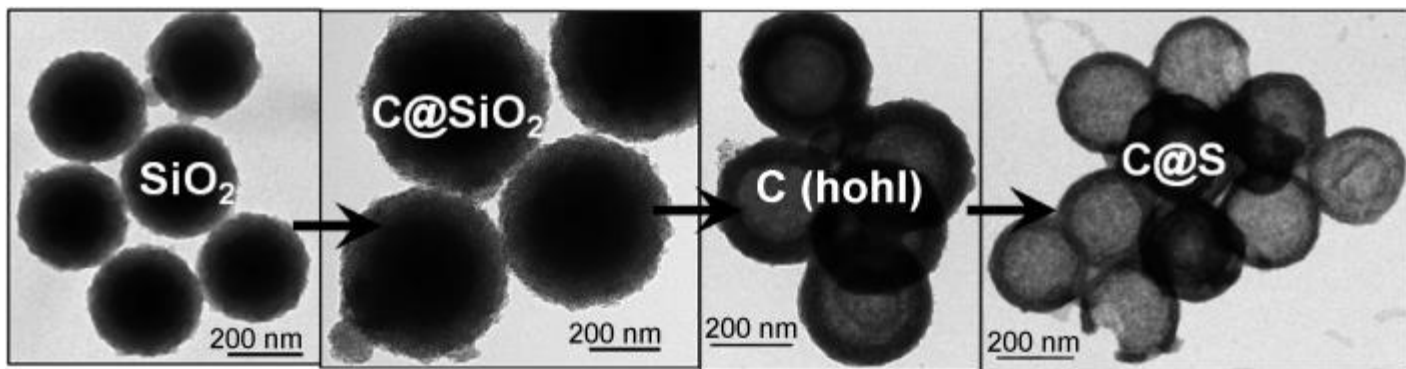


New (nano)solutions for old battery chemistries

Cathode material	Specific capacity (mAh/g)	Cell voltage (V)	Specific energy (Wh/kg)
LiFePO ₄	140	3.3	578
LiCoO ₂	160	3.7	580
LiMn ₂ O ₄	130	3.9	592
S	1670	2.1	2600

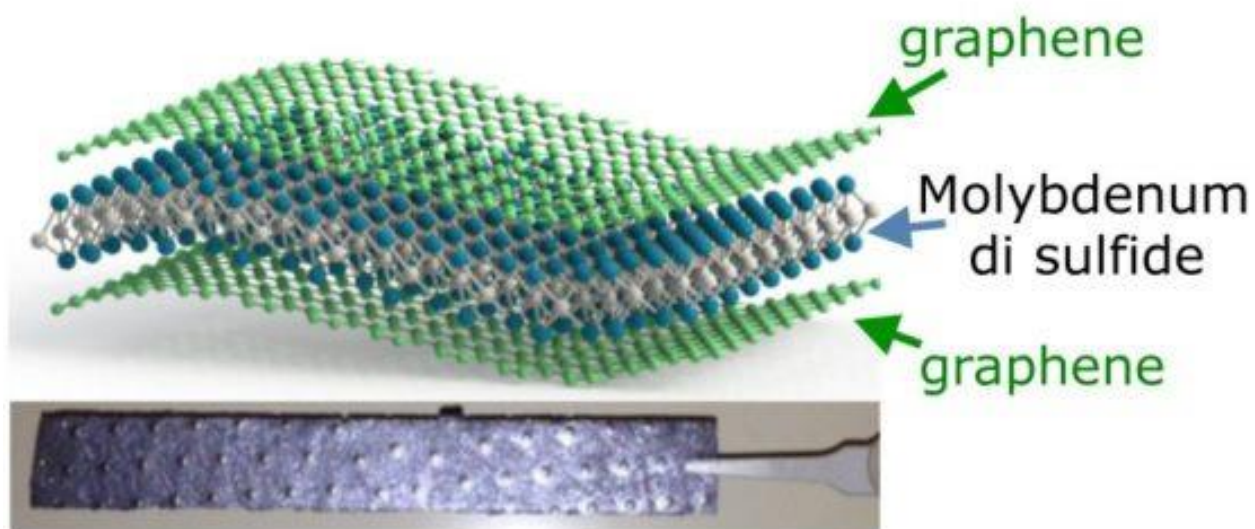
New Composite Cathode Materials for Li/S Batteries: A Review

A. Fedorková*, R. Oriňáková, O. Čech, M. Sedlaříková *Int. J. Electrochem. Sci.*, 8 **(2013)** 10308 - 10319



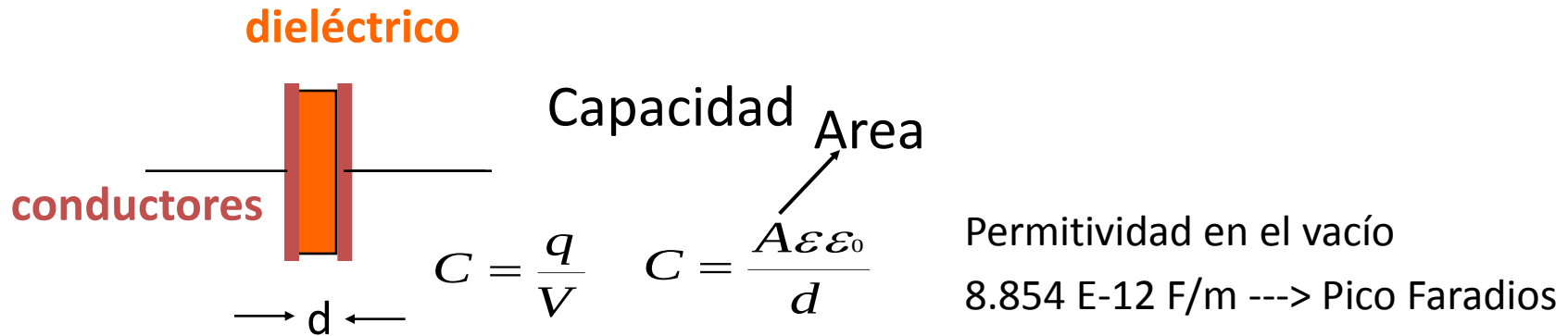
N. Jayaprakash, J. Shen, S. S. Moganty, A. Corona, L. A. Archer, *Angew. Chem.*, 123 (2011) 6026.

De baterías de Litio... a baterías de Sodio.



Lamuel David , Romil Bhandavat , and Gurpreet Singh *
MoS₂/Graphene Composite Paper for Sodium-Ion Battery Electrodes
ACS Nano, 2014, 8 (2), pp 1759–1770

Condensadores (sin súper)



Aumentar A
disminuir d

Supercondensadores

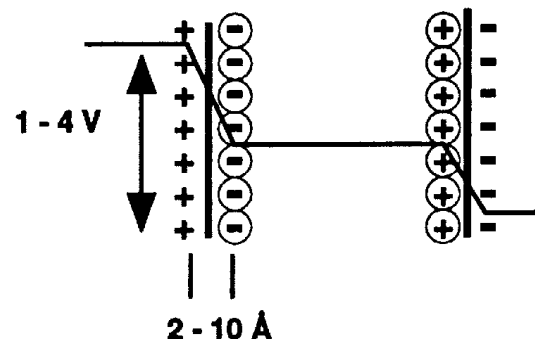
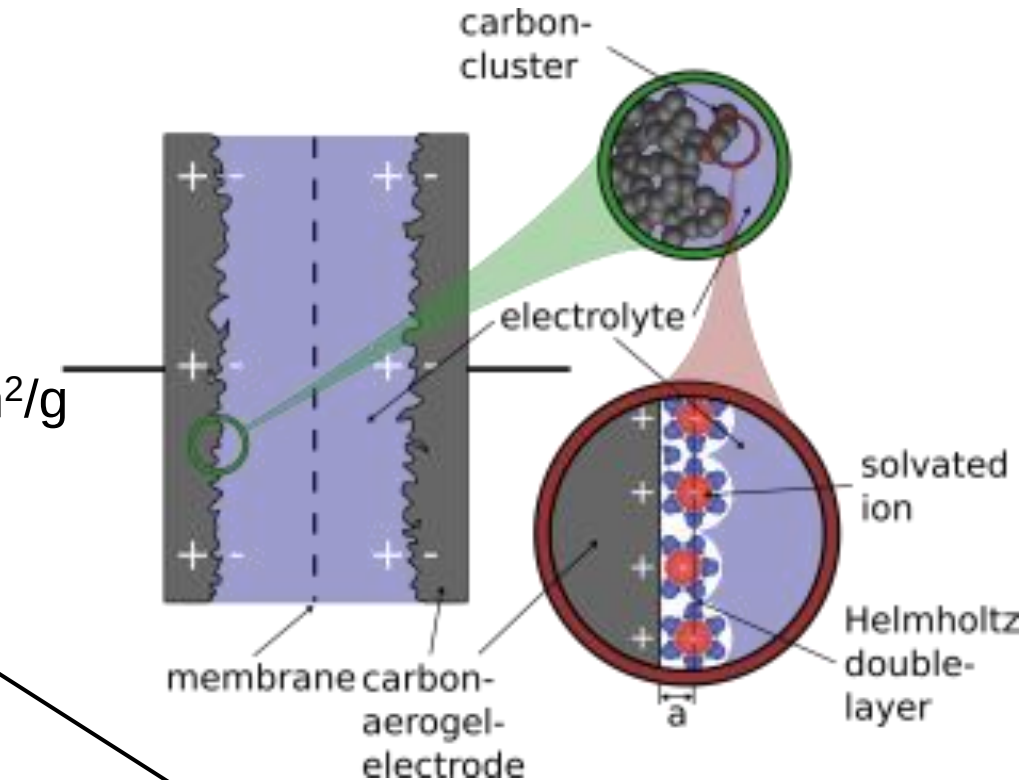
Doble capa eléctrica (d Å)

Supercondensadores de Doble Capa

Capacitance
Farads

$$C = \frac{q}{V} \quad C = \frac{A \epsilon \epsilon_0}{d}$$

Area
 $10^3 \text{ m}^2/\text{g}$



Supercondensadores electroquímicos

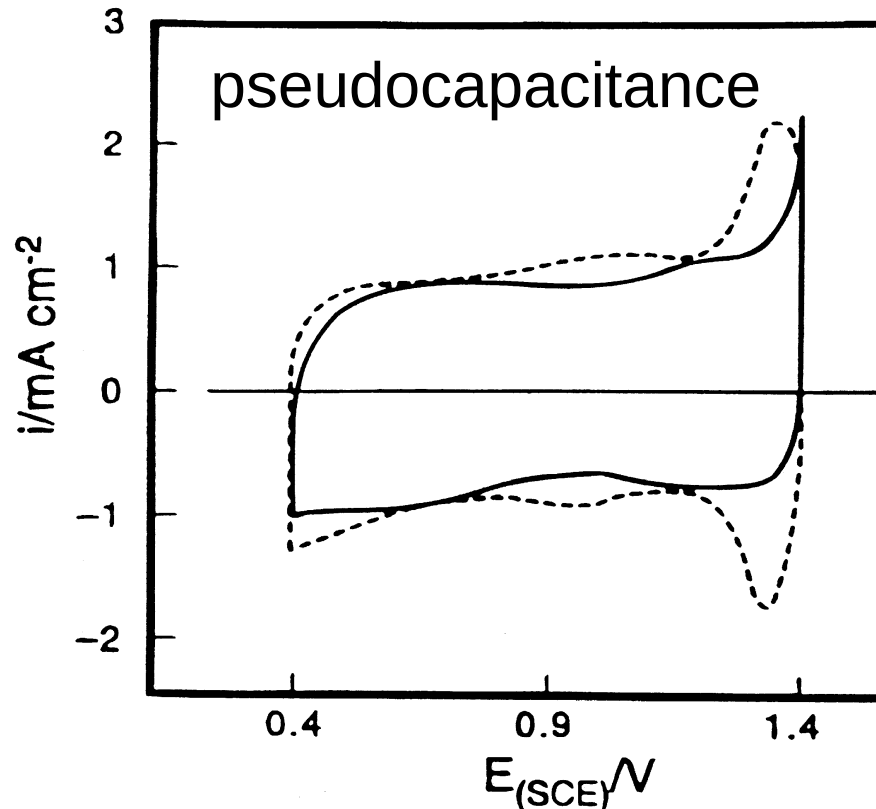
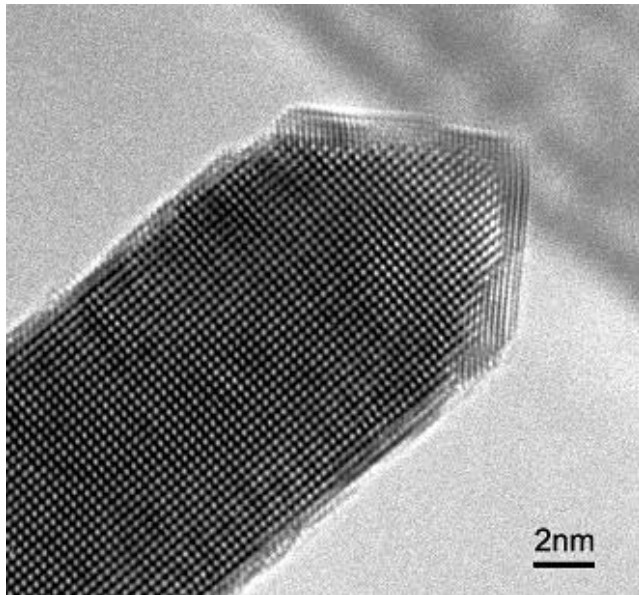
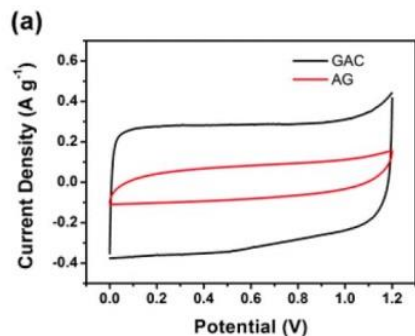


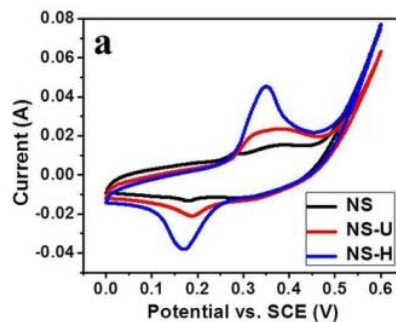
Fig. 5. Typical voltammetric curves at 20 mV s^{-1} of a RuO_2 electrode between 0.4 and 1.4 V (RHE) in 1 mol dm^{-3} solutions (—) HClO_4 and (---) KOH . (From Ref. 9.)

Trasatti 1971
Conway 1975-1980

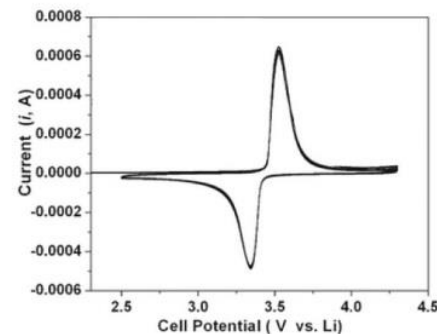
Supercondensadores vs. baterías



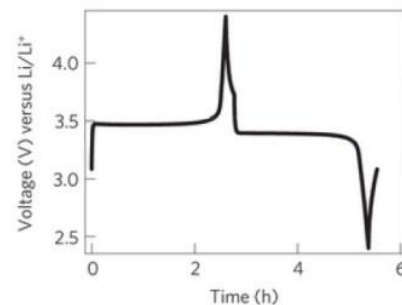
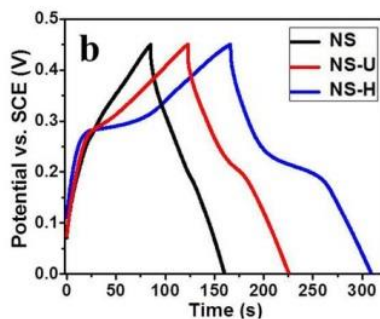
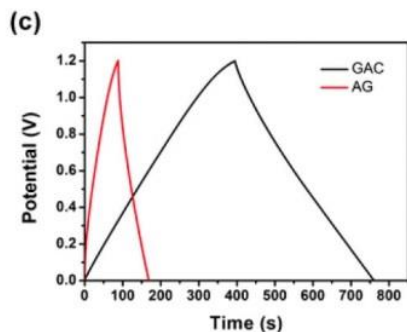
carbon-based EDLC
supercapacitors



Co_3O_4
pseudocapacitor



LiFePO_4 (vs. Li)
battery



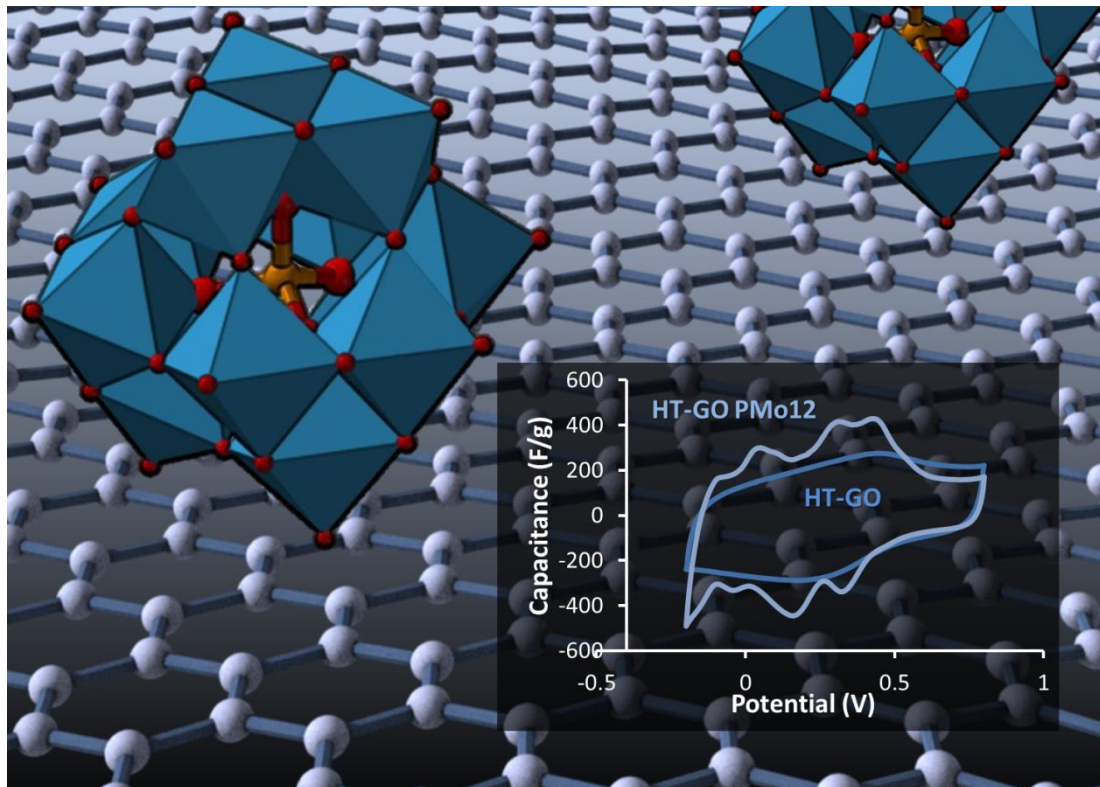
...pero NO un mecanismo híbrido

Hybrid Energy Storage. The merging of battery and supercapacitor chemistries.

D. P. Dubal, O. Ayyad, V. Ruiz, and P. Gomez-Romero* Chemical Society Reviews, **2015**, 44, 1777.

Almacenamiento de energía híbrido

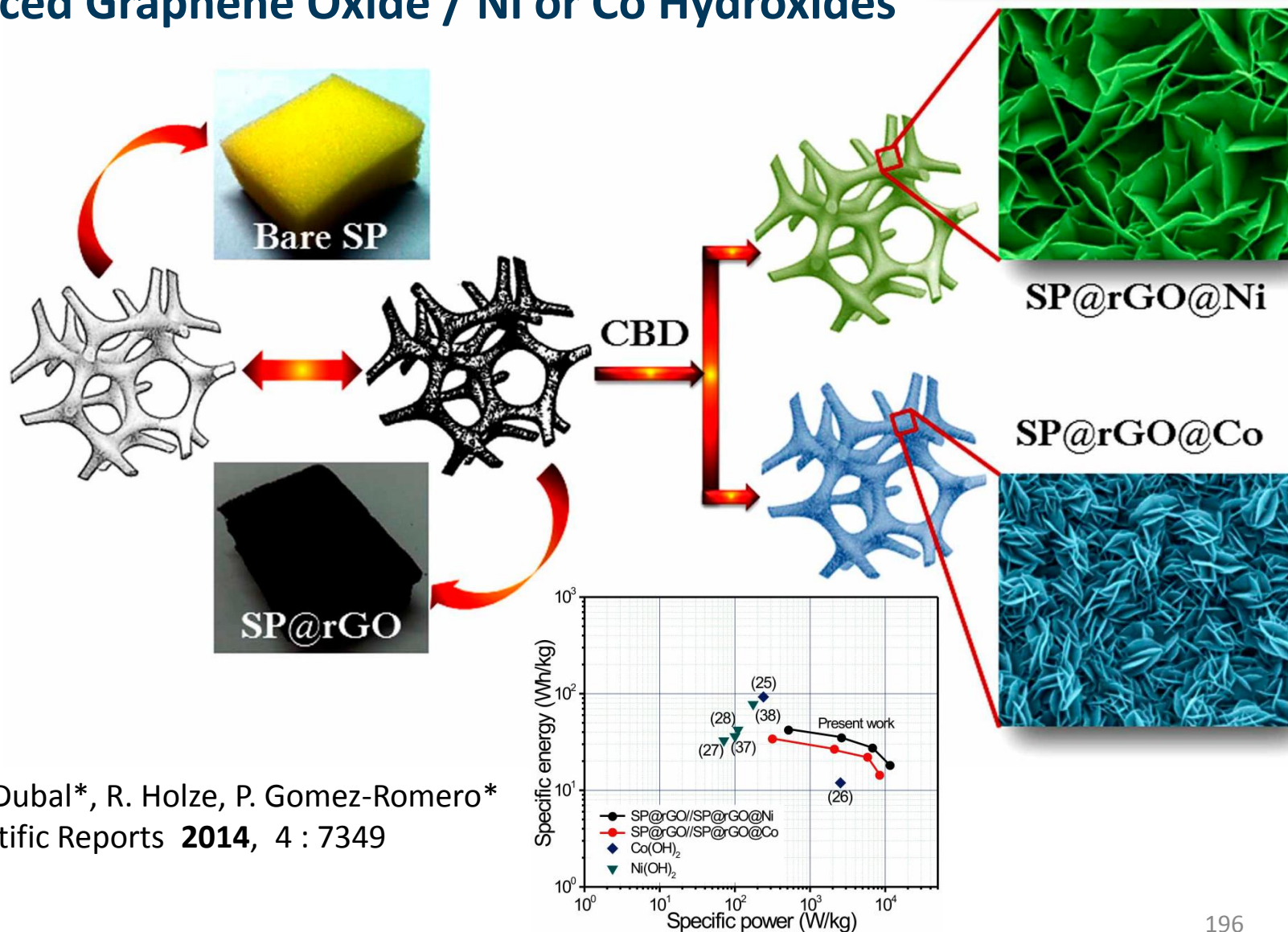
Graphene-Polyoxometalate Hybrid Nanocomposite Material for Supercapacitors



Stable Graphene-Polyoxometalate Nanomaterials for Application in Hybrid Supercapacitors.
J. Suarez-Guevara, V. Ruiz,, P. Gomez-Romero Phys.Chem.Chem.Phys., 2014, 16 (38), 20411.

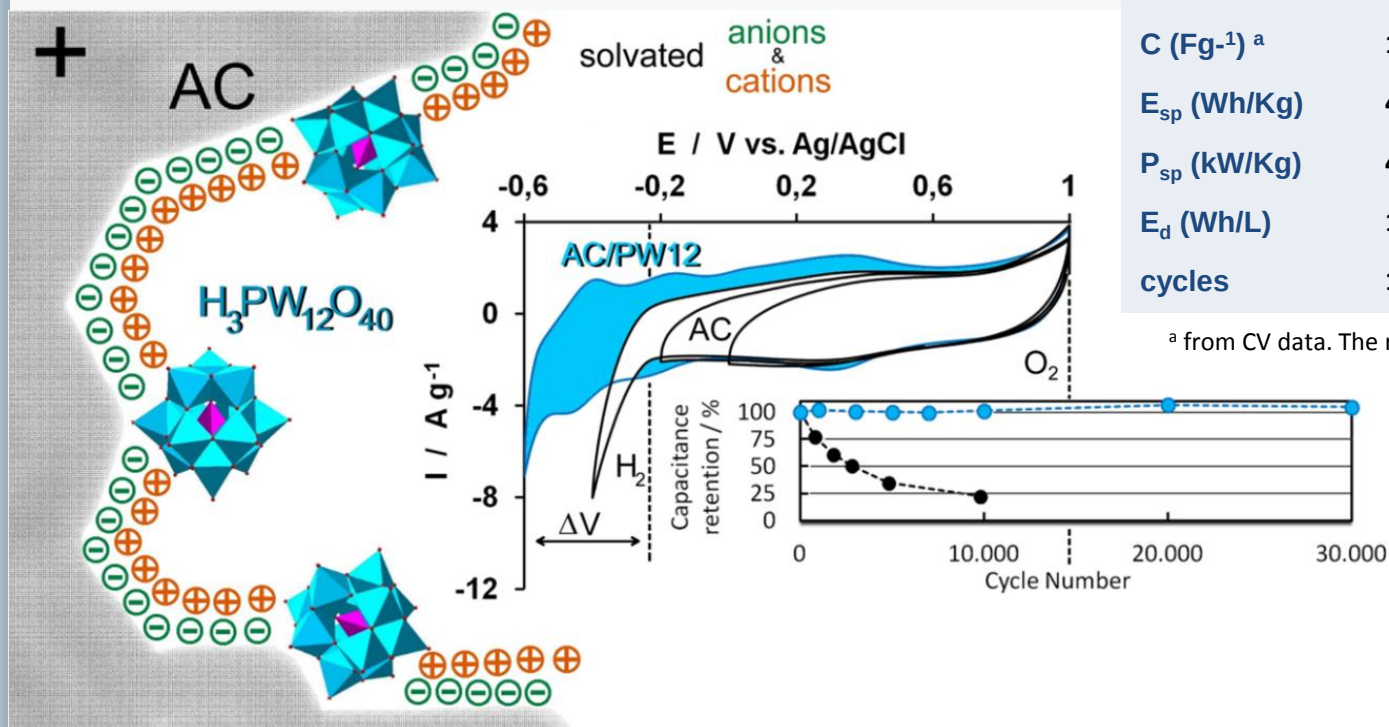
Hybrid Energy Storage:

Reduced Graphene Oxide / Ni or Co Hydroxides



D. P. Dubal*, R. Holze, P. Gomez-Romero*
Scientific Reports **2014**, 4 : 7349

Activated Carbon (AC) - $\text{PW}_{12}\text{O}_{40}$ Hybrid

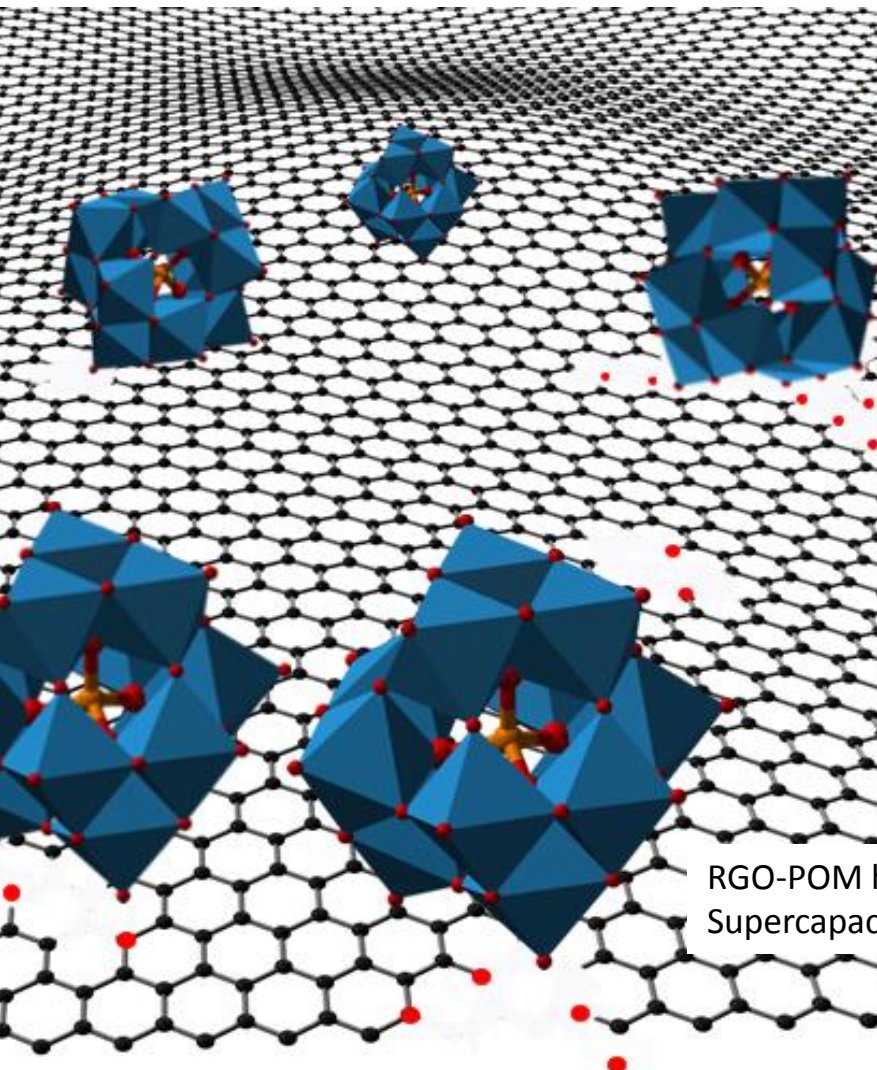


	AC	AC-PW12
$C (\text{Fg}^{-1})^a$	185	254
$E_{\text{sp}} (\text{Wh/Kg})$	4.05	4.96 (1.6 A/g)
$P_{\text{sp}} (\text{kW/Kg})$	45	115
$E_d (\text{Wh/L})$	1.55	2.32
cycles	10,000	> 30,000 (6 A/g)

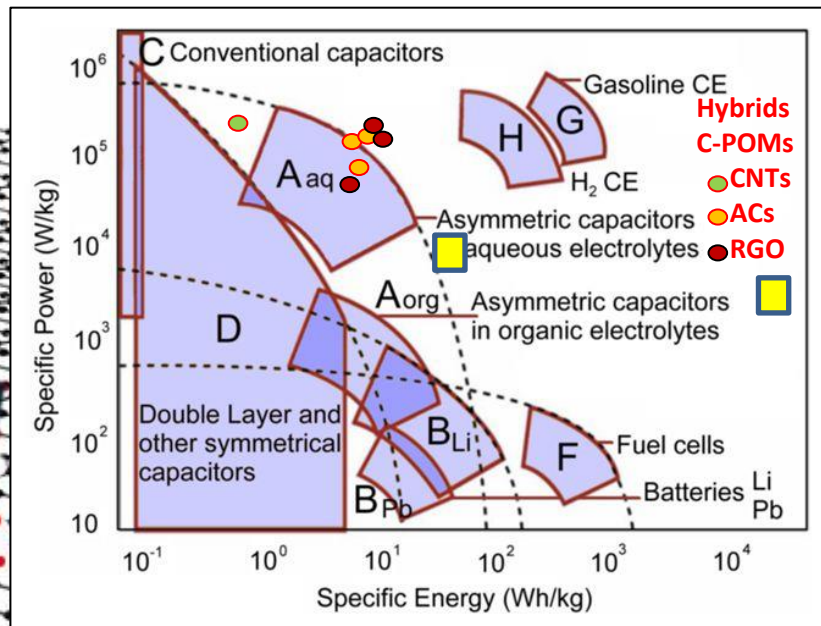
^a from CV data. The rest from charge-discharge data

Hybrid Electrodes Based on Polyoxometalate-Carbon Materials for Electrochemical Supercapacitors
V. Ruiz, J. Suárez-Guevara, P. Gomez-Romero. *Electrochemistry Communications* **2012**, 24, 35-8

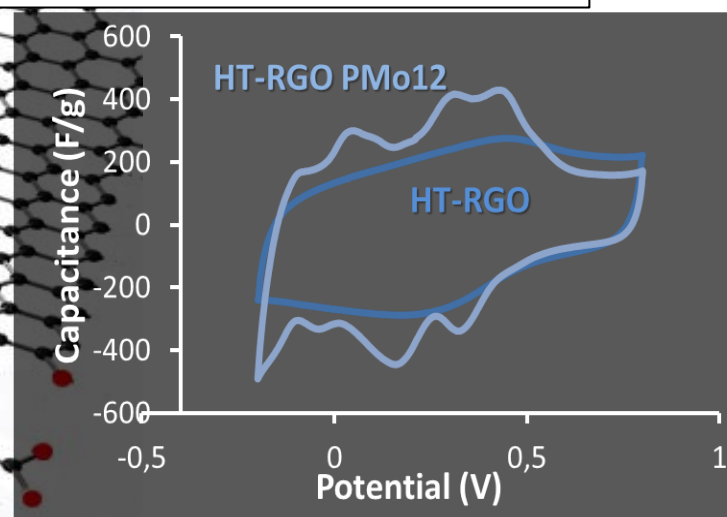
Hybrid Energy Storage: High Voltage Aqueous Supercapacitors based on Activated C-Phosphotungstate Hybrid Materials.
J. Suárez-Guevara, V. Ruiz, P. Gomez-Romero *J. Mater. Chem. A*, **2014**, 2 (4), 1014-1021



RGO-POM hybrid
Supercapacitor electrode



rGO Ni(OH)₂



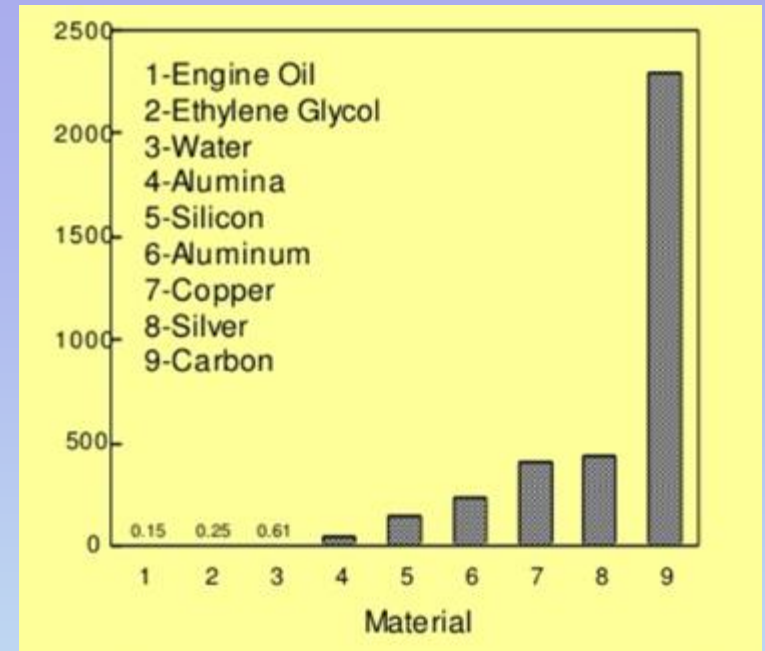
Nanofluids: size matters

dispersions of nanoparticles in base fluids



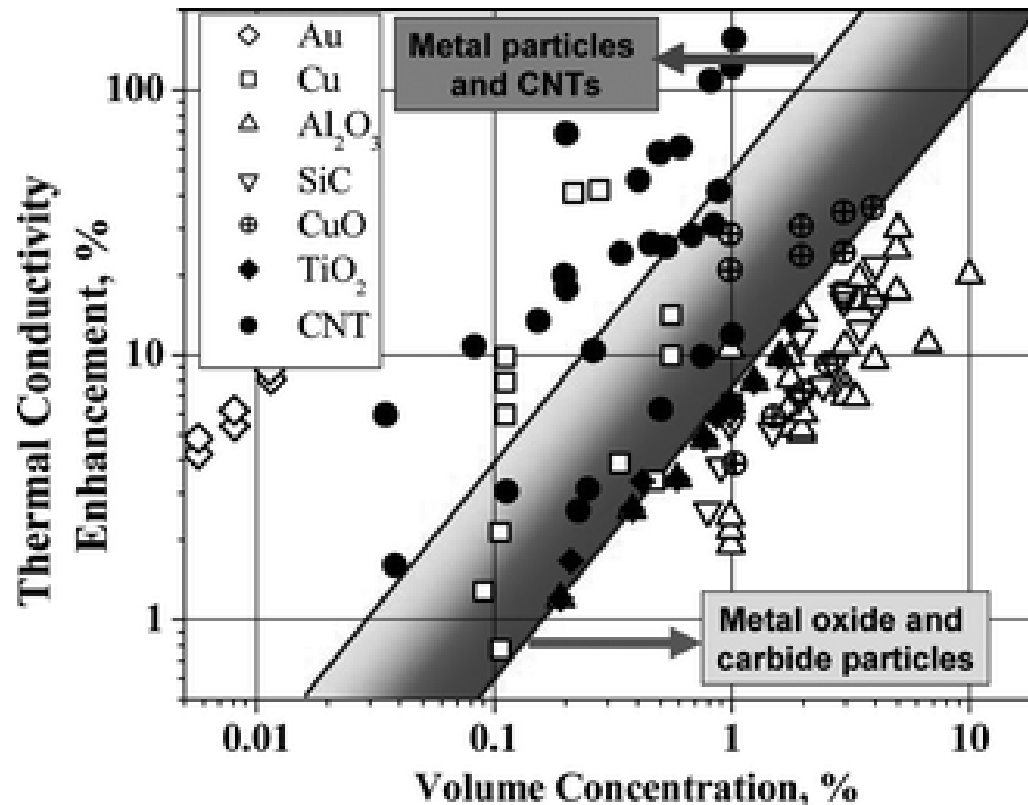
Thermal Nanofluids concept

- Conventional Heat transfer Fluids (HTFs) have poor thermal conductivity compared to solids
- Fluids containing microparticles lead to engineering problems (precipitation, clogging...)
- Nanofluids provide enhanced performance from dispersed solid nanoparticles without those problems



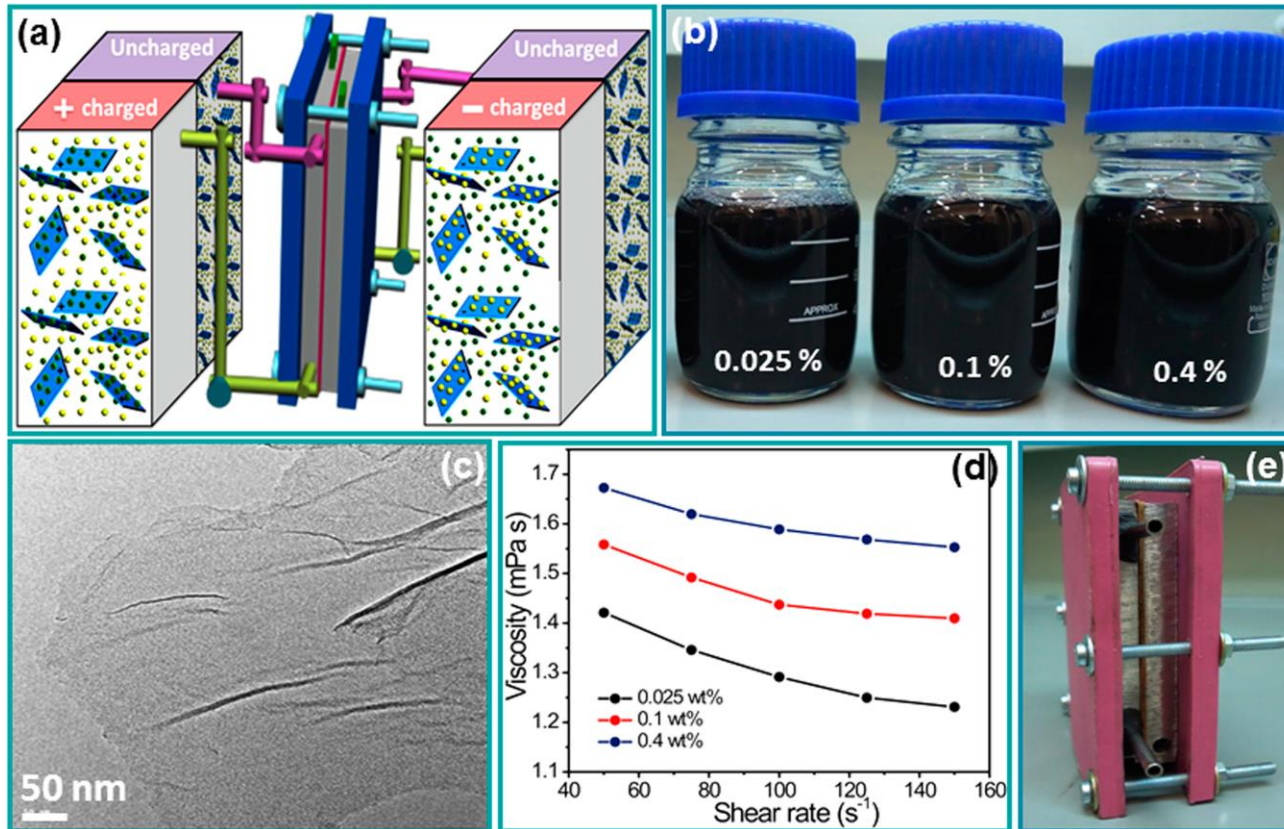
Thermal conductivity of solids:
Orders of magnitude larger than
Those of conventional HTFs

Thermal Nanofluids Performance



The mechanism of heat transfer in nanofluids: state of the art (review). Part 1. Synthesis and properties of nanofluids
V.I. Terekhov, S.V. Kalinina, and V.V. Lemanov. *Thermophysics and Aeromechanics*, 2010, Vol. 17, No. 1, 1-14

Electroactive Nanofluids for New Flow Cell Concepts.



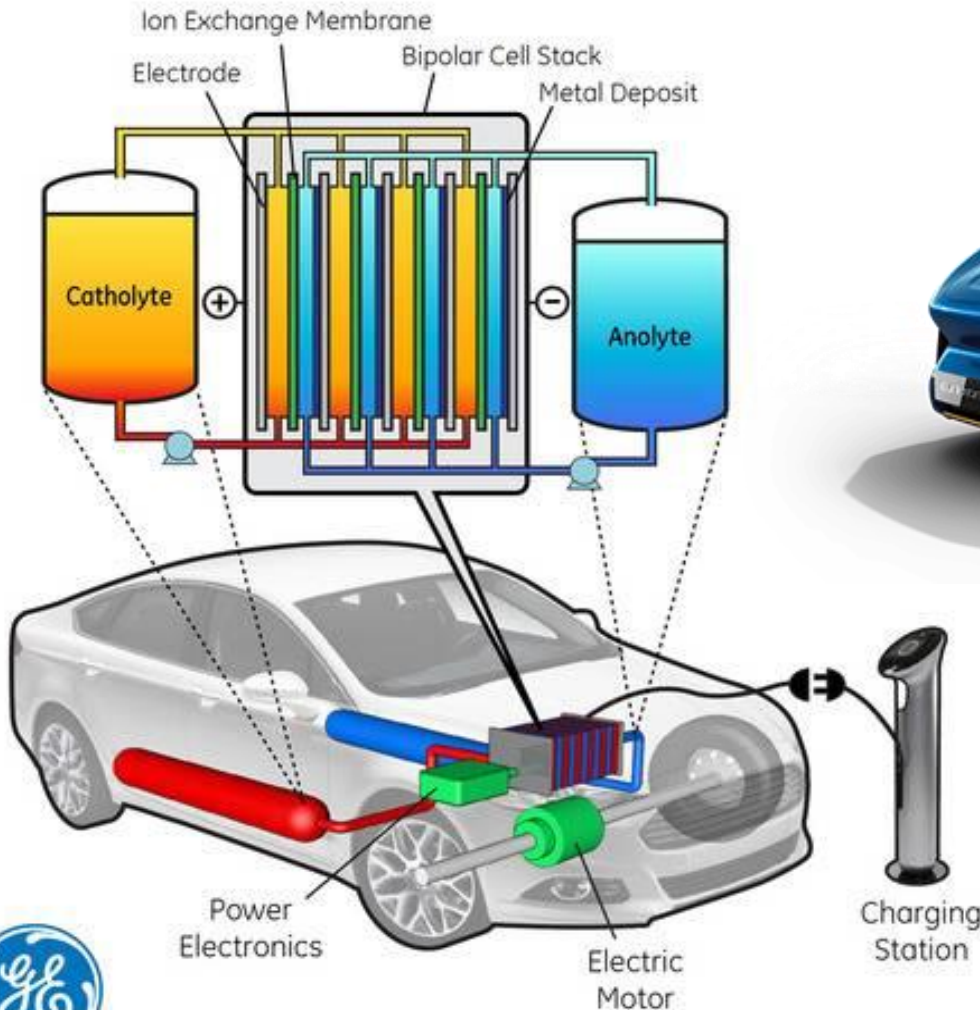
D. P. Dubal, D. Gomez, P. Gómez-Romero, Patent ES1641.1064. "Electroactive nanofluids on graphene-based materials for energy storage in flow cells." 20-05-2015

Electroactive Graphene Nanofluids for Fast Energy Storage.

D.P. Dubal and P. Gomez-Romero 2D-Materials **2016**, 3, 031004

Prototype Electric vehicles with Flow technology

Early Model of Water-Based Flow Battery Designed For Use in Electric Vehicles



Quantino

250-liter tanks to hold liquids that flow past an electrochemical cell

Flowing liquid is undisclosed

News

Press Releases

Feature Stories

Science Highlights

In the News

Media Contacts

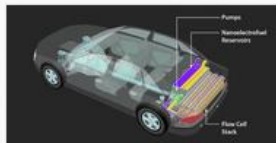
Social Media

Photos

Videos

Fact Sheets, Brochures
and Reports

Summer Science
Writing Internship



The Illinois Institute of Technology will share \$3.4 million to develop a prototype of their "nanoelectrofuel" flow battery. Funding for the prototype project comes from

ARPA-E awards IIT-Argonne team \$3.4 million for breakthrough battery technology

AUGUST 30, 2013



Tweet  Share



CHICAGO – Carlo Segre, Duchossois Leadership Professor of Physics at [Illinois Institute of Technology](#), has received a \$3.4 million award from the [U.S. Department of Energy's](#) Advanced Research Projects Agency (ARPA-E) to develop a breakthrough battery technology that may more than double the current range of electric vehicles (EV), increase safety, reduce costs and simplify recharging.

Segre and his collaborators John Katsoudas, also of IIT, and Elena Timofeeva, Dileep Singh and Michael Duoba of [Argonne National Laboratory](#) will develop a prototype for a rechargeable "nanoelectrofuel" flow battery that may extend the range of EVs to at least 500 miles and provide a straightforward and rapid method of refueling. Current EV ranges are 100-200 miles, with recharging taking up to eight hours.

Flow batteries, which store chemical energy in external tanks instead of within the battery container, are generally low in energy density and therefore not used for transportation applications. The IIT-Argonne nanoelectrofuel flow battery concept will use a high-energy density "liquid" with battery-active nanoparticles to dramatically increase energy density while ensuring stability and low-resistance flow within the battery.

"I am delighted by this award, not only because of the



Researchers (left to right) Dileep Singh, Carlo Segre, Mike Duoba, John Katsoudas, Elena Timofeeva, and Chris Pelliccione stand by one of the plug-in electric vehicles they hope to revolutionize with the IIT-Argonne "nanoelectrofuel" flow battery technology they are developing. [Click to enlarge.](#)

CONTACT US

For more information, contact Angela Hardin at (630-252-5501; media@anl.gov) or Patricia Cronin at (312-567-3132; cronin@iit.edu).

3. NANOMATERIALES PARA EFICIENCIA ENERGÉTICA

Otro campo ignorado por la industria del sector (excepto aplicaciones nicho) y actualmente en despegue:

- LEDs
- OLEDs
- OLEDs poliméricos

Evolución tecnológica de la iluminación



LEDs
Blancos
Brillantes
Baratos, y...

The LED advantage Measured in lumens per Watt

Oil lamp 15,000 BC



Light bulb 19th century



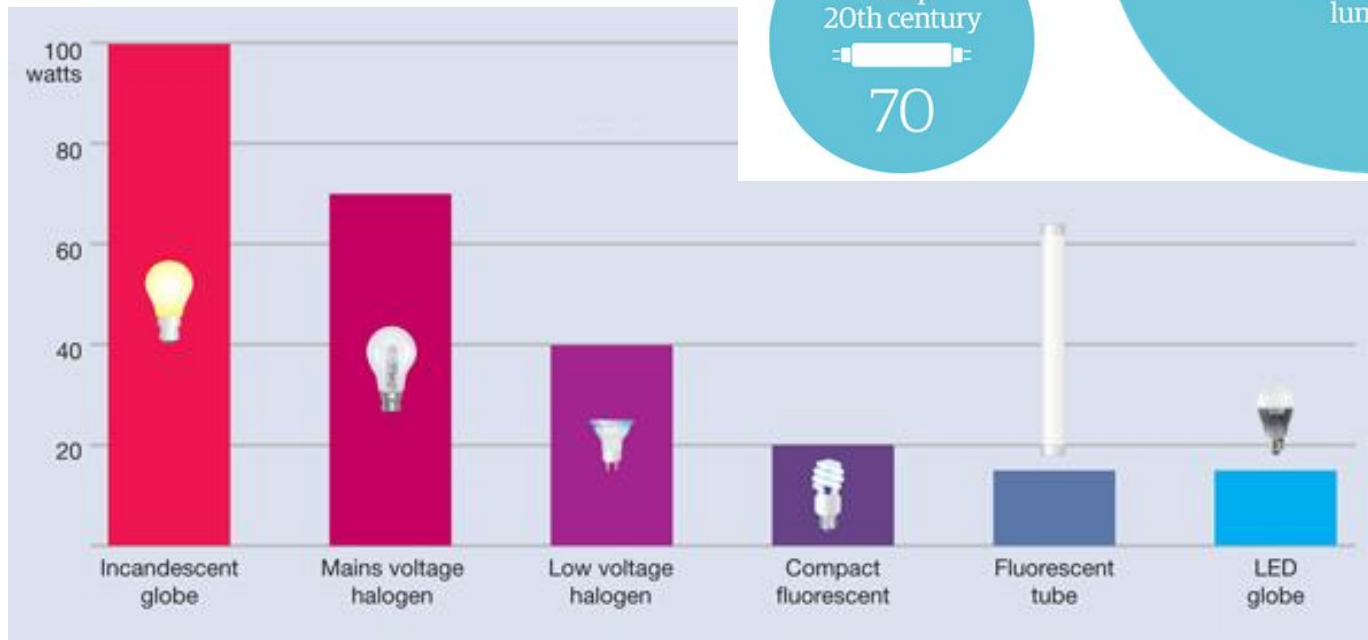
Fluorescent lamp
20th century



LED
21st century



300
lumens



The blue light-emitting diode



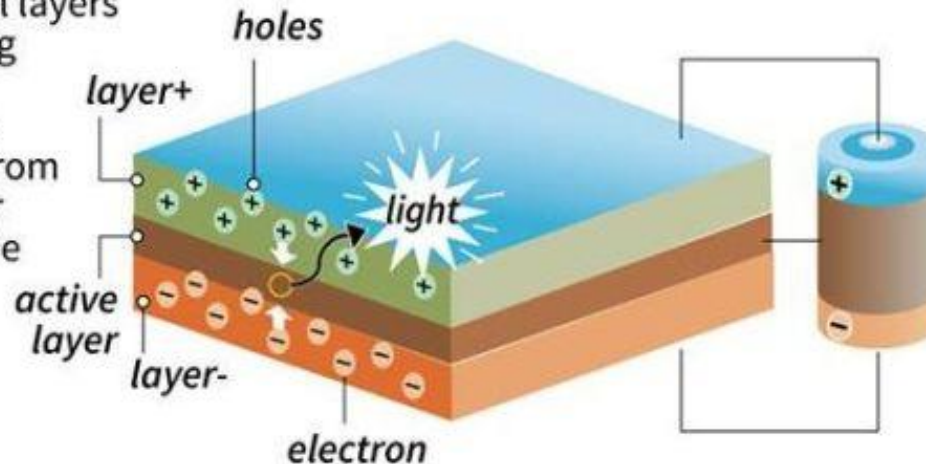
Work by **Isamu Akasaki**, **Hiroshi Amano** and **Shuji Nakamura** led to a low-energy source of light for illuminating homes and computers, as well as a weapon against global warming



A light-emitting diode (LED)

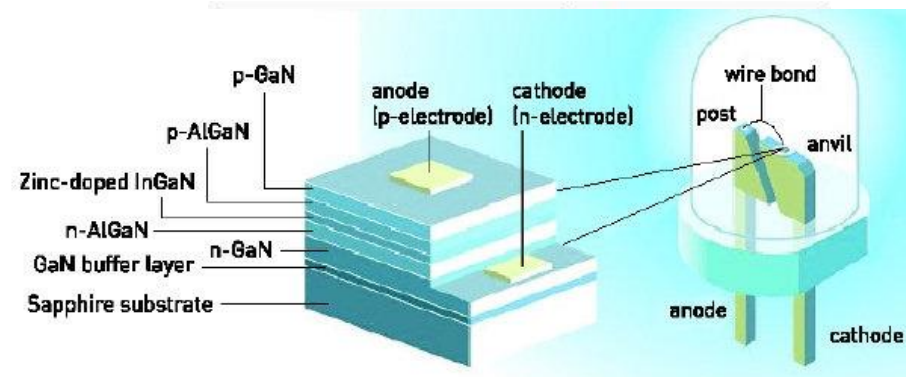
consists of several layers of semiconducting materials.

Electrical voltage drives electrons from the negative layer and holes from the positive layer to the active layer, where they recombine to emit light



GaN → efficient blue LEDs

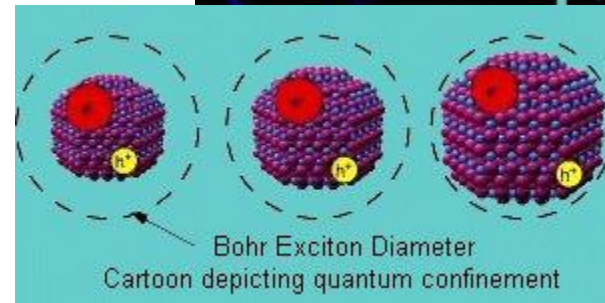
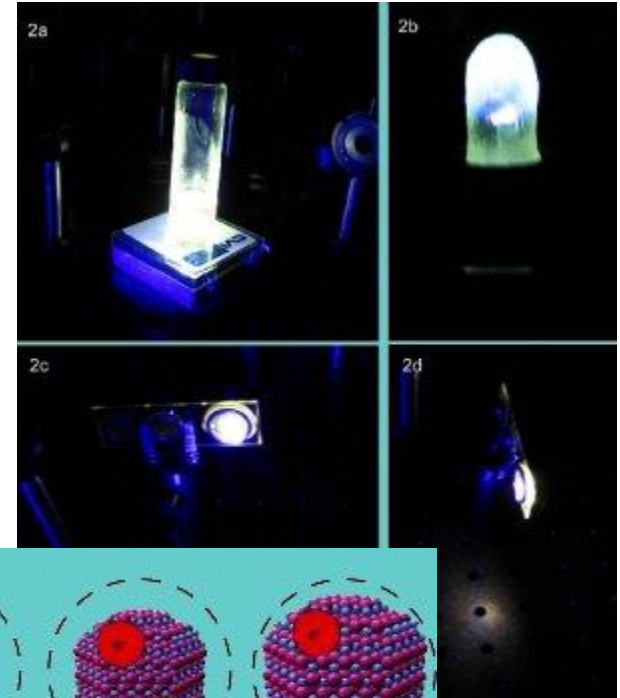
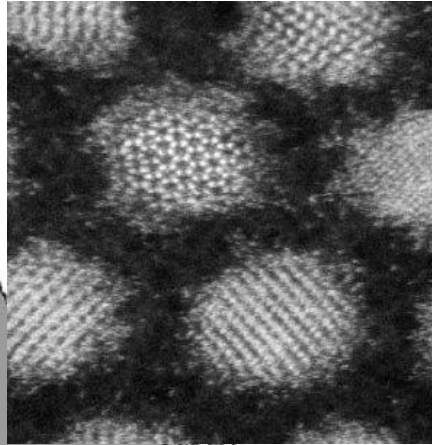
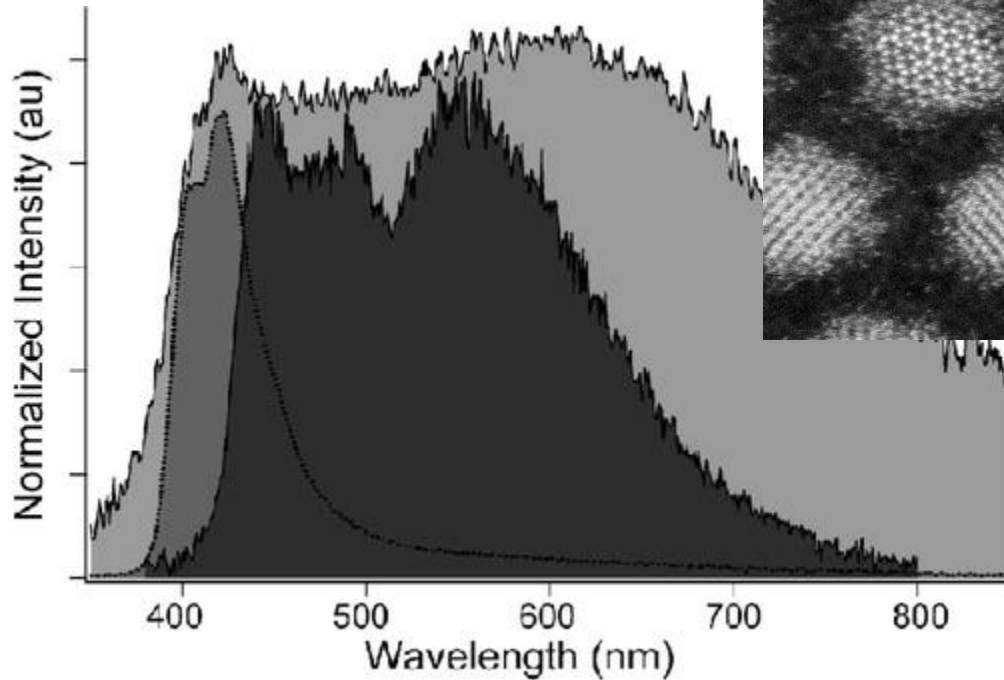
InGaN quantum wells



¿LEDs or Fluorescencia?

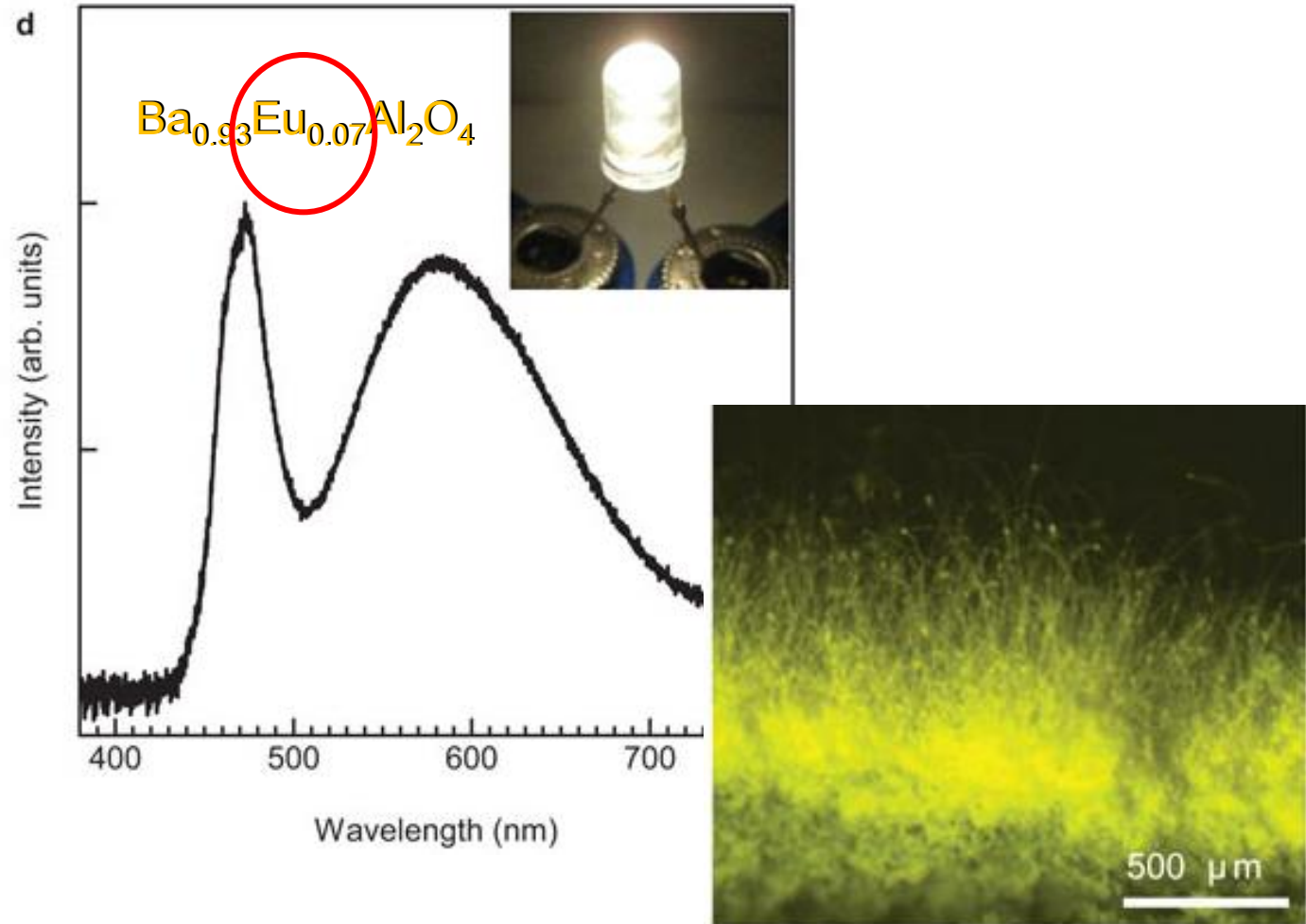
||

||



White Light-Emitting Diodes Based on Ultrasmall [CdSe Nanocrystal](#) Electroluminescence
Michael A. Schreuder et al. (Sandra Rosenthal) *Nano Lett.*, 2010, 10 (2), pp 573–576

Inorganic LEDs



Energy Critical Elements

1 H Hydrogen 1.01																	2 He Helium 4.00														
3 Li Lithium 6.94	4 Be Beryllium 9.01																	5 B Boron 10.81	6 C Carbon 12.01	7 N Nitrogen 14.01	8 O Oxygen 16.00	9 F Fluorine 19.00	10 Ne Neon 20.18								
11 Na Sodium 22.99	12 Mg Magnesium 24.31																	13 Al Aluminum 26.98	14 Si Silicon 28.09	15 P Phosphorus 30.97	16 S Sulfur 32.07	17 Cl Chlorine 35.45	18 Ar Argon 39.95								
19 K Potassium 39.10	20 Ca Calcium 40.08	21 Sc Scandium 44.96	22 Ti Titanium 47.87	23 V Vanadium 50.94	24 Cr Chromium 52.00	25 Mn Manganese 54.94	26 Fe Iron 55.85	27 Co Cobalt 58.93	28 Ni Nickel 58.69	29 Cu Copper 63.55	30 Zn Zinc 65.39	31 Ga Gallium 69.72	32 Ge Germanium 72.61	33 As Arsenic 74.92	34 Se Selenium 78.96	35 Br Bromine 79.90	36 Kr Krypton 83.80														
37 Rb Rubidium 85.47	38 Sr Strontium 87.62	39 Y Yttrium 88.91	40 Zr Zirconium 91.22	41 Nb Niobium 92.91	42 Mo Molybdenum 95.94	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29														
55 Cs Cesium 132.91	56 Ba Barium 137.33	57 La Lanthanum 138.91	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)														
87 Fr Francium (223)	88 Ra Radium (226)	89 Ac Actinium (227)	104 Rf Rutherfordium (261)	105 Db Dubnium (262)	106 Sg Seaborgium (266)	107 Bh Bohrium (264)	108 Hs Hassium (269)	109 Mt Meitnerium (268)																							
																		58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.04	71 Lu Lutetium 174.97
																		90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (262)

Energy Critical Elements: Securing Materials for Emerging Technologies
 APS-MRS. R. Jaffe et al., 2011

Energy Critical Elements



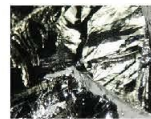
52
Te
Tellurium
127.60



TELLURIUM—brittle, silvery-white metallic element used in solar panels



32
Ge
Germanium
72.61



GERMANIUM—hard, grayish-white element with metallic luster; used in solar panels



78
Pt
Platinum
195.078



PLATINUM—silvery-white, lustrous, ductile and malleable; used in pollution control devices for cars, and in fuel cells



60
Nd
Neodymium
144.24



NEODYMIUM—bright, silvery rare-earth metal element; used in wind turbines and hybrid cars

3
Li
Lithium
6.941



LITHIUM—a soft, silver-white metallic element; used in wind turbines and lithium-ion batteries in hybrid cars



75
Re
Rhenium
186.207



RHENIUM—silvery-white metal with one of the highest melting points of all elements; used to make advanced turbines and jet engine parts



65
Tb
Terbium
158.92534



TERBIUM—a soft, silvery-white rare earth metal; used along with its fellow rare earth europium in compact fluorescent light bulbs to provide an acceptable color balance

ALIEN, our future material(s)

Abundant

Lower costs

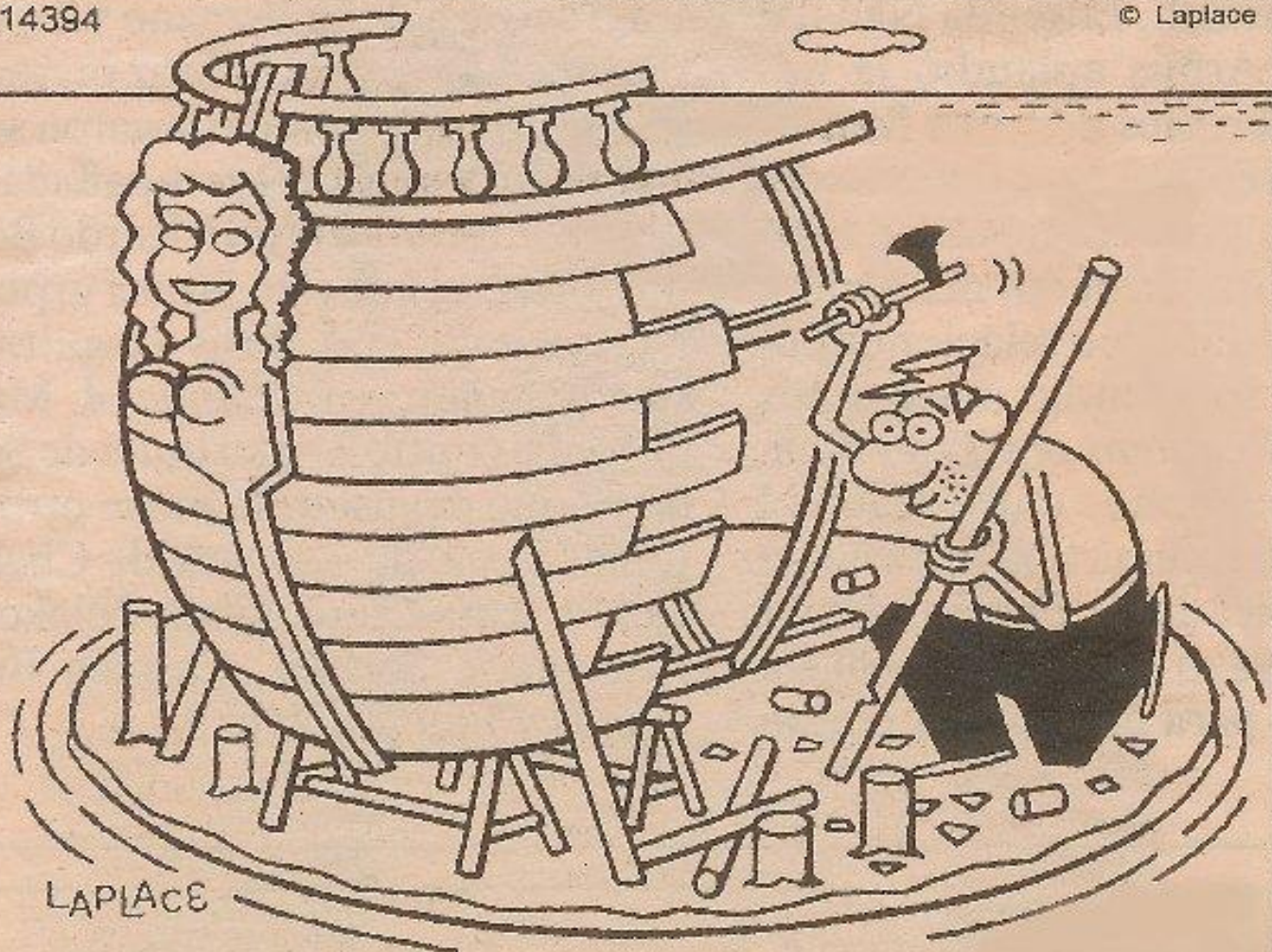
Improved performance

Environmentally friendly

New props tricks and apps

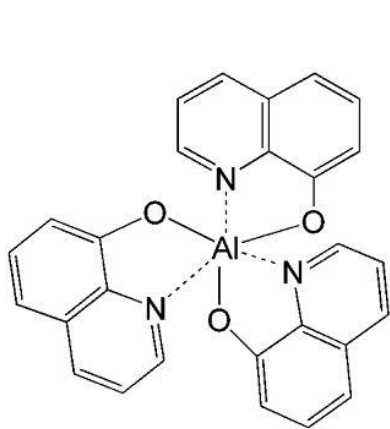
14394

© Laplace

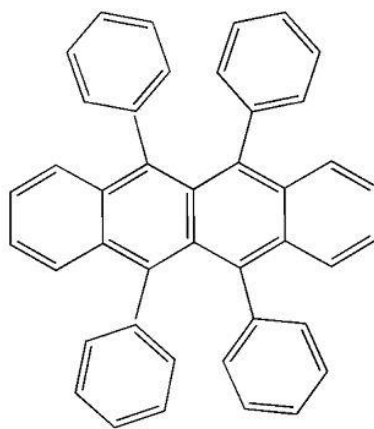


LAPLACE

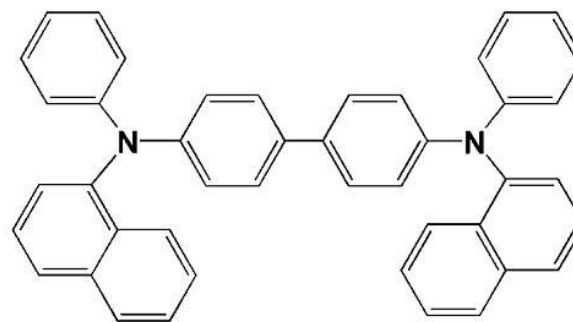
LEDs Orgánicos (OLEDs)



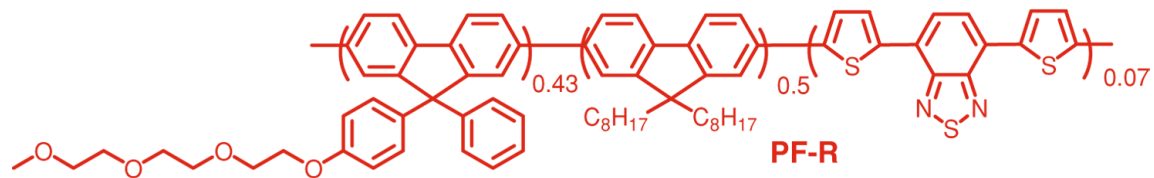
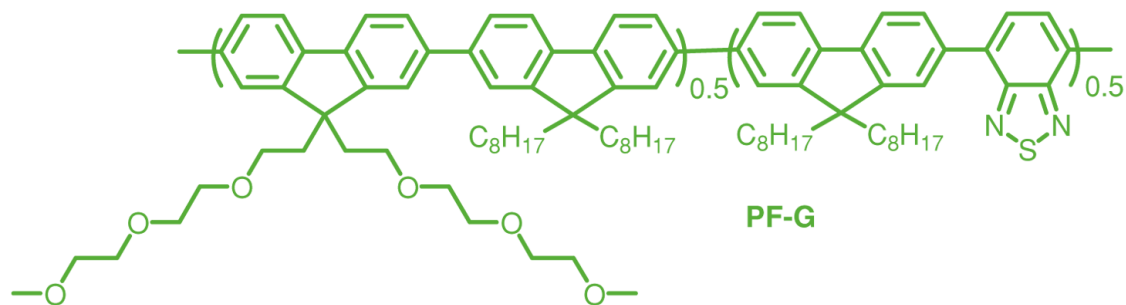
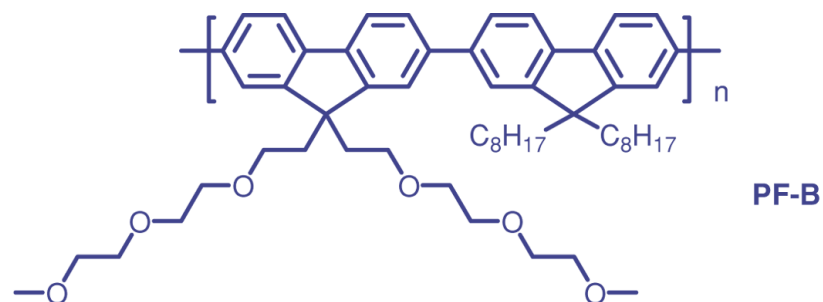
Alq₃



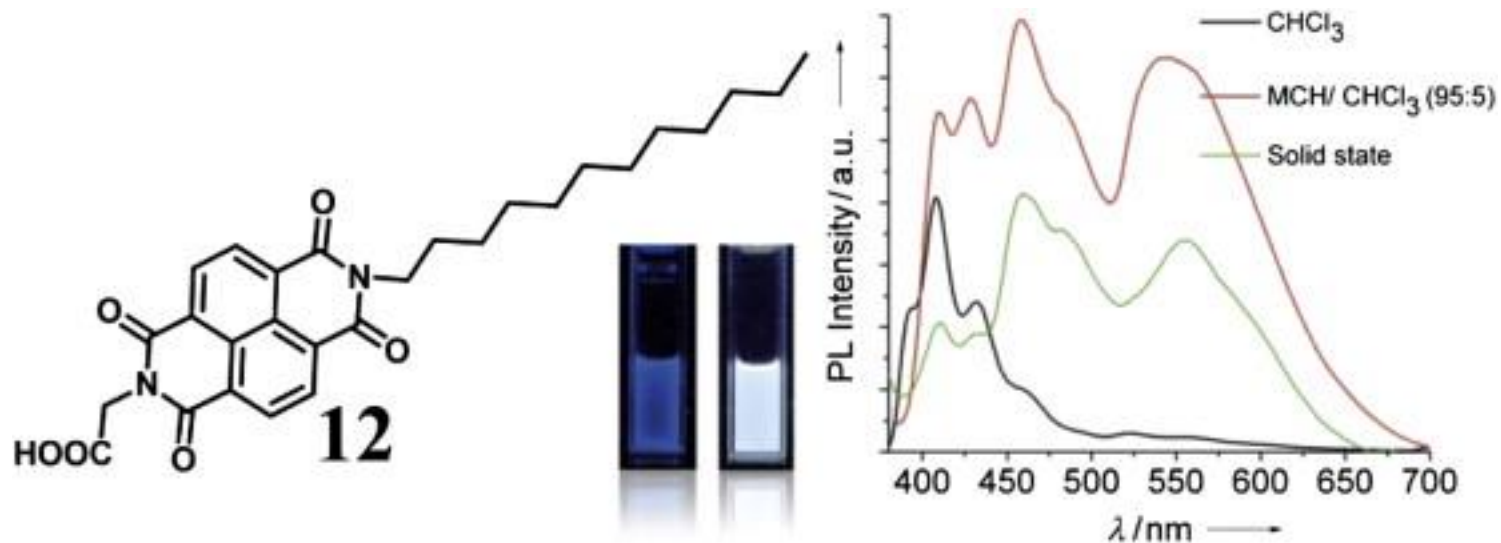
rubrene



α-NPD

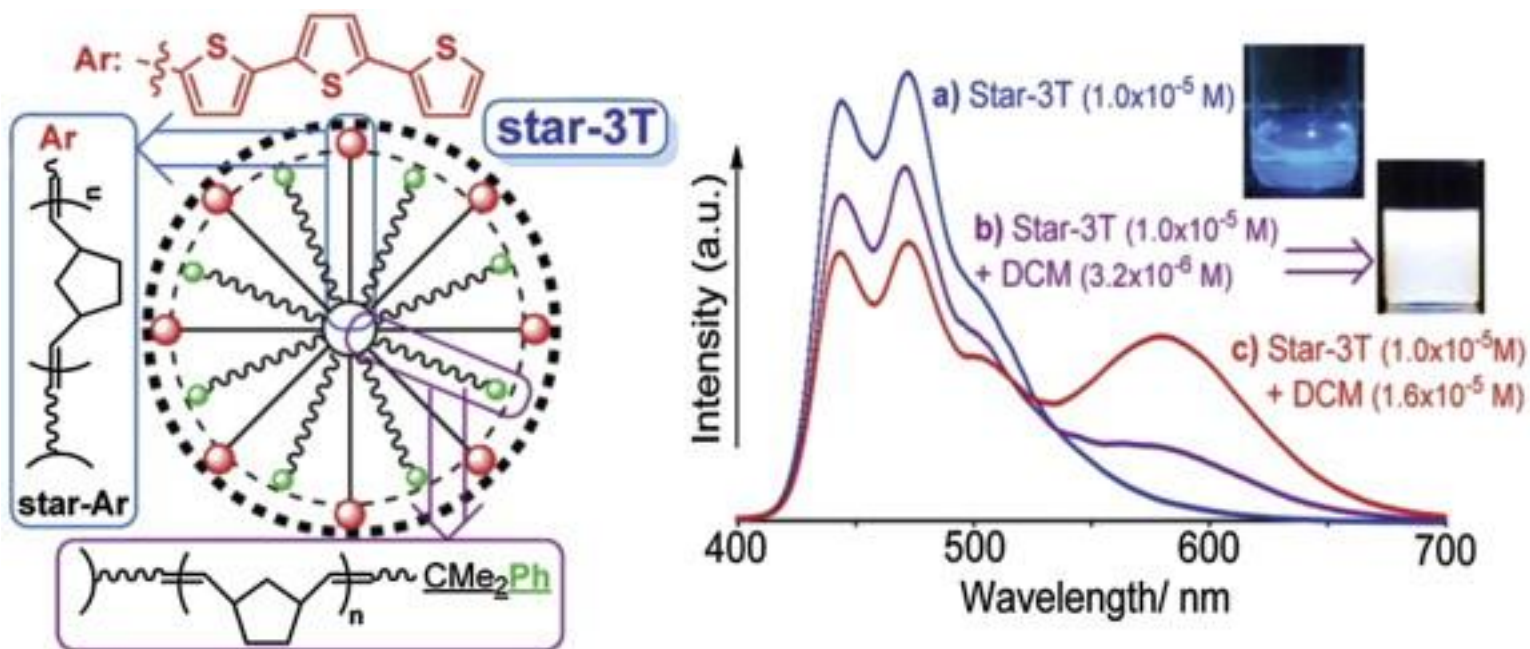


LEDs Orgánicos (OLEDs)



Molecular formula of **12** (left). Photographs of **12** in CHCl_3 and MCH/ CHCl_3 (95:5) solutions (middle). Emission spectra of **12** in different conditions upon excitation at 340 nm (right). Adapted with permission from reference 33. Copyright 2012, John Wiley and Sons

LEDs Orgánicos (OLEDs)



Left; Schematic representation of the star shape structure of the **star-3T** polymer. Right; fluorescence spectra of **star-3T** upon addition of **DCM** in THF at 25 °C with excitation at 380 nm. Adapted with permission from reference 76. Copyright 2012, American Chemical Society. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Conclusión...

Vamos a asistir (estamos ya asistiendo) a una Re-evolución tecnológica hacia un modelo sostenible de generación, almacenamiento y consumo de energía, basada en revoluciones científicas en las que los nanomateriales tendrán mucho que aportar:

Biocombustibles de 2ª y 3ª generación

H₂ Generación sostenible y almacenamiento eficiente

Pilas de combustible baratas, sin Pt

Si solar barato / materiales alternativos / Nanopartículas

Baterías de alta densidad de potencia y recarga rápida

Supercondensadores de alta densidad de energía

LEDs blancos, brillantes y baratos

QUESTIONS

BIBLIOGRAFÍA

GENERAL

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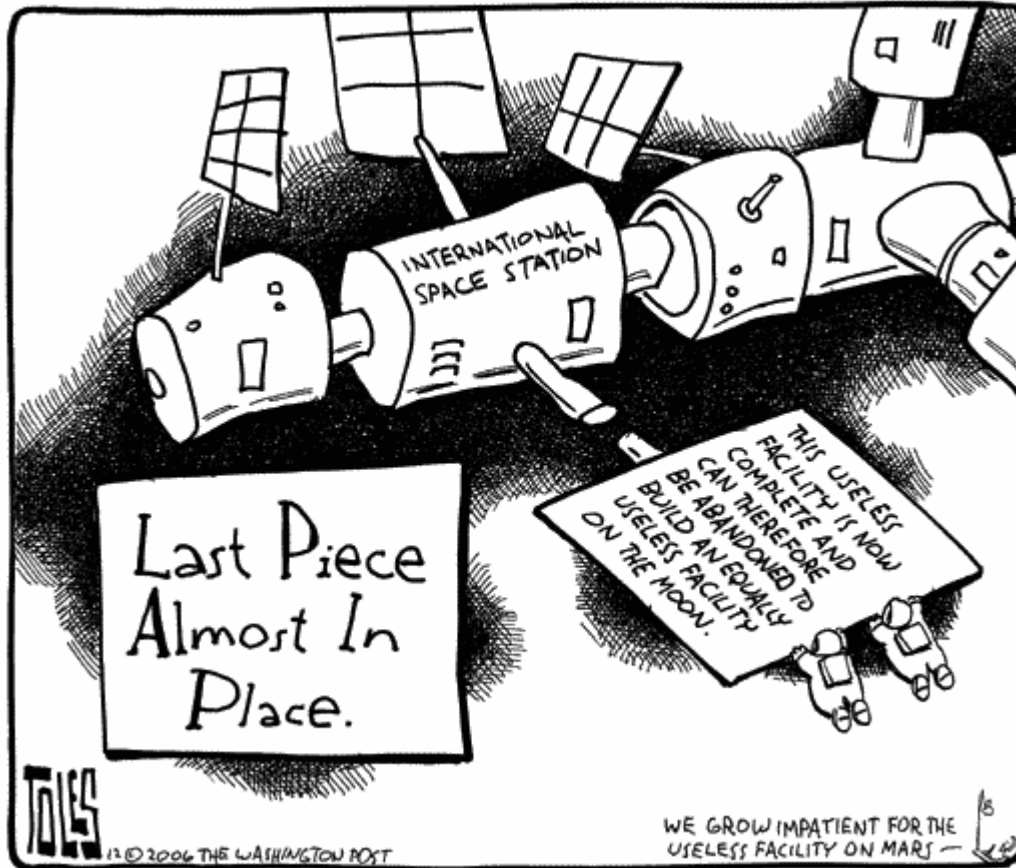
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